

Complex geometric decomposition test case set

Peng Xicheng *pxc* 417@126.com

This document is a collection of test cases for the paper "Complex Number Proof and Quantitative Extension of Geometric Theorems".

To verify the effectiveness of the proposed algorithm, we applied it to solve and expand more than 100 typical geometry problems, including middle school textbooks, college entrance examination questions, and representative questions from domestic and foreign mathematics competitions. Considering that this study involves a large amount of symbolic operations, we chose *Mathematica* as the experimental platform.

The experimental results show that more than 90% of the geometry problems mainly study the properties of special figures. Even if some questions involve quantitative relationships, they are still discussed based on special figures. The study of generalized conclusions is relatively complex and difficult to handle with traditional methods, indicating that there is still a lot of room for research in this field. Surprisingly, although the selected geometry examples are all from formal sources, there are still at least 10 (about 10%) that are debatable, such as redundant conditions and insufficient consideration of degenerate conditions. Another 3 questions failed to generate identities and quantitative expansion failed, mainly because they involved high-order equations and could not identify the geometric meaning of non-degenerate conditions, which required manual assistance and algorithm improvement. Regarding non-degenerate conditions, except for a few geometry propositions (less than 5%) that are generally valid and do not need to consider non-degenerate conditions, other propositions more or less need to consider this condition. Fortunately, most non-degenerate conditions are easy to identify, usually manifested as two points coincident, three points collinear, two lines parallel, etc. In very few problems, the geometric meaning of their degenerate conditions is difficult to find.

Proof: Using the complex number method, use lowercase letters x to represent X the complex number of a point, and use $\bar{x}$ to represent $\bar{X}$ the conjugate complex number of a complex number. The coordinates of point X are $X = (x, \bar{x})$. Solution:

First calculate the coordinates of some points, then establish the identity, and finally interpret the geometric meaning of the identity.

Easy Questions

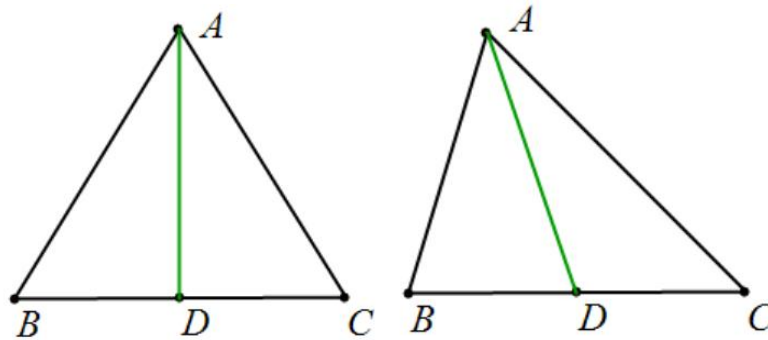


Figure 1

Example 1 : As shown in Figure 1, in $\triangle ABC$, D is the midpoint of BC . If $AB = AC$, prove that: $AD \perp BC$.

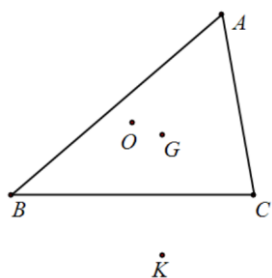
Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, $d = \frac{b+c}{2}$,

generating the identity equation:

$$|b-c|^4 \left(\frac{\frac{b-a}{c-b} - \overline{\left(\frac{b-a}{c-b} \right)}}{\frac{c-a}{c-a}} \right) = \frac{(a-b)(a-c)(b-c)}{abc} |b-c|^2 \left(\frac{a-d}{b-c} + \overline{\left(\frac{a-d}{b-c} \right)} \right),$$

Extended proposition: In $\triangle ABC$, D is the midpoint of BC , prove:

$$BC \sin(B-C) = 4AD \sin A \cos \angle ADB.$$



Example 2 : Given that the symmetric point of the centroid G of triangle ABC about the side BC is K , prove that the necessary and sufficient condition for the four points A, B, K , and C to be cocircular is $AB^2 + AC^2 = 2BC^2$. (The four points being cocircular is symmetric to the centroid)

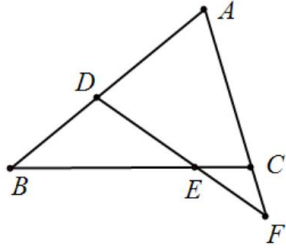
Proof: Let $A = \left(a, \frac{1}{a}\right)$, $B = \left(b, \frac{1}{b}\right)$, $C = \left(c, \frac{1}{c}\right)$, $G = \left(\frac{a+b+c}{3}, \frac{1/a + 1/b + 1/c}{3}\right)$,

$K = \left(\frac{2ab + 2ac - bc}{3a}, \frac{-a + 2b + 2c}{3bc}\right)$, the identity

$$\frac{2}{9} \left(2(c-b)(\overline{c-b}) - (a-b)(\overline{a-b}) - (a-c)(\overline{a-c}) \right) = 1 - k\bar{k},$$

Extended proposition: It is known that the symmetric point of the centroid G of $\triangle$

ABC about the side BC is K , prove: $\frac{2}{9} (2BC^2 - AB^2 - AC^2) = OA^2 - OK^2$.



Example 3 : In triangle ABC , D is on line segment AB , E is on line segment BC , and F is on the extension of line segment AC . If D , E , and F are collinear, prove that: $\frac{AD \cdot BE \cdot CF}{DB \cdot EC \cdot FA} = 1$.

(Complex number identities extend the Menelaos theorem)

Proof: Let $A = \left(a, \frac{1}{a}\right)$, $B = \left(b, \frac{1}{b}\right)$, $C = \left(c, \frac{1}{c}\right)$, $D = \left(d, \frac{a+b-d}{ab}\right)$,

$E = \left(e, \frac{b+c-e}{bc}\right)$, $F = \left(f, \frac{a+c-f}{ac}\right)$,

$$(d-b)(e-c)(f-a) - (a-d)(b-e)(f-c) = abc[DEF],$$

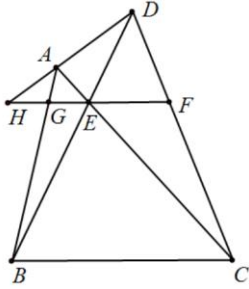
Extended proposition: In $\triangle ABC$, D is on line segment AB , E is on line segment BC , and F is on the extended line segment AC . If D , E , and F are collinear, prove:

$DB \cdot EC \cdot FA - AD \cdot BE \cdot CF = 4R \cdot S_{\triangle DEF}$, where R is the radius of the circumscribed circle of $\triangle ABC$.

Example 4 : As shown in the figure, in quadrilateral $ABCD$, AC and BD intersect at E . Draw $HF \parallel BC$ through E , intersecting CD and AB at F and G respectively, and intersecting the extension of DA at H . Prove that:

$EF = 2 EG \Leftrightarrow EG = HG$. (Complex number identities are mutually extended by the

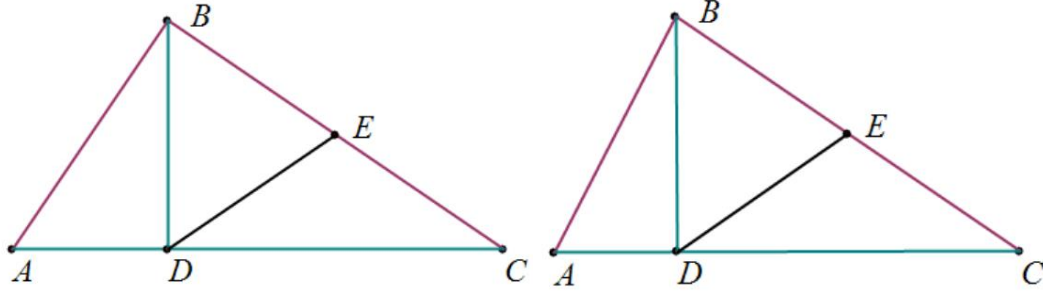
length of the line segment)



Proof: Identity $([ABC] - [DBC])((e - g) - (g - h)) = [ABD](2(e - g) - (f - e))$,

As shown in the figure, in quadrilateral $ABCD$, AC and BD intersect at E . Draw $HF \parallel BC$ through E , which intersects CD and AB at F and G respectively, and intersects the extension line of DA at H . Prove:

$(S_{\triangle ABC} - S_{\triangle DBC})(EG - GH) = S_{\triangle ABD}(FE - 2EG)$.



Example 5 : As shown in the figure, in $\triangle ABC$, $\angle ABC = 90^\circ$, $BD \perp AC$, D is the foot of the perpendicular, E is the midpoint of BC , prove: $\angle EDC = \angle ABD$.
(2016 Jiangsu Province College Entrance Examination Question)

Proof: Let DC be the real axis and DB be the imaginary axis, $A = (a, a)$,

$$B = (b, -b) \quad , \quad C = (c, c) \quad , \quad D = (0, 0) \quad , \quad E = \left(\frac{b+c}{2}, \frac{b-c}{2} \right) \quad , \quad \text{calculate}$$

$$(b-a)\overline{(b-a)} \left(\frac{b-c}{b-a} + \overline{\left(\frac{b-c}{b-a} \right)} \right) = 2(ac-b^2),$$

$$(d-c)(b-d)\overline{(d-c)(b-d)} \left(\frac{\frac{d-e}{b-d} - \overline{\left(\frac{d-e}{b-d} \right)}}{\frac{b-a}{b-d}} \right) = bc(ac-b^2) \quad , \quad \text{generating the}$$

identity:

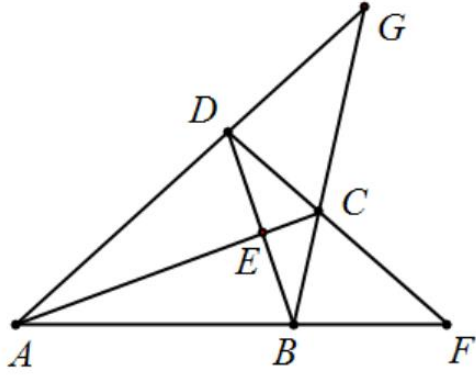
$$bc(b-a)\overline{(b-a)} \left(\frac{b-c}{b-a} + \overline{\left(\frac{b-c}{b-a} \right)} \right) = 2(d-c)(b-d)\overline{(d-c)(b-d)} \left(\frac{\frac{d-e}{b-d} - \overline{\left(\frac{d-e}{b-d} \right)}}{\frac{b-a}{b-d}} \right) ,$$

$$DB \cdot DC \cdot BA \cdot BC \cos B = 2DE \cdot DC \cdot BD \cdot BA \sin(\angle EDC - \angle ABD) \quad , \quad \text{simplified to}$$

$$\cos \angle B = \sin(\angle EDC - \angle ABD).$$

Extended proposition: As shown in the figure, in $\triangle ABC$, $BD \perp AC$, D is the foot of the perpendicular, E is the midpoint of BC , prove:

$$\cos \angle B = \sin(\angle EDC - \angle ABD).$$



Example 6 : Given quadrilateral $ABCD$, no three points of its four vertices are collinear, AC intersects BD at E , AB intersects CD at F , and AD intersects BC at G . Prove that: E , F , and G are not collinear .

Proof: Let $A = (0, 0)$, $B = (1, 1)$, $C = (c, \bar{c})$, $D = (d, \bar{d})$

$$E = \left(-\frac{c(d - \bar{d})}{-c + \bar{c} - \bar{c}d + c\bar{d}}, \frac{\bar{c}(d - \bar{d})}{c - \bar{c} + \bar{c}d - c\bar{d}} \right) , \quad F = \left(\frac{-\bar{c}d + c\bar{d}}{c - \bar{c} - d + \bar{d}}, \frac{-\bar{c}d + c\bar{d}}{c - \bar{c} - d + \bar{d}} \right) ,$$

$$G = \left(\frac{(c - \bar{c})d}{d - \bar{c}d - \bar{d} + c\bar{d}}, \frac{(c - \bar{c})\bar{d}}{d - \bar{c}d - \bar{d} + c\bar{d}} \right),$$

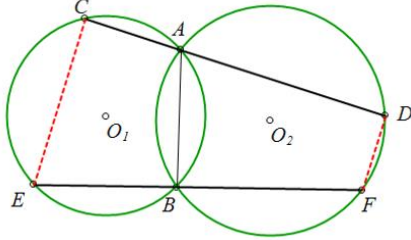
$$[EFG]([ABC] - [ABD])([BCD] - [BCA])([ACD] + [ABC])$$

$$= 2[ABC][ABD][BCD][ACD].$$

Extended proposition: In quadrilateral $ABCD$, the four vertices do not have three collinear points. AC intersects BD at E , AB intersects CD at F , and AD intersects BC at G . Prove that: $S_{\triangle EFG}(S_{\triangle ABC} - S_{\triangle ABD})(S_{\triangle BCD} - S_{\triangle BCA})(S_{\triangle ACD} + S_{\triangle ABC})$

$$= 2S_{\triangle ABC}S_{\triangle ABD}S_{\triangle BCD}S_{\triangle ACD}.$$

Example 7 : Two circles intersect at points A and B . Draw straight lines through A and B that intersect the two circles at points C , D , E , and F respectively. Prove that: $CE \parallel DF$.



Proof: Let the circumscribed circle of $\triangle ABF$ be the unit circle,

$$A = \left(a, \frac{1}{a}\right), B = \left(b, \frac{1}{b}\right), C = \left(c, \frac{a-c+d}{ad}\right), D = \left(d, \frac{1}{d}\right),$$

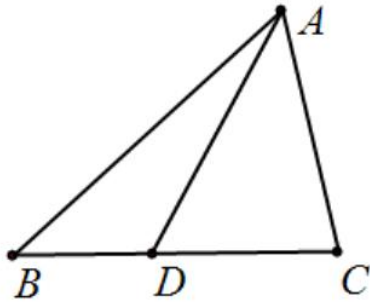
$$E = \left(e, \frac{b-e+f}{bf}\right), F = \left(f, \frac{1}{f}\right),$$

$$(f-d)\overline{(f-d)}\left(\frac{e-c}{f-d} - \overline{\left(\frac{e-c}{f-d}\right)}\right)(a-b)(a-c)(b-e) = [ABEC]ab(d-f),$$

Extended proposition: In the circle inscribed quadrilateral $ABFD$, take point E on the extension line of FB and point C on the extension line of DA , prove:

$$2|S_{\triangle EFD} - S_{\triangle CFD}| = DF \cdot EC |\sin(\angle BAC + \angle BEC)|.$$

Note: Further simplify it to, $\angle BAC + \angle BEC = \pi + \angle(FD, EC)$, turns a circle and returns to the starting point.



Example 8 : In $\triangle ABC$, AD is the angle bisector, then $\frac{AB}{AC} = \frac{BD}{DC}$.

Assume BC as the real axis and point A on the imaginary axis, generating the

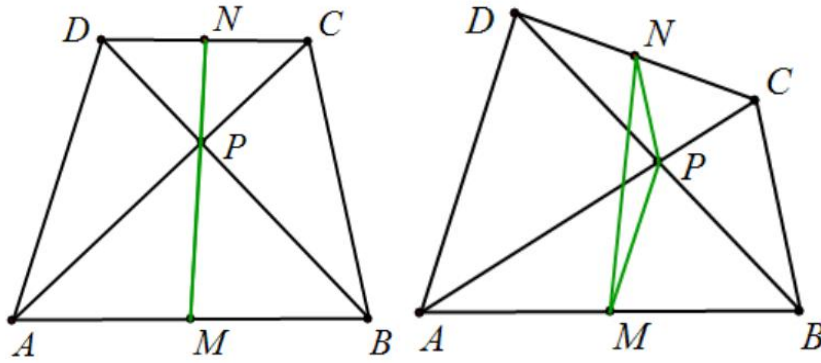
identity equation, $(b-c)^2(a-d)^2(a+d)^2 \left(\frac{\frac{a-c}{a-d}}{\frac{a-b}{a-d}} - \overline{\left(\frac{\frac{a-c}{a-d}}{\frac{a-b}{a-d}} \right)} \right)$

$$= [ABC]((a-c)(-a-c)(d-b)(d-b) - (a-b)(-a-b)(d-c)(d-c)).$$

Extended proposition: In $\triangle ABC$, D is a point on line segment BC , then

$$AD^2 \cdot BC^2 \sin(\angle CAD - \angle DAB) = \sin A (AB^2 \cdot CD^2 - AC^2 \cdot BD^2).$$

The angle in the conclusion is a directed angle, and when AD is the exterior angle bisector, the same holds true $\frac{AB}{AC} = \frac{BD}{DC}$.



Example 9 : Quadrilateral $ABCD$, AC intersects BD at P , M and N are the midpoints of AB and CD respectively. If $AB \parallel DC$, prove that: M , N , and P are collinear.

Proof: Use the complex number method, using lowercase letters x to represent X the complex number of a point and using $\bar{x}$ to represent x the conjugate complex number of the complex number.

Let P be the origin, AC be the real axis, $\bar{d} = \frac{\bar{b}d}{b}$, $m = \frac{a+b}{2}$, $n = \frac{c+d}{2}$,

$$[DAB] - [CAB] = -\frac{(b - \bar{b})(bc - ad)}{b}, \quad [MPN] = -\frac{(b - \bar{b})(bc - ad)}{4b}, \quad \text{so}$$

$$[DAB] - [CAB] = 4[MPN].$$

Extended proposition: Quadrilateral $ABCD$, AC intersects BD at P , M and N are the midpoints of AB and CD respectively, prove: $S_{\triangle DAB} - S_{\triangle CAB} = 4S_{\triangle MPN}$.

Example 1 0 : As shown in the figure, in $\triangle ABC$, D is a point on the line segment AB . Prove: The necessary and sufficient condition for $(DA \cdot BC)^2 + (DB \cdot AC)^2 = (AB \cdot DC)^2$ $CA \perp CB$ is .

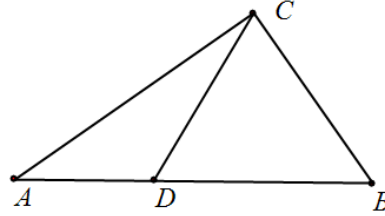


Fig. 32

Explanation: When D is at the midpoint of AB , it is the Pythagorean theorem.

Proof: Let line AD be the real axis and point C be on the imaginary axis

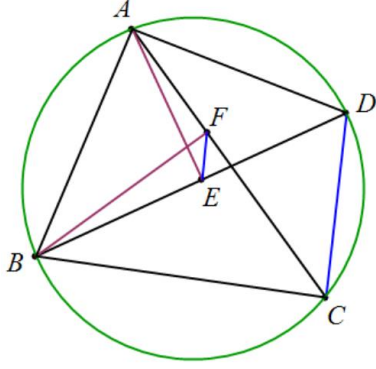
$$(d-a)(d-a)(b-c)(b+c) + (d-b)(d-b)(a-c)(a+c) - (b-a)(b-a)(d-c)(d+c)$$

$$= (a-d)(b-d)(c-a)(-c-a) \left(\frac{c-b}{c-a} + \frac{-c-b}{-c-a} \right).$$

$$(DA \cdot BC)^2 + (DB \cdot AC)^2 - (AB \cdot DC)^2 + 2CA \cdot CB \cdot DA \cdot DB \cos \angle ACB = 0.$$

When D is on line segment AB ,

$$(DA \cdot BC)^2 + (DB \cdot AC)^2 - (AB \cdot DC)^2 = 2CA \cdot CB \cdot DA \cdot DB \cos \angle ACB.$$



Example 11 : As shown in the figure, inscribed in the circle is quadrilateral $ABCD$, $AE \perp BD$, $BF \perp AC$, prove that: $EF \perp CD$.

$$A = (a, \bar{a}) \quad , \quad B = (b, \bar{b}) \quad , \quad C = (c, \bar{c}) \quad , \quad D = (d, \bar{d}) \quad ,$$

$$E = \left(\frac{\bar{a}b + a\bar{b} - \bar{a}d + \bar{b}d - a\bar{d} - b\bar{d}}{2(\bar{b} - \bar{d})}, \frac{\bar{a}b + a\bar{b} - \bar{a}d - \bar{b}d - a\bar{d} + b\bar{d}}{2(b - d)} \right) \quad ,$$

$$F = \left(\frac{\bar{a}b + a\bar{b} + \bar{a}c - \bar{b}c - a\bar{c} - b\bar{c}}{2(\bar{a} - \bar{c})}, \frac{\bar{a}b + a\bar{b} - \bar{a}c - \bar{b}c + a\bar{c} - b\bar{c}}{2(a - c)} \right) \quad , \quad \text{identities}$$

$$|a - c|^2 [ABD] ([ECD] - [FCD]) = [ACE] |a - b|^2 |c - d|^2 \left(\frac{a - d}{a - b} \frac{c - b}{c - d} - \overline{\left(\frac{a - d}{a - b} \frac{c - b}{c - d} \right)} \right).$$

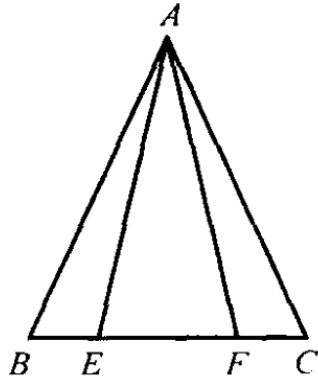
Extended proposition: As shown in the figure, quadrilateral $ABCD$, $AE \perp BD$, $BF \perp AC$, prove: $2AC^2 \cdot S_{\triangle ABD} |S_{\triangle ECD} - S_{\triangle FCD}| = S_{\triangle ACE} AB \cdot BC \cdot CD \cdot DA |\sin(A + C)|$.

Example 12 : In triangle ABC , $AB = AC$, $\angle ACB = 90^\circ$, $\angle EAF = 45^\circ$, E and F are

two points on the base BC (E is between B and F). Prove: $EF^2 = BE^2 + FC^2$.

triangle ABC , $AB = AC$, E and F are two points on the base BC (E is between B and F). Prove that: $\angle EAF = \frac{\angle BAC}{2}$ The necessary and sufficient condition for

is $EF^2 = BE^2 + FC^2 + 2BE \cdot FC \cos \angle BAC$.

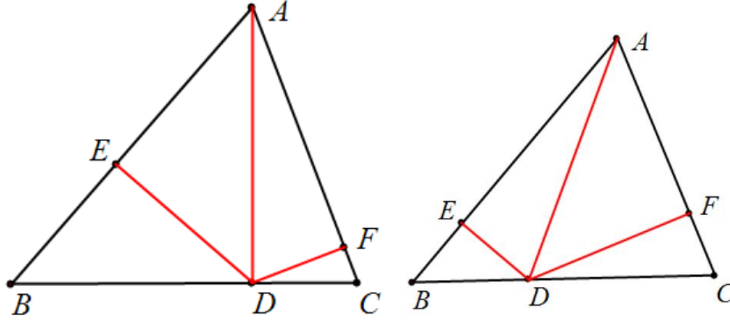


$$(e-f)^2 - (b-e)^2 - (f-c)^2 - 2(b-e)(f-c) \left(\frac{\frac{a-c}{a-b} + \frac{-a-c}{-a-b}}{2} \right)$$

$$= -\frac{2c}{a(a-c)(a+c)} (a-e)(-a-e)(a-c)(-a-c) \left(\frac{\frac{a-f}{a-c}}{\frac{a-0}{a-0}} - \frac{\overline{\frac{a-f}{a-c}}}{\overline{\frac{a-0}{a-0}}} \right),$$

triangle ABC , $AB = AC$, E and F are two points on the base BC (E is between B and F). Prove:

$$EF^2 - (BE^2 + FC^2 + 2BE \cdot FC \cos \angle BAC) = 4AF \cdot AE \cos C \sin \left(\angle EAF - \frac{\angle BAC}{2} \right).$$



Example 13 : In triangle ABC , AD is the altitude, point D is on BC , draw $DE \perp AB$ at E from D , and draw $DF \perp AC$ at F .

Prove that: Points B, C, F and E form a common circle.

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle,

$$D = \left(d, \frac{b+c-d}{bc} \right), \quad E = \left(\frac{-ab+bc+ad+cd}{2c}, \frac{ab+2ac+bc-ad-cd}{2abc} \right),$$

$$F = \left(\frac{-ac+bc+ad+bd}{2b}, \frac{2ab+ac+bc-ad-bd}{2abc} \right),$$

Identity:

$$\begin{aligned} & -\frac{8bc}{(b-d)(c-d)}(e-b)(c-f)\overline{(e-b)(c-f)} \left(\frac{e-f}{e-b} \frac{c-b}{c-f} - \overline{\left(\frac{e-f}{e-b} \frac{c-b}{c-f} \right)} \right) \\ &= \left(\frac{\frac{b-c}{d-a} + \overline{\left(\frac{b-c}{d-a} \right)}}{\frac{b-c}{d-a} - \overline{\left(\frac{b-c}{d-a} \right)}} \right) |b-c|^2 \left(\frac{b-a}{b-c} + \overline{\left(\frac{b-a}{b-c} \right)} \right) |c-a|^2 \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a} \right)} \right), \end{aligned}$$

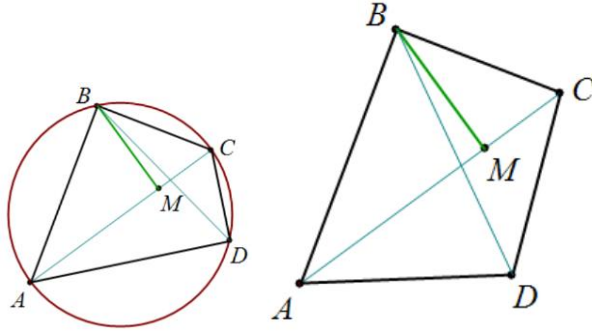
Extended proposition: In acute angle $\triangle ABC$, point D is on BC . From D , draw $DE \perp AB$ at E , and draw $DF \perp AC$ at F .

Verification: $EF \sin(\angle BEF + \angle C) = BC \sin B \sin C \cot \angle ADB$.

Example 1 4 : As shown in Figure 1, inscribed in a circle is a quadrilateral $ABCD$, $AB = BD$, $BM \perp AC$, and the foot of the perpendicular is M . Proof: $AM = DC + CM$. (Archimedes' Broken Line Theorem)

Proof: Let circle O be the unit circle, $b=1$, $d=1/a$,
 $m = \frac{a+b+c-ac/b}{2} = \frac{1}{2}(1+a+c-ac)$,

$$|c-d| = \left| \frac{-1+ac}{a} \right| = |-1+ac| = |(a-m)-(m-c)|.$$



Assume $b=0$, $\bar{b}=0$, $\bar{a}=a$, $d=\frac{a^2}{d}$, there is an identity

$$\begin{aligned} & (b-a)(d-c)\overline{(b-a)(d-c)} \left(\frac{b-c}{d-c} - \overline{\left(\frac{b-c}{d-c} \right)} \right) \begin{vmatrix} a & \bar{a} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \\ &= (a-d)(a-c)\overline{(a-c)(a-d)} \left((a-m-(m-c))\overline{(a-m-(m-c))} - (d-c)\overline{(d-c)} \right), \end{aligned}$$

As shown in the figure, quadrilateral $ABCD$, $AB = BD$, $BM \perp AC$, the foot of the perpendicular is M . Proof:

$$8S_{\triangle ACD} \cdot AB \cdot BC \cdot CD \sin(\angle ABC + \angle ADC) = AD \cdot AC^2 \left((AM - MC)^2 - DC^2 \right).$$

Example 15 : In $\triangle ABC$, AD is the angle bisector, then $AD^2 = AB \cdot AC - BD \cdot DC$.

Assume BC as the real axis and point A on the imaginary axis, generating the

identity equation,

$$(a-d)^2 (a+d)^2 \left(\frac{\frac{a-c}{a-d} - \overline{\left(\frac{a-c}{a-d} \right)}}{\frac{a-b}{a-d} - \overline{\left(\frac{a-b}{a-d} \right)}} \right)$$

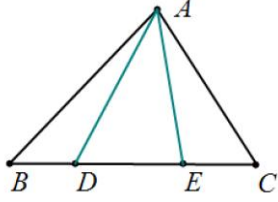
$$= \frac{2a}{b+c} \left(|a-b|^2 |a-c|^2 - \left(|a-d|^2 + (b-d)(d-c) \right)^2 \right).$$

Extended proposition: In $\triangle ABC$, M is the midpoint, D is a point on the line segment BC , then

$$2AD^2 \cdot AB \cdot AC \sin(\angle CAD - \angle DAB) = \tan \angle AMC \left((AD^2 + BD \cdot DC)^2 - AB^2 \cdot AC^2 \right).$$

Example 16 : In $\triangle ABC$, points D and E are on BC , and $\angle CAE = \angle DAB$, prove

$$\frac{AB^2}{AC^2} = \frac{BD \cdot BE}{CD \cdot CE}.$$

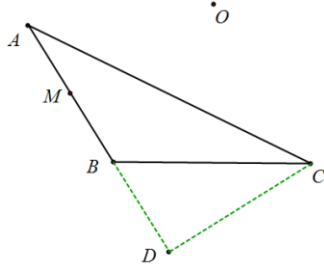


Proof: Let point A be on the imaginary axis, BC be the real axis,

$$\begin{aligned} & (a-b)\overline{(a-b)}(c-d)(c-e) - (a-c)\overline{(a-c)}(b-d)(b-e) \\ & + \frac{b-c}{2a}(a-e)\overline{(a-e)}(a-d)\overline{(a-d)} \left(\frac{a-c}{a-e} / \frac{a-d}{a-b} - \overline{\left(\frac{a-c}{a-e} / \frac{a-d}{a-b} \right)} \right) = 0. \end{aligned}$$

In $\triangle ABC$, points D and E are on BC . Prove that

$$\sin A (AB^2 \cdot CD \cdot CE - AC^2 \cdot BD \cdot BE) = BC^2 \cdot AD \cdot AE \sin(\angle CAE - \angle DAB).$$

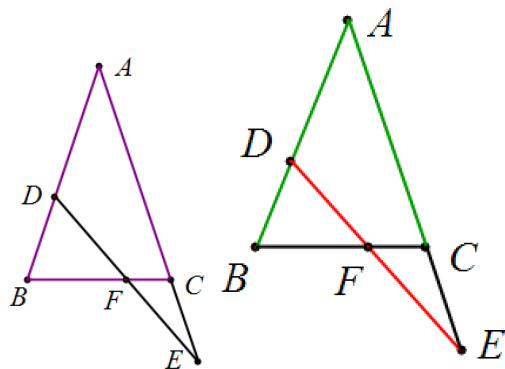


Example 17 : In $\triangle ABC$, O is the circumcenter, CD is the height, and M is the midpoint of AB . Prove that the necessary and sufficient condition for $DM = OA$ is that $\angle A$ and $\angle B$ differ by 90° .

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle,

$$\begin{aligned} & \left(\frac{a+b}{2} - \frac{a+b+c-\frac{ab}{c}}{2} \right) \left(\frac{1/a+1/b}{2} - \frac{1/a+1/b+1/c-\frac{c}{ab}}{2} \right) - 1 \\ &= -\frac{a^3 b^3 c^2}{4(a-b)^4 (a-c)^2 (b-c)^2} \left((a-b)(\overline{a-b})(b-a)(\overline{b-a}) \left(\frac{\frac{a-c}{b-a}}{\frac{b-c}{b-a}} + \frac{\overline{\left(\frac{a-c}{b-a} \right)}}{\overline{\left(\frac{b-c}{b-a} \right)}} \right) \right)^2, \end{aligned}$$

In $\triangle ABC$, O is the circumcenter, CD is the height, and M is the midpoint of AB . Prove: $OA^2 - DM^2 = OA^2 \cos^2(B-A)$.



Example 18 : In $AB=AC \triangle ABC$, take point D on AB , extend AC to E so that $CE = BD$, DE intersects BC at F , prove: $FD = FE$.

Proof: Let A be the origin, AB be the real axis, $a=0$, $e=\frac{\bar{c}\bar{e}}{c}$,

$$f = \frac{\bar{b}\bar{c}\bar{d} - b\bar{c}\bar{e} - b\bar{d}\bar{e} + c\bar{d}\bar{e}}{\bar{c}\bar{d} - b\bar{e}}, \left((d-b)^2 - (c-e)(\overline{c-e}) \right) - (b^2 - c\bar{c}) \left(1 - \frac{e}{c} \right)^2$$

$$= \left((d-f)(\overline{d-f}) - (f-e)(\overline{f-e}) \right) \frac{(\bar{c}\bar{d} - b\bar{e})^2}{c^2 (d-e)(\overline{d-e})}.$$

In $\triangle ABC$, take point D on line AB and point E on line AC . DE intersect BC at F .
Prove that:

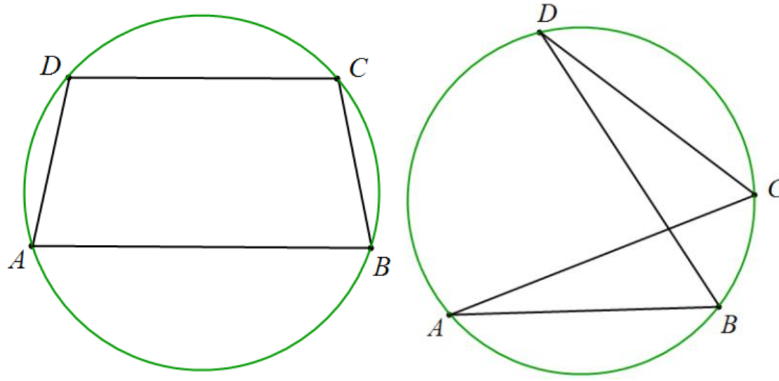
$$(BD^2 - CE^2) - (BA^2 - CA^2) \left(1 - \frac{AE}{AC} \right)^2 = (DF^2 - EF^2) \left(\frac{AC \cdot AD - AB \cdot AE}{AC \cdot DE} \right)^2.$$

Note 1: Note that $AC \cdot AD - AB \cdot AE = 0$ F does not exist at this time.

Explanation 2 : Assume $AD = AB - DB$, $AE = AC + CE$, $DE = DF + FE$,

$$-\frac{2(AB \cdot CE + AC \cdot DB)(AB \cdot CE \cdot DF - AC \cdot DB \cdot FE)}{AC^2 (DF + FE)} = 0, \text{ then } \frac{AC}{CE} \frac{EF}{FD} \frac{DB}{BA} = 1.$$

Its geometric meaning is that the line BC intersects $\triangle ADE$, and Menelaos' theorem is applied. The three equal conditions of the three line segments involved in the question are combined in such a simple form. With hard work, you may also come up with a clever solution.



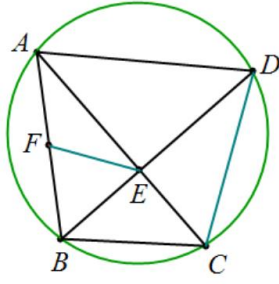
Example 1 9 : As shown in Figure 1, a quadrilateral $ABCD$ is inscribed in a circle . If $AC = BD$, prove that: $AB \parallel CD$ or $AD \parallel BC$. Four points share a circle. Diagonals are equal. Two sets of parallel lines. Full decomposition. Partial decomposition.

Proof: Let the circle be the unit circle,

$$(a-b)(b-c)(c-d)(d-a)\left((a-c)\overline{(a-c)}-(b-d)\overline{(b-d)}\right) \\ = abcd\left([CDA]-[BDA]\right)\left([DAB]-[CAB]\right)$$

Inscribed in the circle is quadrilateral $ABCD$. Prove that:

$$AB \cdot BC \cdot CD \cdot DA \cdot (BD^2 - AC^2) = 16R^2 (S_{\triangle CDA} - S_{\triangle BDA}) \cdot (S_{\triangle DAB} - S_{\triangle CAB}).$$



Example 20 : In a quadrilateral $ABCD$ inscribed in a circle , $AC \perp BD$, AC intersects BD at E , F is the midpoint of AB , prove that $EF \perp CD$. (Brahmagupta Theorem)

In the following, we assume that we discard the concircle and $EF \perp CD$. Let $E = 0$, ED be the real axis, and EA be the imaginary axis.

$$e=0, \quad \bar{b}=b, \quad \bar{d}=d, \quad \bar{a}=-a, \quad \bar{c}=-c, \quad f=\frac{a+b}{2}, \quad \bar{f}=\frac{-a+b}{2},$$

$$\begin{aligned} \text{Identity: } & (a-b)(\overline{a-b})(c-d)(\overline{c-d}) \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d} \right)} \right) \\ & = -2(a-c)(b-d) \left((c-d)(\overline{c-d}) \left(\frac{e-f}{c-d} + \frac{\overline{e-f}}{c-d} \right) \right). \end{aligned}$$

Geometric meaning: Quadrilateral $ABCD$, $AC \perp BD$, AC intersects BD at E , F is the midpoint of AB , prove that $AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD) = 2AC \cdot BD \cdot CD \cdot EF \cos \alpha$, where α is the angle through which the straight line CD turns counterclockwise to EF .

Example 21 : As shown in Figure 1 6 , in triangle ABC , $BC = 2 AC$, D is the midpoint of BC , and E is the midpoint of DC . $\angle BAD = \angle DAE$. Prove that: .

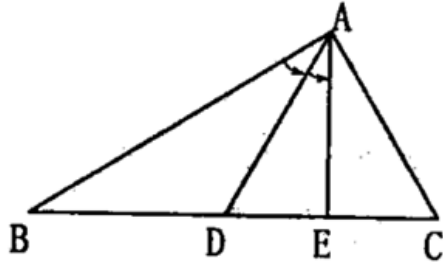


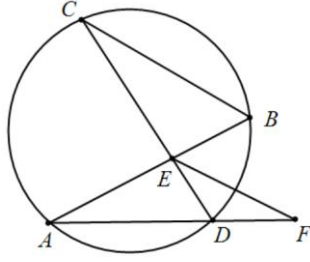
Figure 1 6

$$\text{prove: } 16 \left(\left(a - \frac{b+c}{2} \right) \left(\overline{a - \frac{b+c}{2}} \right) \right)^2 \left(\begin{array}{c} \frac{a - \frac{b+3c}{4}}{a - \frac{b+c}{2}} \\ \frac{\frac{a - \frac{b+3c}{4}}{a - \frac{b+c}{2}}}{\frac{b+c}{a-b}} \end{array} - \begin{array}{c} \overline{\frac{a - \frac{b+3c}{4}}{a - \frac{b+c}{2}}} \\ \overline{\frac{\frac{a - \frac{b+3c}{4}}{a - \frac{b+c}{2}}}{\frac{b+c}{a-b}}} \end{array} \right)$$

$$= \begin{vmatrix} a & \overline{a} & 1 \\ b & \overline{b} & 1 \\ c & \overline{c} & 1 \end{vmatrix} \left(4(a-c)(\overline{a-c}) - (b-c)(\overline{b-c}) \right),$$

In $\triangle ABC$, D is the midpoint of BC , and E is the midpoint of DC . Prove that:

$$8AD^2 \cdot AE \cdot AB \sin(\angle BAD - \angle DAE) = S_{\triangle ABC} (4CA^2 - CB^2).$$



Example 22 : It is known that E is the intersection of the two chords AB and CD in a circle . The straight line $EF \parallel CB$ intersects the extension of AD at F .

Prove that: $FE^2 = FA \cdot FD$. Complex number identities expand the relationship between the product of the inscribed quadrilateral in the circle and the line segment

$$O = (0,0) \quad , \quad E = \left(\frac{abc + abd - acd - bcd}{ab - cd}, \frac{a + b - c - d}{ab - cd} \right) \quad , \quad F = \left(f, \frac{a + d - f}{ad} \right) \quad ,$$

$$ad[ABC](b-c) \left[(e-f)\overline{(e-f)} - (f-a)\overline{(f-d)} \right]$$

$$= ([FBC] - [EBC])(a-b)(a-d)(c-e) \quad ,$$

Given that E is the intersection of two chords AB and CD in a circle . F is a point on the extension line of AD . Prove that:

$$(EF^2 - FA \cdot FD) \cdot S_{\triangle ABC} \cdot BC = (S_{\triangle FBC} - S_{\triangle EBC}) AB \cdot CE \cdot AD.$$

Note: When F runs from one side of the straight line BC to the other side,

$S_{\triangle FBC} - S_{\triangle EBC}$ it should be changed to $S_{\triangle FBC} + S_{\triangle EBC}$.

Example 23 : In $\frac{AB}{\sin C} = \frac{AC}{\sin B} \triangle ABC, .$

There are two ways to expand:

Extension 1:

Let the circumcircle of $\triangle ABC$ be the unit circle, then

$$\frac{abc}{(a-b)(b-c)(a-c)}(c-a)(b-a)\overline{(c-a)}\overline{(b-a)}\left(\frac{\frac{c-b}{b-a}}{\frac{b-c}{b-a}} - \frac{\overline{\left(\frac{c-b}{b-a}\right)}}{\overline{\left(\frac{b-c}{b-a}\right)}}\right)$$

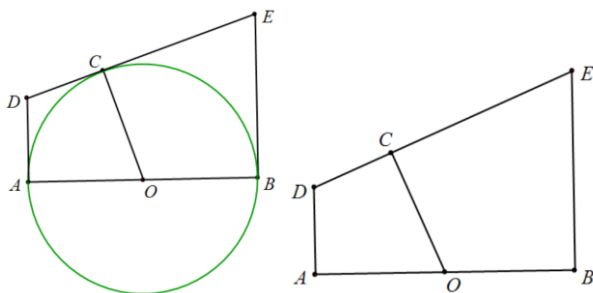
$$= (a-b)\overline{(a-b)} - (a-c)\overline{(a-c)},$$

Geometric meaning $2R \cdot BC \sin(C-B) = AB^2 - AC^2,$

The corresponding identity $\sin(C+B)\sin(C-B) = \sin^2 C - \sin^2 B.$

Extension 2 : $(c-a)\overline{(c-a)}\left(\frac{c-b}{c-a} - \overline{\left(\frac{c-b}{c-a}\right)}\right) = (b-c)\overline{(b-c)}\left(\frac{b-a}{b-c} - \overline{\left(\frac{b-a}{b-c}\right)}\right),$

geometric meaning $AC \cdot \sin \angle ACB = AB \cdot \sin \angle ABC.$



Example 24 : C is an arbitrary point on the circle O with AB as diameter ; the tangent line through point C intersects the tangent lines through points A and B at points D and E respectively . Prove $CO^2 = CD \cdot CE$.

Note: The original question is very simple. It has been proved that $OD \perp OE$, which can be obtained according to the projection theorem. But after generalization, it becomes somewhat difficult.

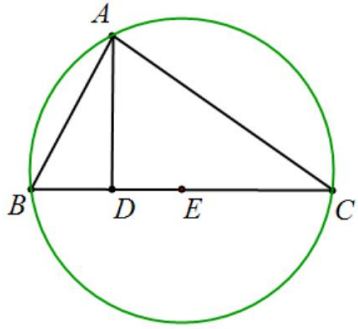
Proof: Let $o=0$ the line AB be the real axis, $a=-b$, $\bar{d}=2a-d$, $\bar{e}=2b-e$,

solve the equation $\frac{c}{c-d} = -\overline{\left(\frac{c}{c-d}\right)}$, $\frac{c-d}{c-e} = \overline{\left(\frac{c-d}{c-e}\right)}$, and get $c = \frac{b(d+e)}{4b+d-e}$,

$$\frac{\bar{c}}{c} = \frac{b(d+e)}{d-e} ,$$

$$2(a-b)^2 \left(c\bar{c} - (d-c)\overline{(c-e)} \right) = (d-e)\overline{(d-e)}(c-a)\overline{(c-a)} \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a} \right)} \right) .$$

Quadrilateral $ABED$, $AD \perp AB$, $BE \perp AB$, O is the midpoint of AB , $OC \perp DE$ at C , and C is on the line segment DE . Prove $AB^2(CO^2 - CD \cdot CE) = DE^2 \cdot CA \cdot CB \cdot \cos \angle ACB$.

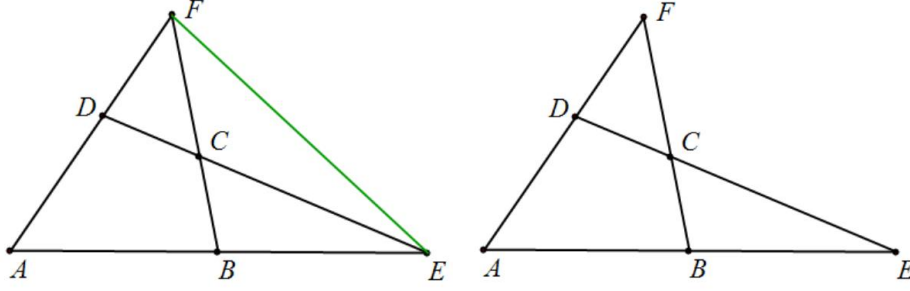


Example 25 : $\triangle ABC$ In , $\angle B = 2\angle C$, E is the midpoint of BC , and $AD \perp BC$ intersects BC at D . Prove that : $AB = 2DE$.

$$(a-b)\overline{(a-b)} - 4(d-e)\overline{(d-e)} = 2 \frac{b(a-b)(a-c)^3}{a^2(b-c)^3} \left(\frac{\left(\frac{c-b}{c-a}\right)^2}{\frac{b-a}{b-c}} - \overline{\left(\frac{\left(\frac{c-b}{c-a}\right)^2}{\frac{b-a}{b-c}} \right)} \right),$$

in Figure 1, $\triangle ABC$ E is the midpoint of BC , $AD \perp BC$ intersects BC at D , prove that :

$$(AB^2 - 4DE^2) \sin B = CA^2 \sin(2C - B).$$



Example 26 : As shown in Figure 1, quadrilateral $ABCD$, AB intersects DC at E , AD intersects BC at F . If A , B , C , and D are cocircular, prove that: $EF^2 = EA \cdot EB + FA \cdot FD$. (Complex number identities extend the relationship between four points cocircular and the side length of a complete quadrilateral)

Proof: Let $A = (0, 0)$, $B = (1, 1)$, $C = (c, \bar{c})$, $D = (d, \bar{d})$,

$$E = \left(\frac{-\bar{c}d + c\bar{d}}{c - \bar{c} - d + \bar{d}}, \frac{-\bar{c}d + c\bar{d}}{c - \bar{c} - d + \bar{d}} \right), F = \left(\frac{(c - \bar{c})d}{d - \bar{c}d - \bar{d} + c\bar{d}}, \frac{(c - \bar{c})\bar{d}}{d - \bar{c}d - \bar{d} + c\bar{d}} \right),$$

Generate the identity: $-[ABCD][ABD]$

$$= ([BCD] - [BCA])([DAB] - [CAB])((e - f)(\overline{e - f}) - (e - a)(\overline{e - b}) - (f - a)(\overline{f - d}))$$

, Expand the proposition: As shown in Figure 1, quadrilateral $ABCD$, AB intersects DC at E , AD intersects BC at F , prove: $\sin E \sin F (EF^2 - EA \cdot EB - FA \cdot FD) = 2S_{\triangle ABD} \sin(B + D)$.

Example 27 : There is a point P inside rectangle $ABCD$. Prove that:

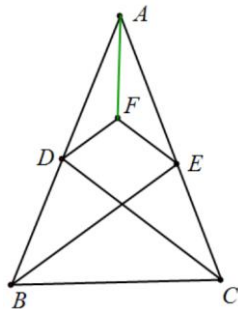
$$PA^2 + PC^2 = PB^2 + PD^2.$$

Proof: Let P be the origin, $a\bar{a} + c\bar{c} - b\bar{b} - (a+c-b)\overline{(a+c-b)}$

$$= (a-b)\overline{(a-b)} \left(\frac{a-(a+c-b)}{a-b} + \overline{\left(\frac{a-(a+c-b)}{a-b} \right)} \right),$$

P inside parallelogram $ABCD$. Prove: $PA^2 + PC^2 - PB^2 - PD^2 = 2AB \cdot AD \cos A$.

Example 28 : As shown in Figure 1, in triangle ABC , D and E are on AB and AC respectively, $DF \parallel BE$, $EF \parallel CD$, if $BD = CE$, prove that AF is the angle bisector of $\angle A$. (Complex number identities extend equal length line segments and angle bisectors)



Proof: Suppose $A = \left(a, \frac{1}{a}\right)$, $B = \left(b, \frac{1}{b}\right)$, $C = \left(c, \frac{1}{c}\right)$, $D = \left(d, \frac{a+b-d}{ab}\right)$,

$E = \left(e, \frac{a+c-e}{ac}\right)$, it can be solved $F = (f, \bar{f})$.

$$[AEF][ADF][ABC]((b-d)\overline{(b-d)} - (c-e)\overline{(c-e)})$$

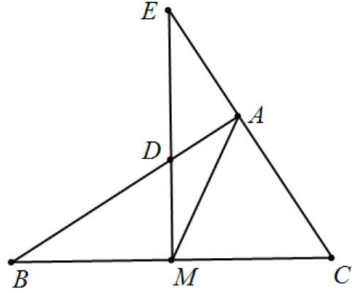
$$= [BDE][CDE] \left| a - f \right|^4 \left(\frac{\frac{a-c}{a-f}}{\frac{a-f}{a-b}} - \overline{\left(\frac{\frac{a-c}{a-f}}{\frac{a-f}{a-b}} \right)} \right),$$

Extended proposition: As shown in Figure 1, in $\triangle ABC$, D and E are on AB and AC respectively, $DF \parallel BE$, $EF \parallel CD$, prove:

$$S_{\triangle AEF} \cdot S_{\triangle ADF} \sin A (BD^2 - CE^2) = S_{\triangle BDE} \cdot S_{\triangle CDE} \cdot AF^2 \sin(\angle CAF - \angle BAF).$$

Note: When D is on the extended line of line segment AB , E is on line segment AC , and AF is the exterior angle bisector of $\angle A$.

Example 29 : As shown in Figure 1, in triangle ABC , M is the midpoint of BC , $MD \perp BC$, angle AB at D , intersecting the extended line of CA at E , if $AB \perp AC$, prove: $MD \cdot ME = MA^2$. (Complex number identities extend the perpendicular relationship between line segments and perpendicularity)



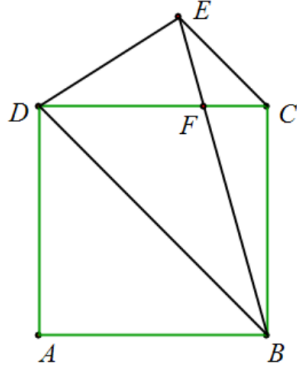
Proof: Assume $M = (0, 0)$, $C = (c, c)$, $B = -C$, $D = (d, -d)$, $E = (e, -e)$, we can

$$\text{find } A = \left(\frac{-cd + ce + 2de}{d + e}, -\frac{cd - ce + 2de}{d + e} \right),$$

$$8de(d(-e) - a\bar{a}) = (d - e)^2(a - b)(\overline{a - b}) \left(\frac{a - c}{a - b} + \overline{\left(\frac{a - c}{a - b} \right)} \right),$$

Extended proposition: As shown in Figure 1, in $\triangle ABC$, M is the midpoint of BC , $MD \perp BC$ intersects AB at D , and intersects the extended line of CA at E . Prove:

$$4MD \cdot ME (MD \cdot ME - MA^2) = DE^2 \cdot AB \cdot AC \cos A.$$



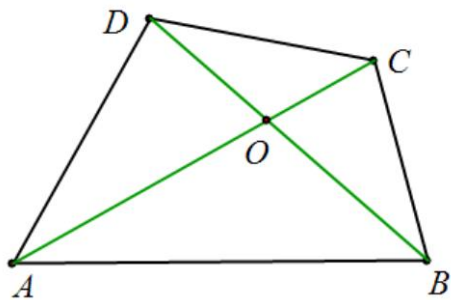
Example 30 : Square $ABCD$, $CE \parallel BD$, and $BD = BE$, F is the intersection of straight lines BE and CD , prove that: $DE = DF$.

Proof: Let C be the origin, $b = -i$, $d = -1$, solve for $e = \frac{(1-i)f}{1+f}$, $\bar{e} = \frac{(1+i)f}{1+f}$,

and get the identity:

$$d^2 \left((d-f)(\overline{d-f}) - (d-e)(\overline{d-e}) \right) = f^2 \left((b-d)(\overline{b-d}) - (b-e)(\overline{b-e}) \right),$$

Extended proposition: Square $ABCD$, $CE \parallel BD$, F is the intersection of straight lines BE and CD , prove: $CD^2 (DF^2 - DE^2) = CF^2 (BD^2 - BE^2)$.



Example 3 1 : Quadrilateral $ABCD$, AC intersects BD at O . If $S_{\triangle OAB} S_{\triangle OCD} = S_{\triangle OBC}^2$, prove:
 $AB \parallel DC$. (Area relationship and parallelism of quadrilaterals)

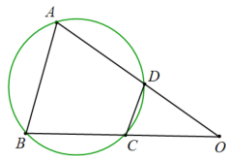
Proof: Let O be the origin, OC be the real axis, $\bar{d} = \frac{db}{b}$,

$$[OAB][OCD] - [OBC]^2 = ([DAB] - [CAB])[OBC],$$

Extended proposition: Quadrilateral $ABCD$, AC intersects BD at O , prove:

$$S_{\triangle OAB} S_{\triangle OCD} - S_{\triangle OBC}^2 = (S_{\triangle DAB} - S_{\triangle CAB}) S_{\triangle OBC}.$$

Example 32 : Quadrilateral $ABCD$, straight line AD intersects BC at O . Prove that $OA \cdot OD = OB \cdot OC$ the necessary and sufficient condition is that the four points A , B , C , and D are concentric circles.



Proof: Let O be the origin and line OB be the real axis.

(1) Since points O , A , and D are collinear, we have $\frac{a}{d} = \overline{\left(\frac{a}{d}\right)}$, that is $\bar{d} = \frac{\bar{a}d}{a}$.

Substituting this into the following two equations, we find that there are common factors $bc - \bar{a}d = 0$.

(2) $OA \cdot OD = OB \cdot OC$, that is $bc - \bar{a}d = 0$.

(3) The four points A , B , C and D are concentric circles, that is

$$\frac{\frac{a-c}{a-b} - \overline{\left(\frac{a-c}{a-b}\right)}}{\frac{d-c}{d-b} - \overline{\left(\frac{d-c}{d-b}\right)}} = \frac{(a-\bar{a})(b-c)(a-d)(bc-\bar{a}d)}{(a-b)(b-\bar{a})(c-d)(ac-\bar{a}d)} = 0.$$

$$(a-b)(\bar{a}-b)(c-d)\left(c - \frac{\bar{a}d}{a}\right)\left(\frac{\frac{a-c}{a-b} - \overline{\left(\frac{a-c}{a-b}\right)}}{\frac{d-c}{d-b} - \overline{\left(\frac{d-c}{d-b}\right)}}\right) + \frac{(a-\bar{a})(b-c)(a-d)}{a}(bc-\bar{a}d) = 0$$

, so we get a new proposition: quadrilateral $ABCD$, straight line AD intersects BC at O , prove:

$$AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD) = 2(S_{\triangle ABC} - S_{\triangle DBC})(OA \cdot OD - OB \cdot OC).$$

Note: This conclusion is a generalization of the circle power theorem, which quantifies the circle power theorem. From this conclusion, we can know that $OA \cdot OD = OB \cdot OC \Leftrightarrow$ the four points A , B , C , and D are concentric circles.

This is a very beautiful conclusion. The equation is rich in geometric elements, involving line segment length, angle, area, collinearity and cocircle. The left side of the conclusion $AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD)$ is the product of the four sides of the quadrilateral and the trigonometric function, $\sin(\angle BAD + \angle BCD)$ the value of which determines whether the four points are cocircular. The right side of the conclusion

$2(S_{\triangle ABC} - S_{\triangle DBC})(OA \cdot OD - OB \cdot OC)$ is the product of the difference in the area

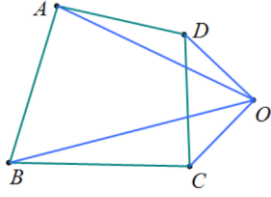
of the two triangles and the difference in the product of the line segments,
which $S_{\triangle ABC} - S_{\triangle DBC}$ determines whether the lines AD and BC are parallel or
intersecting. The known condition that the line AD intersects BC at O was not
taken seriously in the past. In fact, this is an indispensable condition for
the circle power theorem to hold, and it should be reflected in the conclusion
expression.

The ultimate ideal is to generate an identity that allows us to see at a
glance:

$ABCO$ on the plane, BCO and ADO are collinear, and four points
 $OA \cdot OD = OB \cdot OC$, A, B, C and D are in a circle. If you know any three of the four
conditions, you can deduce the fourth one.

There are five points A, B, C, D and O on a plane. Points O, A and D are
collinear. Points O, B and C are collinear. Points $OA \cdot OD = OB \cdot OC$, A, B, C
and D are in a circle. Can we establish an identity and study the relationship
between these conditions?

$$\begin{aligned}
& 2(a-b)\overline{(a-b)}(c-d)\overline{(c-d)}\left(\frac{a-d}{a-b}\frac{c-b}{c-d}-\overline{\left(\frac{a-d}{a-b}\frac{c-b}{c-d}\right)}\right) \\
& + \left(\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} - \begin{vmatrix} d & \bar{d} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}\right) \left((o-a)\overline{(o-a)}\left(\frac{o-d}{o-a}+\overline{\left(\frac{o-d}{o-a}\right)}\right) - (o-b)\overline{(o-b)}\left(\frac{o-c}{o-b}+\overline{\left(\frac{o-c}{o-b}\right)}\right) \right) \\
& - \begin{vmatrix} o & \bar{o} & 1 \\ d & \bar{d} & 1 \\ a & \bar{a} & 1 \end{vmatrix} \left((b-c)\overline{(b-c)}\left(\frac{b-a}{b-c}+\overline{\left(\frac{b-a}{b-c}\right)}\right) - (c-d)\overline{(c-d)}\left(\frac{c-b}{c-d}+\overline{\left(\frac{c-b}{c-d}\right)}\right) \right) \\
& - \begin{vmatrix} o & \bar{o} & 1 \\ c & \bar{c} & 1 \\ b & \bar{b} & 1 \end{vmatrix} \left((a-b)\overline{(a-b)}\left(\frac{a-d}{a-b}+\overline{\left(\frac{a-d}{a-b}\right)}\right) - (d-a)\overline{(d-a)}\left(\frac{d-c}{d-a}+\overline{\left(\frac{d-c}{d-a}\right)}\right) \right) = 0
\end{aligned}$$

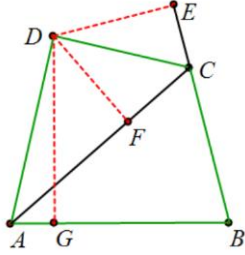


Translate the complex number identity into common geometric quantities. Considering that directed line segments, directed angles, and directed areas differ from general line segments, angles, and areas by a positive or negative sign, the choice of positive or negative sign depends on the relative positions of the points. For example, in this problem, the positions that can be formed by the five free points are very complex. The following relationship is valid in the state shown in Figure 3.

$$\begin{aligned}
& \frac{1}{2} AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD) \\
& - (S_{\triangle ABC} - S_{\triangle DBC})(OA \cdot OD \cos \angle AOD - OB \cdot OC \cos \angle BOC) \\
& - S_{\triangle ODA}(BA \cdot BC \cos \angle ABC - CB \cdot CD \cos \angle BCD) \\
& - S_{\triangle OCB}(AD \cdot AB \cos \angle BAD - DC \cdot DA \cos \angle CDA) = 0.
\end{aligned}$$

problem

Example 1 : Simson's Theorem



Solve the equation $\frac{c-e}{c-b} = \overline{\left(\frac{c-e}{c-b}\right)}$ and $\frac{d-e}{c-b} = -\overline{\left(\frac{d-e}{c-b}\right)}$ get

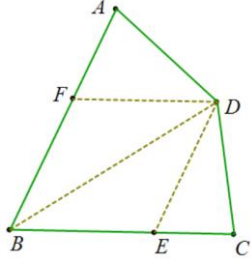
$$\{e, \bar{e}\} = \left\{ \frac{\bar{b}c - b\bar{c} + \bar{b}d - \bar{c}d + b\bar{d} - c\bar{d}}{2(\bar{b} - \bar{c})}, \frac{-\bar{b}c + b\bar{c} + \bar{b}d - \bar{c}d + b\bar{d} - c\bar{d}}{2(b - c)} \right\}, \text{ and similarly ,}$$

$$[A, B, C]^2 (b - c) \overline{(b - c)} (a - d) \overline{(a - d)} \left(\frac{d - c}{d - a} \frac{b - a}{b - c} - \overline{\left(\frac{d - c}{d - a} \frac{b - a}{b - c} \right)} \right) \text{ get } \{f, \bar{f}\} ,$$

$$\{g, \bar{g}\}$$

$$= -4(a - b) \overline{(a - b)} (b - c) \overline{(b - c)} (c - a) \overline{(c - a)} [E, F, G].$$

Extended proposition: Convex quadrilateral $ABCD$, draw $DE \perp BC$ through D , $DF \perp CA$, $DG \perp AB$, R is the radius of the circumscribed circle of $\triangle ABC$, prove: $AB \cdot BC \cdot CD \cdot DA \sin(\angle ADC + \angle ABC) = 8R^2 \cdot S_{\triangle EFG}$.



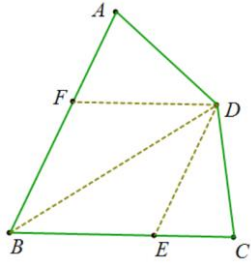
Example 2 : Convex quadrilateral $ABCD$, take points E and F on BC and BA to form parallelogram $BEDF$. Prove that the necessary and sufficient condition for the four points A, B, C , and D to be cocircular is $BD^2 = BC \cdot BE + BA \cdot BF$.

Proof: Let B be the origin and BC be the real axis, then $\bar{f} = \frac{\bar{a}f}{a}$, $d = (e + f)$,

$$(a-b)\overline{(a-b)}(c-d)\overline{(c-d)}\left(\frac{a-d}{a-b}\frac{c-b}{c-d}-\overline{\left(\frac{a-d}{a-b}\frac{c-b}{c-d}\right)}\right)=-[A,B,C](ec+f\bar{a}-d\bar{d}).$$

Extended proposition: Convex quadrilateral $ABCD$, take points E and F on BC and BA to form parallelogram $BEDF$, prove:

$$AB \cdot BC \cdot CD \cdot DA \sin(A+C) = 2S_{\triangle ABC} (BC \cdot BE + BA \cdot BF - BD^2).$$



Example 3 : Convex quadrilateral $ABCD$, take points E and F on BC and BA to form parallelogram $BEDF$. Prove that the necessary and sufficient condition for the

four points $\cos B = \frac{1}{2} \left(\frac{EC}{BF} + \frac{FA}{BE} \right)$ A, B, C , and D to be cocircular is .

Proof: $\cos B = \frac{1}{2} \left(\frac{EC}{BF} + \frac{FA}{BE} \right)$ This is equivalent to

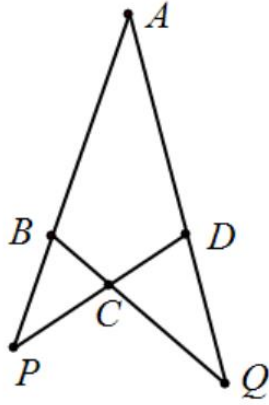
$$2BE \cdot BF \cdot \cos B = BE \cdot EC + BF \cdot FA,$$

Let B be the origin and BC be the real axis, then $\bar{f} = \frac{\bar{a}f}{a}$, $d = (e + f)$,

$$\begin{aligned} & (a-b)(\overline{a-b})(c-d)(\overline{c-d}) \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d} \right)} \right) \\ &= -[A, B, C] \left(e(c-e) + f(\overline{a-f}) - e \overline{e} \left(\frac{b-f}{b-e} + \overline{\left(\frac{b-f}{b-e} \right)} \right) \right). \end{aligned}$$

Extended proposition: Convex quadrilateral $ABCD$, take points E and F on BC and BA to form parallelogram $BEDF$, prove:

$$AB \cdot BC \cdot CD \cdot DA \sin(A+C) = 2S_{\triangle ABC} (BE \cdot EC + BF \cdot FA - 2BE \cdot BF \cdot \cos B).$$



Example 4 : Convex quadrilateral $ABCD$, AB intersects CD at P , BC intersects AD at Q . Prove that the necessary and sufficient condition for the four points

A , B , C , and D to be cocircular is $BD^2 = AB^2 \cdot AP \cdot \frac{DQ}{AQ} + AD^2 \cdot \frac{BP}{AP}$.

prove: $S_1 = |b-d|^2 (a-p) \overline{(a-q)} - |a-b|^2 (a-p) \overline{(d-q)} - |a-d|^2 (b-p) \overline{(a-q)}$,

$$S_2 = (a-b)(c-d) \overline{(a-b)(c-d)} \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d} \right)} \right) ,$$

$$([ABC] - [DBC])([BCD] - [ACD]) S_1 = -[ABD](a-b) \overline{(a-d)} S_2 ,$$

Extended proposition: Convex quadrilateral $ABCD$, take points E and F on BC and BA to form parallelogram $BEDF$. Prove that:

$$(S_{\triangle ABC} - S_{\triangle DBC})(S_{\triangle BCD} - S_{\triangle ACD})(BD^2 \cdot AP \cdot AQ - AB^2 \cdot AP \cdot DQ - AD^2 \cdot BP \cdot AQ)$$

$$= AB^3 \cdot AD^3 \cdot BC \cdot CD \sin A \sin(A+C)$$

Example 5 : Ptolemy's theorem $AC \cdot BD = AB \cdot CD + BC \cdot AD$.

Considering that it is inconvenient to use complex numbers to represent line segments, we will $AB \cdot CD + BC \cdot AD - AC \cdot BD = 0$ find a dual formula $AB \cdot CD + BC \cdot AD + AC \cdot BD = 0$ and multiply the two formulas to get

$$AB^2 \cdot CD^2 + BC^2 \cdot AD^2 + 2AB \cdot BC \cdot CD \cdot DA - AC^2 \cdot BD^2 = 0.$$

$$(a-b)(c-d)\overline{(a-b)(c-d)} + (b-c)(d-a)\overline{(b-c)(d-a)}$$

$$-2(a-b)(c-d)\overline{(a-b)(c-d)} \frac{\frac{a-d}{a-b} \frac{c-b}{c-d} + \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d}\right)}}{2} - (a-c)(b-d)\overline{(a-c)(b-d)} = 0$$

.

$$AB^2 \cdot CD^2 + BC^2 \cdot AD^2 - 2AB \cdot BC \cdot CD \cdot DA \cos(A+C) - AC^2 \cdot BD^2 = 0,$$

$A+C = \pi$ At that time, $\cos(A+C) = -1$, so there was

$$AC \cdot BD = AB \cdot CD + BC \cdot AD.$$

$A+C = \frac{\pi}{2}$ At that time, $\cos(A+C) = 0$, so there was

$$AB^2 \cdot CD^2 + BC^2 \cdot AD^2 = AC^2 \cdot BD^2.$$

$$AB^2 \cdot CD^2 + BC^2 \cdot AD^2 - 2AB \cdot BC \cdot CD \cdot DA \left(2 \cos^2 \frac{A+C}{2} - 1\right) - AC^2 \cdot BD^2 = 0,$$

$$AB^2 \cdot CD^2 + BC^2 \cdot AD^2 + 2AB \cdot BC \cdot CD \cdot DA - AC^2 \cdot BD^2 = 4AB \cdot BC \cdot CD \cdot DA \cos^2 \frac{A+C}{2}$$

$$(AB \cdot CD + BC \cdot AD - AC \cdot BD)(AB \cdot CD + BC \cdot AD + AC \cdot BD) = 4AB \cdot BC \cdot CD \cdot DA \cos^2 \frac{A+C}{2}$$

.

That is to say, at the beginning, in order to express the common circle information, we use it $\cos(A+C) = -1$, and after generating the identity equation, we re-understand it and get a new proposition.

$$\frac{1}{2}(a-b)(a-b) \frac{\frac{a-d}{a-b} - \frac{a-\bar{d}}{a-\bar{b}}}{2} + \frac{1}{2}(c-b)(\bar{c}-b) \frac{\frac{c-d}{c-b} - \frac{\bar{c}-\bar{d}}{\bar{c}-\bar{b}}}{2}$$

$$= \frac{1}{4}(-bc + b\bar{c} - ad + 2bd - \bar{c}d + a\bar{d} - 2b\bar{d} + c\bar{d}),$$

$$\left((a-b)(a-b) + (b-c)(b-\bar{c}) + (c-d)(\bar{c}-\bar{d}) + (d-a)(\bar{d}-a)\right)^2$$

$$\begin{aligned}
& -8(a-b)(a-b)(c-b)(\bar{c}-b) \frac{\frac{a-d}{a-b} \frac{c-d}{c-b} + \frac{a-\bar{d}}{a-b} \frac{\bar{c}-\bar{d}}{\bar{c}-b}}{2} \\
& -2\left((a-b)^2(a-b)^2 + (b-c)^2(b-\bar{c})^2 + (c-d)^2(\bar{c}-\bar{d})^2 + (d-a)^2(\bar{d}-a)^2\right) \\
& = -\left(bc - b\bar{c} + ad - 2bd + \bar{c}d - a\bar{d} + 2b\bar{d} - cd\right)^2,
\end{aligned}$$

For any quadrilateral, $\left(AB^2 + BC^2 + CD^2 + DA^2\right)^2 - 8AB \cdot BC \cdot CD \cdot DA \cos(A+C)$

$$-2\left(AB^4 + BC^4 + CD^4 + DA^4\right) = 16S_{ABCD}^2.$$

$$\left(AB^2 + BC^2 + CD^2 + DA^2\right)^2 - 8AB \cdot BC \cdot CD \cdot DA \left(2\cos^2 \frac{A+C}{2} - 1\right)$$

$$-2\left(AB^4 + BC^4 + CD^4 + DA^4\right) = 16S_{ABCD}^2,$$

$$\left(AB^2 + BC^2 + CD^2 + DA^2\right)^2 + 8AB \cdot BC \cdot CD \cdot DA$$

$$-2\left(AB^4 + BC^4 + CD^4 + DA^4\right) - 16AB \cdot BC \cdot CD \cdot DA \cos^2 \frac{A+C}{2} = 16S_{ABCD}^2,$$

$$S_{ABCD} = \sqrt{(-AB + BC + CD + DA)(AB - BC + CD + DA)(AB + BC - CD + DA)(AB + BC + CD - DA) - AB \cdot BC \cdot CD \cdot DA \cos^2 \frac{A+C}{2}}$$

$$S_{ABCD} = \sqrt{-\frac{1}{4}(AB \cdot CD + BC \cdot AD - AC \cdot BD)(AB \cdot CD + BC \cdot AD + AC \cdot BD)}$$

Example 6 : Let A , B , C , and D be four points on a circle, and O be any point. Prove that:

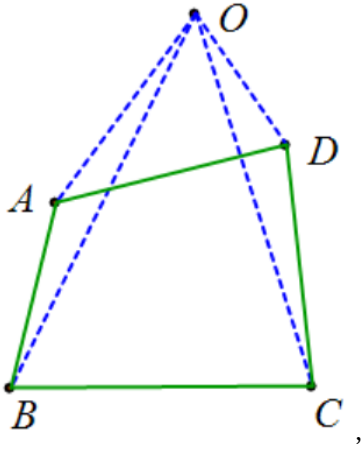
$$OA^2 \cdot S_{\triangle BCD} - OB^2 \cdot S_{\triangle CDA} + OC^2 \cdot S_{\triangle DAB} - OD^2 \cdot S_{\triangle ABC} = 0.$$

Proof 1: Take O as the origin. Since A , B , C , and D are all in the same

circle, then
$$\begin{vmatrix} x_A^2 + y_A^2 & x_A & y_A & 1 \\ x_B^2 + y_B^2 & x_B & y_B & 1 \\ x_C^2 + y_C^2 & x_C & y_C & 1 \\ x_D^2 + y_D^2 & x_D & y_D & 1 \end{vmatrix} = 0$$
 expand the first column and notice

$x_A^2 + y_A^2 = OA^2$ the determinant formula of the area. The proposition is proved.

The above proof comes from Professor Shan Xu's "Techniques of Analytic Geometry". Mr. Shan commented: This question seems difficult, indeed, it is very tricky without using analytic geometry.



Identity:
$$a\bar{a} \begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} - b\bar{b} \begin{vmatrix} c & \bar{c} & 1 \\ d & \bar{d} & 1 \\ a & \bar{a} & 1 \end{vmatrix} + c\bar{c} \begin{vmatrix} d & \bar{d} & 1 \\ a & \bar{a} & 1 \\ b & \bar{b} & 1 \end{vmatrix} - d\bar{d} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}$$

$$+ (a-b)(\overline{a-b})(c-d)(\overline{c-d}) \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\frac{a-d}{a-b} \frac{c-b}{c-d}} \right) = 0.$$

Extended proposition: Given quadrilateral $ABCD$ and any point O , prove that:

$$OA^2 \cdot S_{\triangle BCD} - OB^2 \cdot S_{\triangle CDA} + OC^2 \cdot S_{\triangle DAB} - OD^2 \cdot S_{\triangle ABC}$$

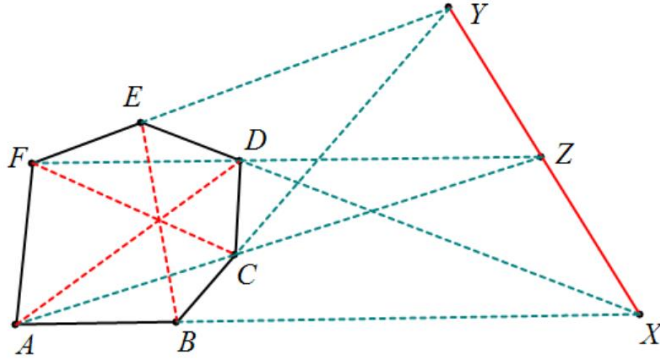
$$= \frac{1}{2} AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD).$$

Proof: Use the complex number method, using lowercase letters x to represent X the complex number of a point and using $\bar{x}$ to represent x the conjugate complex number of the complex number.

$$\begin{aligned}
& \frac{1}{2} AB \cdot BC \cdot CD \cdot DA \sin(\angle BAD + \angle BCD) \\
&= \frac{1}{2} |(a-b)(b-c)(c-d)(d-a)| \frac{\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d}\right)}}{2 \left| \frac{a-d}{a-b} \frac{c-b}{c-d} \right| i} \\
&= \frac{1}{4} |(a-b)\overline{(a-b)}(c-d)\overline{(c-d)}| \frac{\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d}\right)}}{i} \\
&= \frac{i}{4} \left(\overline{(a-d)(c-b)}(a-b)(c-d) - \overline{(a-b)(c-d)}(a-d)(c-b) \right);
\end{aligned}$$

Let O be the origin, $OA^2 \cdot S_{\triangle BCD} - OB^2 \cdot S_{\triangle CDA} + OC^2 \cdot S_{\triangle DAB} - OD^2 \cdot S_{\triangle ABC}$

$$\begin{aligned}
&= \frac{i}{4} \left(a\bar{a} \begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} - b\bar{b} \begin{vmatrix} c & \bar{c} & 1 \\ a & \bar{a} & 1 \\ d & \bar{d} & 1 \end{vmatrix} + c\bar{c} \begin{vmatrix} d & \bar{d} & 1 \\ a & \bar{a} & 1 \\ b & \bar{b} & 1 \end{vmatrix} - d\bar{d} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} \right) \\
&= \frac{i}{4} (a\bar{a}(b\bar{c} + d\bar{b} + c\bar{d} - d\bar{c} - b\bar{d} - c\bar{b}) - b\bar{b}(c\bar{d} + a\bar{c} + d\bar{a} - a\bar{d} - c\bar{a} - d\bar{c}) \\
&\quad + c\bar{c}(d\bar{a} + b\bar{d} + a\bar{b} - b\bar{a} - d\bar{b} - a\bar{d}) - d\bar{d}(a\bar{b} + b\bar{a} + b\bar{c} - c\bar{b} - a\bar{c} - b\bar{a})) \\
&= \frac{i}{4} \left(\overline{(a-d)(c-b)}(a-b)(c-d) - \overline{(a-b)(c-d)}(a-d)(c-b) \right);
\end{aligned}$$



Example 7 : Hexagon $ABCDEF$, straight line AB intersects DE at X , straight line BC intersects EF at Y , straight line AC intersects DF at Z . Prove that the necessary and sufficient condition for straight lines AD , BE , and CF to intersect at a point is that the three points X , Y , and Z are collinear.

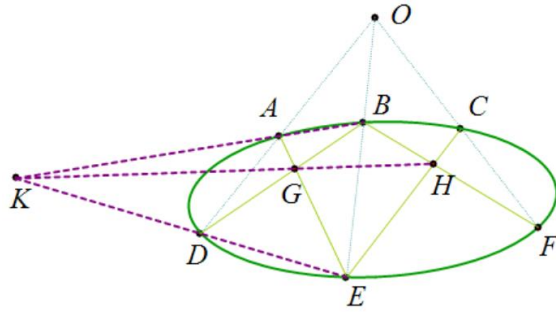
Identity: $[ABC][DEF]([FAD][CBE] - [FBE][CAD])$

$$= ([EAB] - [DAB])([FAC] - [DAC])([FBC] - [EBC])[XYZ].$$

Extended proposition: Hexagon $ABCDEF$, straight line AB intersects DE at X , straight line BC intersects EF at Y , straight line AC intersects DF at Z , prove:

$$[ABC][DEF]([FAD][CBE] - [FBE][CAD])$$

$$= ([EAB] - [DAB])([FAC] - [DAC])([FBC] - [EBC])[XYZ].$$



Example 8 : Given six points $A, B, C, D, E,$ and F on a quadratic curve , straight lines $DA, EB,$ and FC intersect at a point O , straight line AE intersects BD at G , and straight line CE intersects BF at H . Prove that straight lines $AB, GH,$ and DE intersect at a point.

Extended proposition: Given six points $A, B, C, D, E,$ and F , straight lines $DA, EB,$ and FC intersect at a point O , straight line AE intersects BD at G , and straight line CE intersects BF at H . Prove that:

$$S_{\triangle ACF}^2 \cdot S_{\triangle ABD}^2 \cdot S_{\triangle BCE} \cdot S_{\triangle ACE} (S_{\triangle GBA} \cdot S_{\triangle HDE} - S_{\triangle HBA} \cdot S_{\triangle GDE})$$

$$= S_{\triangle ABE} \cdot S_{\triangle ABG} \cdot S_{\triangle ACH} \cdot S_{\triangle ACD} (S_{\triangle ACD} \cdot S_{\triangle BCF} \cdot S_{\triangle AEF} \cdot S_{\triangle BDE} - S_{\triangle BEF} \cdot S_{\triangle ACF} \cdot S_{\triangle BCD} \cdot S_{\triangle ADE}).$$

The six-point conic section is equivalent to

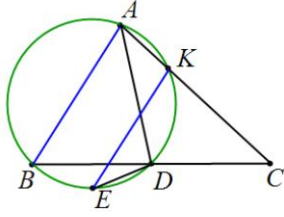
$$S_{\triangle ACD} \cdot S_{\triangle BCF} \cdot S_{\triangle AEF} \cdot S_{\triangle BDE} = S_{\triangle BEF} \cdot S_{\triangle ACF} \cdot S_{\triangle BCD} \cdot S_{\triangle ADE}.$$

Straight lines $AB, GH,$ and DE intersect at one point, which is equivalent to

$$S_{\triangle GBA} \cdot S_{\triangle HDE} = S_{\triangle HBA} \cdot S_{\triangle GDE}.$$

The straight lines $DA, EB,$ and FC intersect at one point, which is equivalent

$$\text{to } S_{\triangle BAD} \cdot S_{\triangle ECF} = S_{\triangle BCF} \cdot S_{\triangle EAD}.$$



Example 9 : As shown in the figure, D is the midpoint of side BC in triangle ABC , K is the intersection of AC and the circumscribed circle O of triangle ABD , EK is parallel to AB and intersects circle O at E . If $AD = DE$, prove that: $AB + AK = KC$. (2019 Jiangsu High School Mathematics Competition)

$c = 2d - b$, solve the equation $\frac{a-k}{a-c} = \overline{\left(\frac{a-k}{a-c}\right)}$, get $k = \frac{bd(a+b-2d)}{2ab-ad-bd}$, $e = \frac{ab}{k}$,

$$(a-b)\overline{(a-b)} - (k-c-\overline{(a-k)})\overline{(k-c-\overline{(a-k)})}$$

$$= \frac{\left(\frac{a-b}{a-d}\right)^2}{\frac{b}{d}} \left((d-a)\overline{(d-a)} - (d-e)\overline{(d-e)} \right),$$

Extended proposition: As shown in the figure, D is the midpoint of side BC in triangle ABC , K is the intersection point of AC and the circumscribed circle O of triangle ABD , EK is parallel to AB and intersects circle O at E . Prove:

$$AB^2 - (CK - KA)^2 = \left(\frac{AB}{AD}\right)^2 \cdot (AD^2 - ED^2).$$

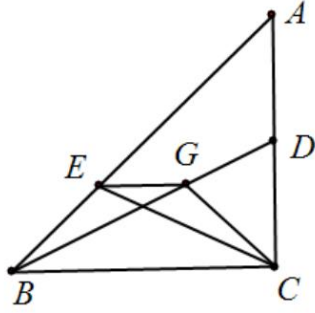
General complex number solution: $(a-b)\overline{(a-b)} - (k-c-\overline{(a-k)})\overline{(k-c-\overline{(a-k)})}$

$$= \frac{2(a-b)^2(b-d)(2a^2b - a^2d + abd - ad^2 + bd^2 - 2d^3)}{abd(a+b-2d)(2ab-ad-bd)} ;$$

$$(d-a)\overline{(d-a)} - (d-e)\overline{(d-e)} = \frac{2(a-d)^2(b-d)(2a^2b - a^2d + abd - ad^2 + bd^2 - 2d^3)}{ad^2(a+b-2d)(2ab-ad-bd)} ; \text{ This means that } AD = DE$$

and $AB + AK = KC$ are equivalent $2a^2b - a^2d + abd - ad^2 + bd^2 - 2d^3 = 0$, so it is proved that:

$$AD = DE \Leftrightarrow AB + AK = KC.$$



Example 10 : Given that G is the centroid of triangle ABC , D is the midpoint of side CA , and a straight line passing through point G and parallel to base BC intersects AB at point E . *Proof:* $\angle AEC = \angle DGC$ if and only if $\angle ACB = 90^\circ$.
(2010 British Mathematical Olympiad)

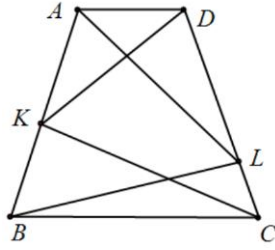
Proof: Let $A = (a, \bar{a})$, $B = (b, b)$, $C = (0, 0)$, $D = \left(\frac{a}{2}, \frac{\bar{a}}{2}\right)$, $G = \left(\frac{a+b}{3}, \frac{\bar{a}+b}{3}\right)$,

$e = \left(\frac{a+2b}{3}, \frac{\bar{a}+2b}{3}\right)$, the identity

$$27|e-c|^2|g-d|^2 \left(\frac{\frac{e-a}{g-d} - \overline{\left(\frac{e-a}{g-d}\right)}}{g-c} \right) = [ABC]|c-a|^2 \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a}\right)} \right),$$

Extended proposition: Given that G is the centroid of triangle ABC , D is the midpoint of side CA , and the straight line passing through point G and parallel to base BC intersects AB at point E . *Proof:*

$$3\sin(\angle AEC - \angle DGC) = 2\sin \angle AEC \sin \angle DGC \cot C.$$



Example 11 : As shown in Figure 1, in trapezoid $ABCD$, it is known that $AD \parallel BC$. K and L are points on AB and CD respectively , satisfying $\angle BAL = \angle CDK$. Proof: $\angle BLA = \angle CKD$. (2008 Singapore Mathematical Olympiad)

$$A = (a, \bar{a}) \quad , \quad B = (0, 0) \quad , \quad C = (1, 1) \quad , \quad D = (d, d - (a - \bar{a})) \quad , \quad K = \left(k, \frac{2\bar{a}}{a} \right) \quad ,$$

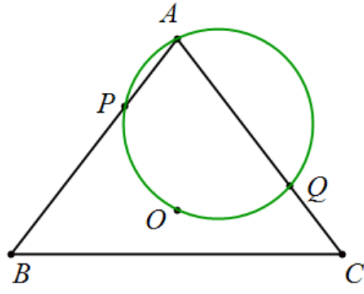
$$L = \left(l, \frac{a - \bar{a} - l - al + \bar{a}l + dl}{-1 + d} \right) ,$$

$$([AKC] - [LKC])(a - b)(\overline{a - b})(d - c)(\overline{d - c}) \left(\frac{\frac{a - l}{d - c}}{\frac{d - k}{d - c}} - \frac{\overline{\frac{a - l}{d - c}}}{\overline{\frac{d - k}{d - c}}} \right)$$

$$= [ABD](l - a)(\overline{l - a})(k - d)(\overline{k - d}) \left(\frac{\frac{l - b}{k - d}}{\frac{k - c}{k - d}} - \frac{\overline{\frac{l - b}{k - d}}}{\overline{\frac{k - c}{k - d}}} \right) ,$$

Extended proposition: As shown in Figure 1, in trapezoid $ABCD$, it is known that $AD \parallel BC$. K and L are points on AB and CD respectively . Prove that:

$$(S_{\triangle AKC} - S_{\triangle LKC})AB \cdot DC \sin(BAL - CDK) = S_{\triangle ABD} \cdot LB \cdot KC \sin(\angle BLA - \angle CKD) .$$



Example 12 : In triangle ABC , $AB = AC$, points P and Q are on sides AB and AC respectively and do not coincide with the vertices of the triangle. Prove: The circumcircle of triangle APQ passes through the circumcenter of ABC if and only if $AP = CQ$. (2012 Estonian National Team Selection Exam)

Proof: Assume $A = (1, 1)$ that $B = \left(b, \frac{1}{b}\right)$, $C = \left(\frac{1}{b}, b\right)$, $P = \left(p, \frac{a+b-p}{ab}\right)$,

$$Q = \left(q, \frac{a+c-q}{ac}\right), \quad O = (0, 0), \quad [APOQ]([APC] + [BCQ])$$

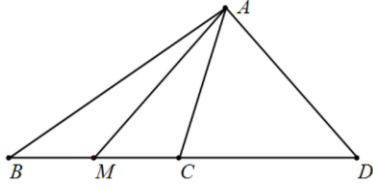
$$= \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a}\right)}\right) \overline{(a-p)}(a-b)\overline{(a-q)}(a-c)\left((c-q)\overline{(c-q)} - (a-p)\overline{(a-p)}\right),$$

Extended proposition: In $\triangle ABC$, it is known that $AB = AC$, points P and Q are on sides AB and AC respectively and do not coincide with the vertices of the triangle.

Proof: $4(S_{\triangle APC} + S_{\triangle BCQ}) \cdot OP \cdot OQ \sin(\angle A + \angle POQ) = (CQ^2 - AP^2) \cdot CB \cdot CA \cos C$.

Example 13 : In $2\angle ACB = 180^\circ + \angle CBA$ triangle ABC , M is the midpoint of side BC .

Draw a circle with vertex A as the center and intersect the line BC at points M and D . Prove:
 $MD = AB$. (2010 Croatian Mathematical Competition Final)



Proof: Assume $A = (a, -a)$, $B = (b, b)$, $C = (c, c)$, $M = \frac{B+C}{2}$, $D = -M$,

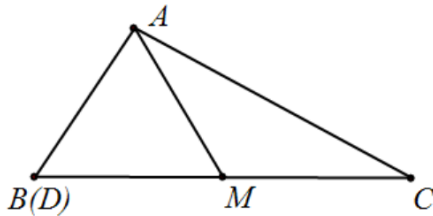
$$(c-a)^2 (c+a)^2 (b-a)(b+a) \left(\frac{\left(\frac{c-b}{c-a} \right)^2}{\frac{b-a}{b-c}} - \frac{\left(\frac{c-b}{c+a} \right)^2}{\frac{b+a}{b-c}} \right)$$

$$= 2a(b-c)^3 ((m-d)(m-d) - (a-b)(-a-b)) ,$$

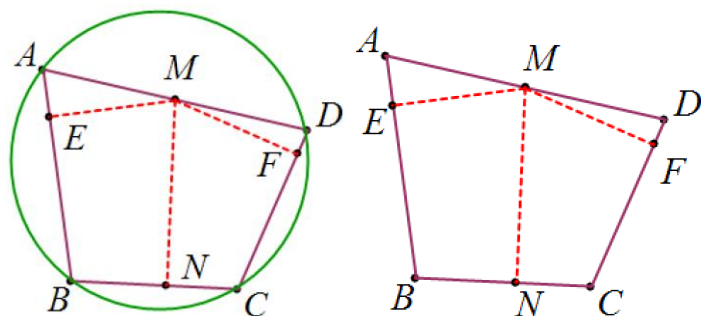
Extended proposition: In $\triangle ABC$, M is the midpoint of side BC . Draw a circle with vertex A as the center that intersects the straight line BC at points M and D . Proof:

$$-CA^2 \sin(2\angle ACB - \angle CBA) = (MD^2 - AB^2) \sin \angle ABC .$$

Note: This identity also corresponds to the classic figure, $\angle A = 90^\circ$, $\angle B = 60^\circ$, $\angle C = 30^\circ$, at this time B and D coincide.



Example 14 : As shown in the figure, in the inscribed quadrilateral $ABCD$, from the midpoint M of AD , draw $MN \perp BC$, $ME \perp AB$, $MF \perp CD$, N , E , F are the feet of perpendiculars. Proof: MN passes through the midpoint of line segment EF . (2018 Shanxi High School Mathematics Competition)



Abandoning the cocircle of points A , B , C , and D , explore the relationship between the cocircle of points A , B , C , and D and MN passing through the midpoint of line segment EF . Among them, the cocircle of points $\angle A + \angle C = 180^\circ$, A , B , C , and D is equivalent to , and MN passing through the midpoint of line segment $S_{\triangle MEN} = S_{\triangle MFN}$ is equivalent to .

Establish the identity:

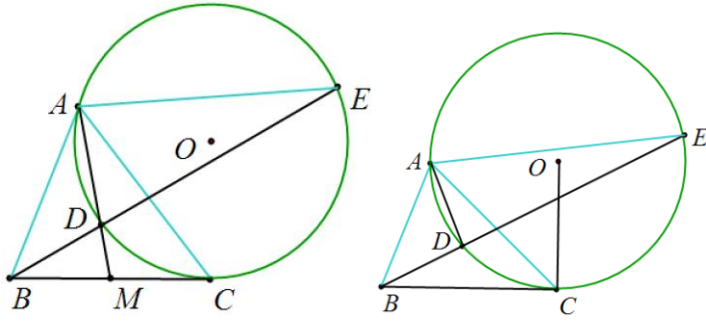
$$\frac{\begin{vmatrix} m & \bar{m} & 1 \\ e & \bar{e} & 1 \\ n & \bar{n} & 1 \end{vmatrix} + \begin{vmatrix} m & \bar{m} & 1 \\ f & \bar{f} & 1 \\ n & \bar{n} & 1 \end{vmatrix}}{(a-b)(\overline{a-b}) \left(\frac{a-d}{c-d} - \overline{\left(\frac{a-d}{c-d} \right)} \right)}$$

$$= \frac{\begin{vmatrix} m & \bar{m} & 1 \\ e & \bar{e} & 1 \\ n & \bar{n} & 1 \end{vmatrix} \left(\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \right)}{\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix}},$$

Extended proposition: As shown in the figure, in the convex quadrilateral $ABCD$, from the midpoint M of AD , draw $MN \perp BC$, $ME \perp AB$, $MF \perp CD$, N , E , F are the feet of the perpendicular. Proof:

$$S_{\triangle MEN} \cdot (S_{\triangle ABC} - S_{\triangle ABD}) \cdot AB \cdot BC \cdot DA \sin(\angle A + \angle C) = 2CD \cdot S_{\triangle ABC} \cdot S_{\triangle ABD} \cdot (S_{\triangle MEN} - S_{\triangle MFN})$$

.



Example 15 : As shown in the figure, in the acute angle $\triangle ABC$, M is the midpoint of BC , circle O passes through point A and is tangent to line BC at point C , line AM intersects circle O at another point D , and line BD intersects circle O at another point E . Proof: $\angle EAC = \angle BAC$. (2018 Shaanxi High School Mathematics League)

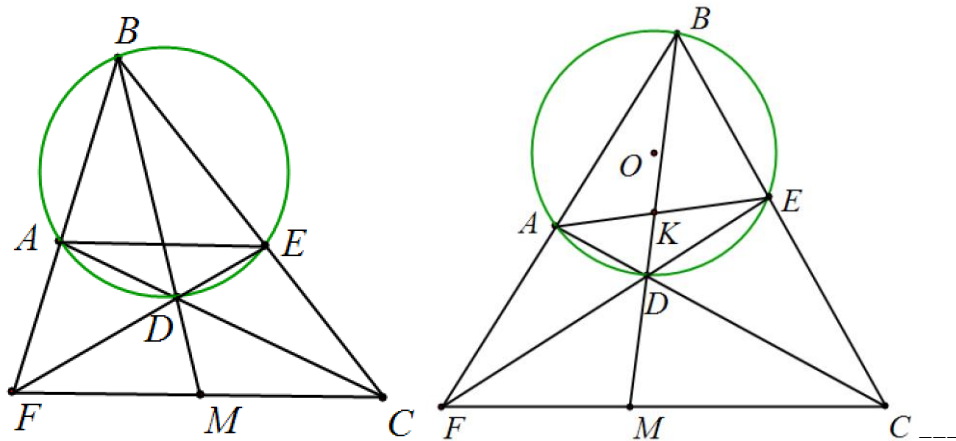
the circumscribed circle of $b = \frac{cc(d+e)-de(c+c)}{cc-de} \triangle ADC$ be the unit circle, ,

$$\frac{\frac{a-e}{a-c} - \overline{\left(\frac{a-e}{a-c}\right)}}{\frac{a-c}{a-b} - \overline{\left(\frac{a-c}{a-b}\right)}} = \frac{acd(a-e)(c-e)}{e(a-c)^2(a-d)(c-d)} \left(\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} + \begin{vmatrix} a & \bar{a} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \right),$$

Extended proposition: As shown in the figure, in the acute angle $\triangle ABC$, the circle O passes through point A and is tangent to the straight line BC at point C . The straight line through B intersects the circle O at D and E . Proof:

$$2CE \cdot (S_{\triangle ABD} - S_{\triangle ACD}) = AB \cdot AD \cdot CD \sin(\angle CAB - \angle EAC).$$

Example 16 : A circle passes through vertices A and B of triangle ABC , intersecting line segments AC and BC at points D and E respectively, straight lines BA and ED intersect at point F , and straight lines BD and CF intersect at point M . Proof: $MF = MC \Leftrightarrow MB \cdot MD = MC^2$. (2003 American Mathematical Olympiad test paper)



$$, \quad f = \frac{ab(d+e) - de(a+b)}{ab - de}, \quad , \quad c = \frac{ad(b+e) - be(a+d)}{ad - be}, \quad ,$$

$$m = \frac{-ab^2d - abd^2 + 2b^2d^2 + ab^2e - b^2de + ad^2e - bd^2e}{-2abd + b^2d + bd^2 + abe + ade - 2bde}, \quad k = \frac{ae(b+d) - bd(a+e)}{ae - bd},$$

$$\frac{(m-c)\overline{(m-c)} - (m-d)\overline{(m-b)}}{(m-c)\overline{(m-c)} - (m-f)\overline{(m-f)}} \cdot \frac{\frac{a-b}{d-b}}{\frac{a-e}{d-e}} \cdot \frac{\left(\begin{vmatrix} a & \bar{a} & 1 \\ e & \bar{e} & 1 \\ b & \bar{b} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ e & \bar{e} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \right)^2}{\left(\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ e & \bar{e} & 1 \end{vmatrix} \right)^2} = 1.$$

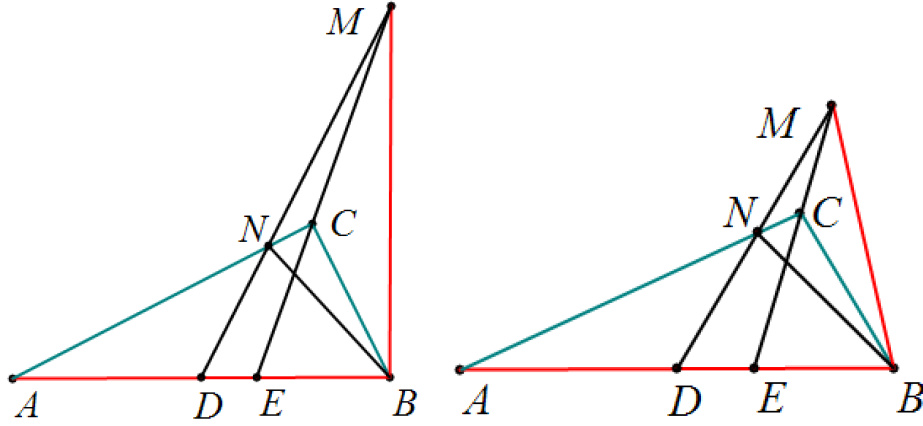
Extended proposition: In the circle O , quadrilateral $ADEB$ is inscribed. Lines BE and AD intersect at point C . Lines BA and ED intersect at point F . Lines AE and BD intersect at point K . Lines BD and CF intersect at point M . Proof:

$$(MC^2 - MB \cdot MD) \cdot OK^2 \cdot AB \cdot DE \cdot \left(\frac{S_{ABCD}}{S_{\triangle ABD} - S_{\triangle ABE}} \right)^2 = AE \cdot DB \cdot R^2 (MC^2 - MF^2).$$

Interpreting diversity, equivalence

$$\frac{a-e}{f-c} - \frac{1/a - 1/e}{f - c} = \frac{(ad - be)(ab - de)(abd + a^2e - 2abe - 2ade + bde + ae^2)}{a(b-d)e(a-b-d+e)(-abd + abe + ade - bde)}, \quad ,$$

$$\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} + \begin{vmatrix} e & \bar{e} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = -\frac{(b-d)(abd+a^2e-2abe-2ade+bde+ae^2)}{abde},$$



Example 17 : As shown in the figure, in the right triangle ABC , point D is the midpoint of the hypotenuse AB , $MB \perp AB$, MD intersects AC at N ; the extension of MC intersects AB at E . Proof: $\angle DBN = \angle BCE$. (2007 China Southeast Mathematical Olympiad)

Proof: Let D be the origin, DB be the real axis, $d = 0$, $a = -b$, $\bar{n} = \frac{\bar{m}n}{m}$,

$$\bar{c} = \frac{bcm - bmn + \bar{b}mn + \bar{c}mn}{m(b+n)},$$

$$\text{Identity: } \begin{vmatrix} d & \bar{d} & 1 \\ b & \bar{b} & 1 \\ m & \bar{m} & 1 \end{vmatrix} n(c-n)(b-m)\overline{(b-m)} \left(\frac{b-d}{b-m} + \overline{\left(\frac{b-d}{b-m} \right)} \right)$$

$$- \frac{1}{2} \left(\begin{vmatrix} d & \bar{d} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} - \begin{vmatrix} d & \bar{d} & 1 \\ b & \bar{b} & 1 \\ m & \bar{m} & 1 \end{vmatrix} \right) m(b+n)^2 \overline{(c-a)} \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a} \right)} \right)$$

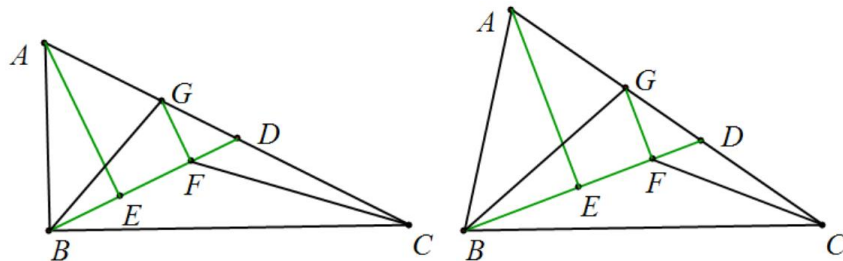
$$+ m(b+n)(b-n)\overline{(b-n)}(c-b)\overline{(c-b)} \left(\frac{\frac{b-d}{c-b}}{\frac{m-c}{m-c}} - \overline{\left(\frac{\frac{b-d}{c-b}}{\frac{m-c}{m-c}} \right)} \right) = 0.$$

Extended proposition: As shown in the figure, in triangle ABC , point D is the midpoint of the hypotenuse AB , MD intersects AC at N ; the extension line of MC intersects AB at E . Proof:

$$4S_{\triangle DBM} \cdot CN \cdot DN \cdot BM \cdot BD \cos \angle MBD = 2(S_{\triangle DBC} - S_{\triangle DBM}) \cdot AN^2 \cdot DM \cdot CB \cos \angle ACB$$

$$+ DM \cdot AN \cdot BD \cdot BN \cdot CB \cdot MC \sin(\angle DBN - \angle BCE).$$

Example 18 : As shown in the figure , in triangle ABC , $\angle ABC = 90^\circ$, D and G are two points on *side* CA , connect BD and BG . Draw perpendicular lines to BD through points A and G , with their feet at E and F , respectively , and connect CF . Given $BE = EF$, prove that: $\angle ABG = \angle DFC$. (2006 China Southeast Mathematical Olympiad Test Question)



Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, solve the

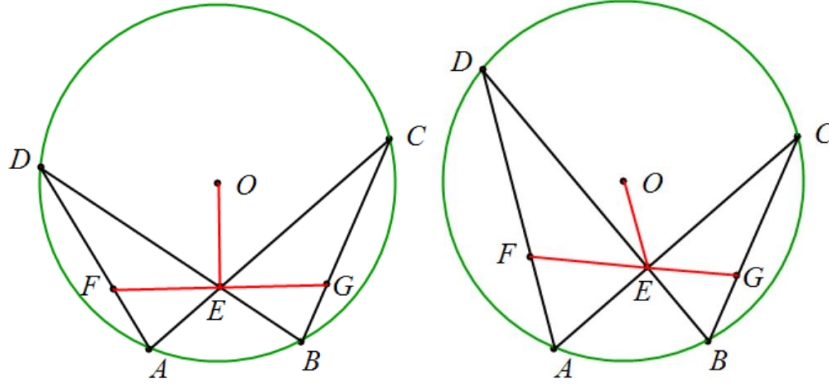
equation $\frac{a-d}{a-c} = \overline{\left(\frac{a-d}{a-c}\right)}$, and we get $\bar{d} = \frac{a+c-d}{ac}$; Similarly $\bar{g} = \frac{a+c-g}{ac}$.

Establish the identity: $abc(d-g)(b-c)\overline{(b-c)}\left(\frac{b-a}{b-c} + \overline{\left(\frac{b-a}{b-c}\right)}\right)$

$$- \frac{2(a-b)(a-c)(b-c)(c-g)}{\left(\begin{vmatrix} a & \bar{a} & 1 \\ c & \bar{c} & 1 \\ b & \bar{b} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ c & \bar{c} & 1 \\ f & \bar{f} & 1 \end{vmatrix}\right)} \left((e-b)\overline{(e-b)} - (e-f)\overline{(e-f)} \right) \\ + \frac{2a^2b^2c^2}{(a-b)(b-c)} (b-g)(b-d)\overline{(b-g)}\overline{(b-d)} \left(\frac{\frac{b-a}{b-g}}{\frac{b-d}{f-c}} - \overline{\left(\frac{\frac{b-a}{b-g}}{\frac{b-d}{f-c}}\right)} \right) = 0.$$

Extended proposition: As shown in the figure , in $\triangle ABC$, D and G are two points on the side CA , connect BD and BG . Draw perpendicular lines to BD through points A and G , with their feet at E and F respectively , and connect CF .

$$\text{Prove: } \frac{S_{\triangle ACB} \cdot CG}{(S_{\triangle ACB} - S_{\triangle ACF})} (EB^2 - EF^2) - DG \cdot BC \cdot BA \cos \angle ABC \\ = BG \cdot BD \cdot FC \frac{\sin(\angle ABG - \angle DFC)}{\sin \angle BAC}.$$



Example 19 : In the circle O , inscribe a quadrilateral $ABCD$. AC intersects BD at E . Draw a straight line through E that intersects AD at F and BC at G . If $OE \perp FG$, prove that $EF = EG$. (Butterfly Theorem)

Proof: Let O be the unit circle, $e = \frac{ac(b+d) - bd(a+c)}{ac - bd}$, solve the equation

$$\frac{a-d}{a-f} = \overline{\left(\frac{a-d}{a-f}\right)}, \text{ and we get } \bar{f} = \frac{a+d-f}{ad}. \text{ Solve the equation } \frac{e-f}{e-g} = \overline{\left(\frac{e-f}{e-g}\right)},$$

$$\frac{b-g}{b-c} = \overline{\left(\frac{b-g}{b-c}\right)}, \quad \text{and} \quad \text{we} \quad \text{get}$$

$$g = \frac{a^2b^2c^2 - 2a^2b^2cd + a^2b^2d^2 - a^2c^2d^2 + 2abc^2d^2 - b^2c^2d^2 - ab^2c^2f + 2ab^2cdf + a^2c^2df - 2abc^2df + b^2c^2df - ab^2d^2f}{a^2bc^2 - 2a^2bcd + a^2bd^2 - a^2cd^2 + 2abcd^2 - b^2cd^2 - abc^2f + a^2cdf + b^2cdf - abd^2f}$$

and

$$\bar{g} = \frac{a^2c^2 - 2a^2cd + 2abd^2 - b^2d^2 - ac^2f + a^2df - 2abdf + b^2df + 2acdf - ad^2f}{a^2bc^2 - 2a^2bcd + a^2bd^2 - a^2cd^2 + 2abcd^2 - b^2cd^2 - abc^2f + a^2cdf + b^2cdf - abd^2f}.$$

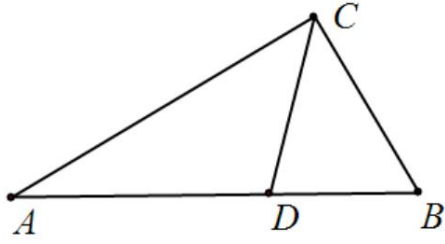
Identity: $(a-b)(b-c)(a-d)(c-d)((e-f)\overline{(e-f)} - (e-g)\overline{(e-g)})$

$$= a^2b^2c^2d^2\overline{(e-f)(e-g)}\left(\frac{f-g}{f-g}\right)\left(\frac{\bar{b}-d}{a-c}\left(\frac{a-c}{b-d} - \overline{\left(\frac{a-c}{b-d}\right)}\right)\right)^2 e\bar{e}\left(\frac{g-f}{e-o} + \overline{\left(\frac{g-f}{e-o}\right)}\right).$$

$ABCD$ is inscribed in the circle O . AC intersects BD at E . Draw a straight line through E that intersects AD at F and BC at G . R is the radius of the circle. Prove

$$AB \cdot BC \cdot CD \cdot DA \cdot (EF^2 - EG^2) = 8R^2 \cdot EF \cdot EG \cdot OE \cdot FG \sin^2 \angle AEB \cos \angle OEF \quad .$$

(Extended version of the butterfly theorem)



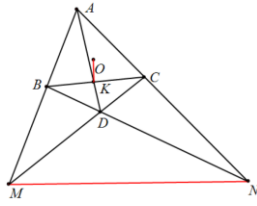
Example 20 : In $\triangle ABC$, $\angle ACB = 90^\circ$, point D is on AB , and $(CD \cdot AB)^2 = 2CA \cdot AD \cdot DB \cdot BC$, prove that: CD bisects $\angle ACB$. (Mathematics Bulletin Call for Solutions 2639)

Proof: Let AB be the real axis and C be on the imaginary axis, then $b = \frac{c^2}{a}$,

$$\begin{aligned} & ((a-b)(a-b)(d-c)(d+c))^2 - 4(c-a)(-c-a)(d-a)(d-a)(d-b)(d-b)(b-c)(b+c) \\ &= \frac{(b-a)^2}{4c^2} \left((c-d)^2 (c+d)^2 \left(\frac{\frac{c-b}{c-d}}{\frac{c-a}{c-d}} - \overline{\left(\frac{\frac{c-b}{c-d}}{\frac{c-a}{c-d}} \right)} \right) \right)^2. \end{aligned}$$

In $\triangle ABC$, $\angle ACB = 90^\circ$, point D is on AB , prove that:

$$(AB \cdot CD)^4 - 4CA^2 \cdot AD^2 \cdot DB^2 \cdot BC^2 = AB^4 \cdot CD^4 \sin^2(\angle BCD - \angle ACD) .$$



Example 2 1 : The circumcenter of acute angle ABC is O , K is a point on side BC (not the midpoint of side BC), D is a point on the extension of line segment AK , straight lines BD and AC intersect at point N , and straight lines CD and AB intersect at point M . Prove: If $OK \perp MN$, then points A , B , D , and C are in the same circle. (2010 National High School Mathematics League)

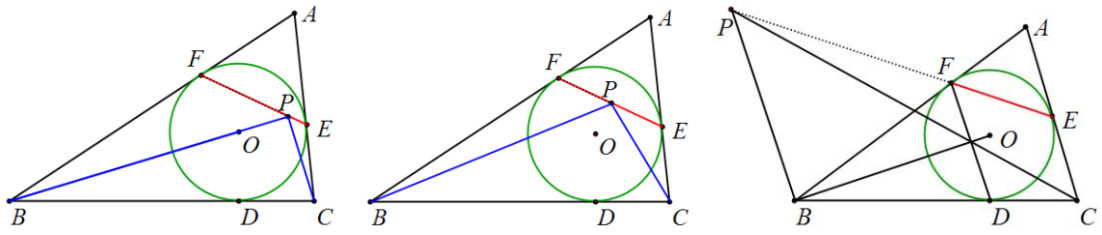
Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle,

$$\begin{aligned} \text{Identity: } & [A, B, D][A, C, D][A, D, O][B, D, O][C, D, O] |m-n|^2 \left(\frac{o-k}{m-n} + \overline{\left(\frac{o-k}{m-n} \right)} \right) \\ & = -([A, B, D] - [A, D, C])[A, K, O][D, N, O] |a-b|^2 |c-d|^2 \left(\frac{a-c}{a-b} \frac{d-b}{d-c} - \overline{\left(\frac{a-c}{a-b} \frac{d-b}{d-c} \right)} \right) \end{aligned}$$

acute angle ABC is O , K is a point on side BC (not the midpoint of side BC), D is a point on the extension of line segment AK , straight lines BD and AC intersect at point N , and straight lines CD and AB intersect at point M . Let the angle between straight lines α OK and MN be , prove:

$$\begin{aligned} & 4S_{\triangle ABD} \cdot S_{\triangle ACD} \cdot S_{\triangle ADO} \cdot S_{\triangle BDO} \cdot S_{\triangle CDO} \cdot MN \cdot OK |\cos \alpha| \\ & = |S_{\triangle ABD} - S_{\triangle ADC}| \cdot S_{\triangle AKO} \cdot S_{\triangle DMO} \cdot S_{\triangle DNO} \cdot AB \cdot BD \cdot DC \cdot CA |\sin (A + D)|. \end{aligned}$$

In the original question, it is emphasized that K is not the midpoint of side BC , which is clearly reflected in the identity. $S_{\triangle ABD} = S_{\triangle ADC}$ Correspondingly, K is the midpoint of side BC .



Example 22 : $\triangle ABC$ has an incircle O . D , E , and F are the points of tangency between the circle and the three sides. P is the intersection of the bisector of $\angle B$ and EF . Prove that: $PB \perp PC$.

Proof: Let O be the unit circle, $b = \frac{2df}{d+f}$, $c = \frac{2de}{d+e}$, $\bar{p} = \frac{e+f-p}{ef}$, if P is a point on line segment $\angle PBF - \angle PBC$ EF , explore the relationship between $\angle BPC$ and .

Calculate $\frac{\frac{b-f}{b-p} - \left(\frac{\overline{b-f}}{\overline{b-p}} \right)}{\frac{b-p}{b-c} + \left(\frac{\overline{p-c}}{\overline{p-b}} \right)}$ and try to express the result in the form of geometric quantities, so there is an identity

$$2 \frac{\begin{vmatrix} c & \bar{c} & 1 \\ 0 & \bar{0} & 1 \\ p & \bar{p} & 1 \\ b & \bar{b} & 1 \end{vmatrix}}{\begin{vmatrix} c & \bar{c} & 1 \\ 0 & \bar{0} & 1 \end{vmatrix}} (p-b) \overline{(p-b)} \left(\frac{b-f}{b-p} - \frac{\overline{b-f}}{\overline{b-p}} \right) = \left(\begin{vmatrix} d & \bar{d} & 1 \\ f & \bar{f} & 1 \\ b & \bar{b} & 1 \end{vmatrix} - \begin{vmatrix} d & \bar{d} & 1 \\ f & \bar{f} & 1 \\ p & \bar{p} & 1 \end{vmatrix} \right) \left(\frac{p-c}{p-b} + \frac{\overline{p-c}}{\overline{p-b}} \right)$$

$\triangle ABC$ has an incircle O , D , E , F are the points of tangency between the circle and the three sides, P is a point on line segment

$$\frac{S_{\triangle COP}}{S_{\triangle BCO}} BF \cdot BC \cdot BP \sin(\angle PBF - \angle PBC) = 2(S_{\triangle DFB} + S_{\triangle DFP}) PC \cos \angle BPC \quad EF, \text{ prove: .}$$

Since traditional Euclidean geometry does not have the concept of "direction", when complex number identities are translated into Euclidean geometry conclusions, it is necessary to determine the specific figures, and sometimes there will be a difference of positive and negative signs.

point P is on the extension line of

$$\frac{S_{\triangle COP}}{S_{\triangle BCO}} BF \cdot BC \cdot BP \sin(\angle PBF + \angle PBC) = 2(S_{\triangle DFB} - S_{\triangle DFP}) PC \cos \angle BPC \quad \text{line}$$

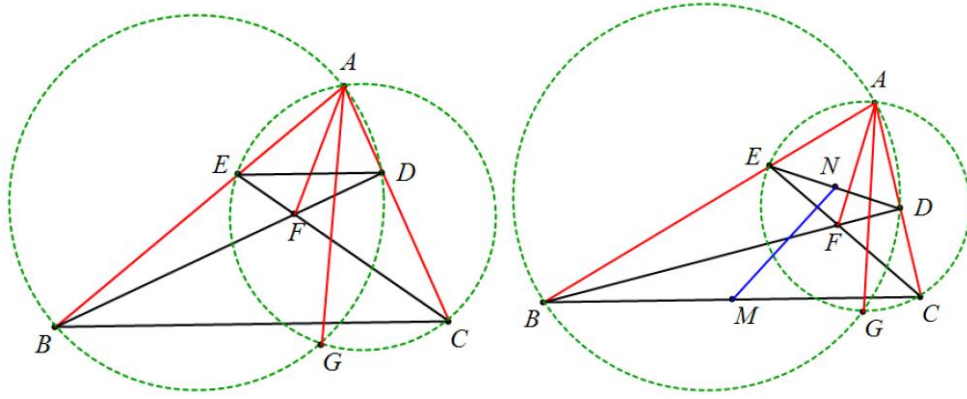
segment EF , the conclusion changes to: .

If $S_{\triangle DFB} = S_{\triangle DFP}$, then $\angle PBF + \angle PBC = 180^\circ$, at this time $BP \parallel DF$, P is on the exterior angle bisector of $\angle B$.

Example 23 : In $\triangle ABC$, $DE \parallel BC$, CE and BD intersect at point F , the circumcircle of $\triangle ADB$ and the circumcircle of $\triangle AEC$ intersect at point G , then $\angle CAG = \angle BAF$.

Proof: $a=0$, $c=kd$, $b=ke$, $f=\frac{b+kd}{1+k}$, solving the equation

$$\frac{\frac{a-g}{d-g}}{\frac{a-b}{d-b}} = \frac{\overline{\left(\frac{a-g}{d-g}\right)}}{\overline{\left(\frac{a-b}{d-b}\right)}}, \quad \frac{\frac{a-g}{e-g}}{\frac{a-c}{e-c}} = \frac{\overline{\left(\frac{a-g}{e-g}\right)}}{\overline{\left(\frac{a-c}{e-c}\right)}} \text{ we get } g = \frac{de(1+k)}{d+e}, \quad \frac{\frac{f}{d}}{\frac{e}{g}} = k.$$



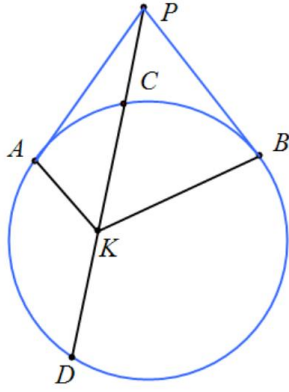
Assume $a=0$, $c=kd$, $b=te$, $f=\frac{dk+et-dkt-ekt}{1-kt}$, $g=\frac{de(1-kt)}{d+e-dk-et}$, then

$$4\left(\frac{c+b}{2}-\frac{d+e}{2}\right)\overline{\left(\frac{c+b}{2}-\frac{d+e}{2}\right)}\left(\frac{\frac{f}{d}}{\frac{e}{g}}-\overline{\left(\frac{\frac{f}{d}}{\frac{e}{g}}\right)}\right)=\left(d\bar{e}-\bar{d}e\right)(1-kt)(k-t),$$

In $\triangle ABC$, CE and BD intersect at point F , and the circumcircle of $\triangle ADB$ intersects the circumcircle of $\triangle AEC$ at point G . *Prove* that:

$$4MN^2 \cdot AF \cdot AG \sin(\angle BAF - \angle CAG) = (AD \cdot AE - AB \cdot AC)(AC \cdot AE - AB \cdot AD) \sin A. \text{ i}$$

$$\text{n } \begin{vmatrix} a & \bar{a} & 1 \\ e & \bar{e} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = \bar{d}e - d\bar{e}.$$



Example 24 : There are four points A , C , B , and D on a circle, PA and PB are tangents, K is the midpoint of CD . The necessary and sufficient condition to prove that P , C , and D are collinear is $\angle AKC = \angle BKC$.

Proof: Let the circle be the unit circle, $p = \frac{2ab}{a+b}$, $\bar{p} = \frac{2}{a+b}$,

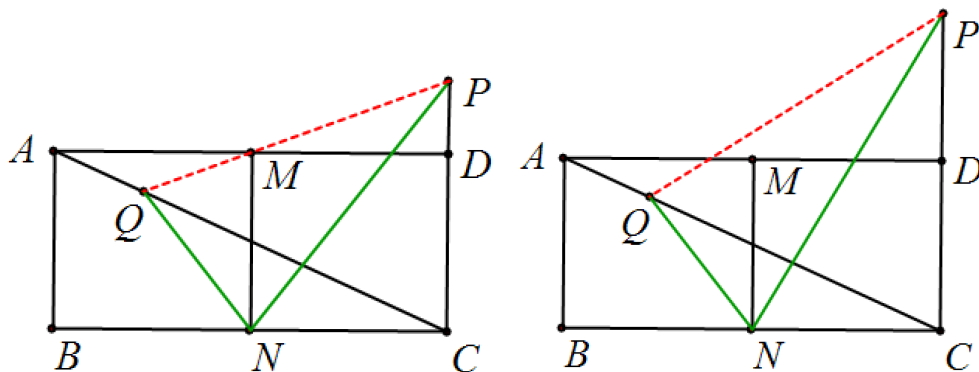
$$(d-c)^2 \overline{(d-c)}^2 \left(\frac{\frac{c+d}{2} - a}{\frac{d-c}{d-c}} - \overline{\left(\frac{\frac{c+d}{2} - a}{\frac{d-c}{d-c}} \right)} \right) +$$

$$\left(\frac{a+b}{a-b} \right) \left(\left| \begin{array}{ccc} a & \bar{a} & 1 \\ d & \bar{d} & 1 \\ c & \bar{c} & 1 \end{array} \right| + \left| \begin{array}{ccc} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{array} \right| \right) \left| \begin{array}{ccc} p & \bar{p} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{array} \right| = 0,$$

four points A , C , B , and D on a circle. PA and PB are tangent lines. K is the midpoint of CD . Prove that:

$$AB \cdot KA \cdot KB \cdot DC^2 \sin(\angle AKC - \angle CKB) = 8OM \cdot S_{\triangle PCD} \cdot (S_{\triangle ADC} + S_{\triangle BCD}).$$

Example 25 : As shown in Figure 31 , in rectangle $ABCD$, M is the midpoint of AD , N is the midpoint of BC , point P is on the extension line of CD , Q is a point on AC , connect PN and QN . Prove that the necessary and sufficient condition for points P , M and Q to be collinear is $\angle QNM = \angle PNM$.



Proof: Let NC be the real axis, NM be the imaginary axis, $n=0$,

$$d = c + m, \quad \bar{d} = c - m, \quad a = 2m - d, \quad \bar{a} = -2m - \bar{d},$$

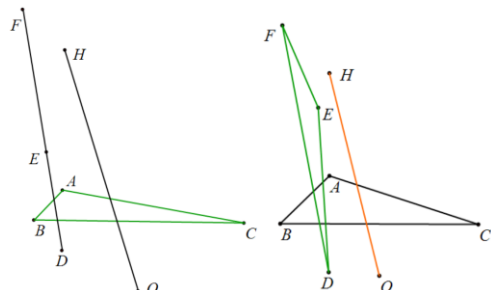
$$\bar{q} = -\frac{2cm - 2cq - mq}{2c - m}, \quad \bar{p} = 2c - p,$$

$$(m - q)(\overline{m - q}) \left(\frac{m - p}{m - q} - \overline{\left(\frac{m - p}{m - q} \right)} \right) = m^2 \left(\frac{\frac{n - q}{n - m}}{n - p} - \overline{\left(\frac{\frac{n - q}{n - m}}{n - p} \right)} \right),$$

rectangle $ABCD$, M is the midpoint of AD , N is the midpoint of BC , P is a point on CD , Q is a point on AC , connect PN , QN . Prove:

$$2S_{\triangle PMQ} = NP \cdot NQ |\sin(\angle QNM - \angle PNM)|.$$

Example 26 : In triangle ABC , H is the orthocenter, O is the circumcenter, R is the radius of the circumscribed circle, let D and A be symmetric about BC , E and B be symmetric about CA , and F and C be symmetric about AB . Prove that D , E and F are collinear if and only if $OH = 2R$. (39th IMO – Shortlist)



Proof: Let the circumcircle of $\triangle ABC$ be the unit circle,
 $d = 2 \frac{a+b+c-bc/a}{2} - a = b+c-bc/a$, $e = a+c-ca/b$, $f = a+b-ab/c$,

$$\frac{[D,E,F]}{[A,B,C]} = 4 - (a+b+c)\overline{(a+b+c)},$$

$\triangle ABC$, H is the orthocenter, O is the circumcenter, R is the radius of the circumscribed circle, let D and A be symmetric about BC , E and B be symmetric about CA , and F and C be symmetric about AB . Proof:

$$\frac{S_{\triangle DEF}}{S_{\triangle ABC}} = 4 - \left(\frac{OH}{R} \right)^2.$$

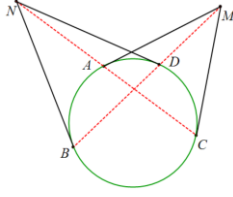


Figure 2 8

Example 27 : As shown in Figure 28 , for quadrilateral $ABCD$ inscribed in a circle , the tangents through A and C intersect at M , and the tangents through B and D intersect at N . The necessary and sufficient condition for N , A , and C to be collinear is that M , B , and D to be collinear.

Proof: Let the circle be the unit circle, $m = \frac{2ac}{a+c}$, $n = \frac{2bd}{b+d}$,

$$\frac{n-a}{a-c} - \overline{\left(\frac{n-a}{a-c}\right)} = \frac{-ab+2ac-bc-ad+2bd-cd}{(a-c)(b+d)},$$

$$\frac{m-b}{b-d} - \overline{\left(\frac{m-b}{b-d}\right)} = \frac{-ab+2ac-bc-ad+2bd-cd}{(a+c)(b-d)}, \text{ so the necessary and sufficient}$$

condition for the three points N , A , and C to be collinear is that the three points M , B , and D to be collinear.

$$\begin{vmatrix} m & \bar{m} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = \frac{(b-d)(ab+bc+cd+da-2ac-2bd)}{bd(a+c)},$$

$$\begin{vmatrix} n & \bar{n} & 1 \\ a & \bar{a} & 1 \\ c & \bar{c} & 1 \end{vmatrix} = \frac{(a-c)(ab+bc+cd+da-2ac-2bd)}{ac(b+d)},$$

Because A , B , C , and D are not equal, the three points N , A , and C are collinear, and the three points M , B , and D are collinear

$ab+bc+cd+da-2ac-2bd=0$. To prove the conclusion, we need to ensure that the other terms in the numerator are $b-d \neq 0$, $a-c \neq 0$ and we generally don't care about the denominator. So far, we have reached the conclusion

$$\begin{vmatrix} m & \bar{m} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = 0 \Leftrightarrow \begin{vmatrix} n & \bar{n} & 1 \\ a & \bar{a} & 1 \\ c & \bar{c} & 1 \end{vmatrix} = 0. \text{ This is a necessary and sufficient condition,}$$

very beautiful.

In fact, if we spend some time to care about the denominator, we will have unexpected gains. In fact, if we spend some time to study the denominator, we will have unexpected gains. Observation shows that

$$\frac{a^2 - c^2}{ac} \begin{vmatrix} m & \bar{m} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = \frac{b^2 - d^2}{bd} \begin{vmatrix} n & \bar{n} & 1 \\ a & \bar{a} & 1 \\ c & \bar{c} & 1 \end{vmatrix} = \frac{(a-c)(b-d)(ab+bc+cd+da-2ac-2bd)}{abcd}$$

$$\text{this is it } \begin{vmatrix} 0 & \bar{0} & 1 \\ a & \bar{a} & 1 \\ c & \bar{c} & 1 \end{vmatrix} \begin{vmatrix} m & \bar{m} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = \begin{vmatrix} 0 & \bar{0} & 1 \\ b & \bar{b} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \begin{vmatrix} n & \bar{n} & 1 \\ a & \bar{a} & 1 \\ c & \bar{c} & 1 \end{vmatrix}.$$

Extended proposition: For quadrilateral $ABCD$ inscribed in circle O , the tangents through A and C intersect at M , and the tangents through B and D

intersect at N . Prove: $S_{\triangle OAC} S_{\triangle MBD} = S_{\triangle OBD} S_{\triangle NAC}$.

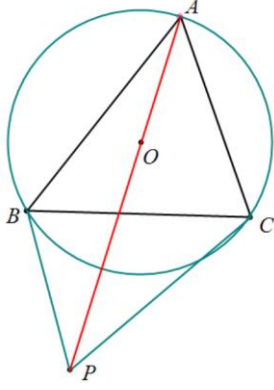


Fig. 40

Example 28 : As shown in Figure 40 , in $\triangle ABC$, O is the circumcenter, $\angle ABP$ and $\angle ABC$ are complementary, $\angle ACP$ and $\angle ACB$ are complementary. Prove that: A , O , and P are collinear.

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, solve the

$$\text{equation } \frac{\frac{P-B}{B-A}}{\frac{B-A}{B-C}} = \frac{\frac{\bar{P}-1/B}{1/B-1/A}}{\frac{1/B-1/A}{1/B-1/C}} \quad \text{and} \quad \frac{\frac{P-C}{C-A}}{\frac{C-A}{C-B}} = \frac{\frac{\bar{P}-1/C}{1/C-1/A}}{\frac{1/C-1/A}{1/C-1/B}} \quad \text{get} \quad P = \frac{A^2 + BC}{B+C},$$

$$\bar{P} = \frac{A^2 + BC}{A^2(B+C)}, \quad \text{so} \quad \frac{P}{\bar{P}} = A^2 = \frac{A}{\frac{1}{A}}.$$

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle,

$$\begin{aligned} & (a-b)^2 c \left(\frac{b-p}{b-a} \frac{b-c}{b-a} - \overline{\left(\frac{b-p}{b-a} \frac{b-c}{b-a} \right)} \right) + (c-a)^2 b \left(\frac{c-p}{c-a} \frac{c-b}{c-a} - \overline{\left(\frac{c-p}{c-a} \frac{c-b}{c-a} \right)} \right) \\ &= (b-c)^2 a \left(\frac{p}{a} - \overline{\left(\frac{p}{a} \right)} \right), \end{aligned}$$

$\triangle ABC$, O is the circumcenter, P is a point inside $\angle BAC$, prove:

$$\left| OC \cdot PB \sin(\angle ABP + \angle ABC) - OB \cdot PC \sin(\angle ACP + \angle ACB) \right| = OP \cdot BC \sin \angle AOP.$$

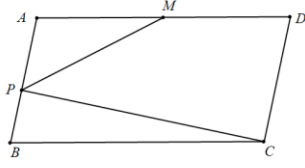


Fig. 28

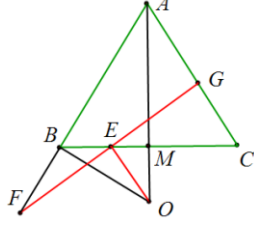
Example 29 : In parallelogram $ABCD$, the midpoint of AD is M , and the projection of C on the side PA is P . Prove that the necessary and sufficient condition for $BC = 2 AB$ is $\angle DAB = 2 \angle APM$.

Proof: Assume $P = 0$,

$$\frac{\frac{a-d}{a-b}}{\left(\frac{p-a}{p-m}\right)^2} - \frac{\frac{a-d}{a-b}}{\left(\frac{p-a}{p-m}\right)^2} = \frac{c}{2a^2(a-b)} \left((b-c)(b-c) - 4(b-a)(b-a) \right),$$

parallelogram $ABCD$, the midpoint of AD is M , and the projection of C on the side PA is P . Prove that: $CP \cdot (DA^2 - 4AB^2) = 4MP^2 \cdot DA \sin(2\angle APM - \angle BAD)$.

Example 30 : In triangle ABC , $AB = AC$. Draw $BO \perp AB$ through B , intersecting the extension of the median AM at point O . Draw a straight line FG through point E on BC , intersecting straight lines AB and AC at points F and G respectively. Prove that $OE \perp FG$ if and only if $FE = EG$. (Question from the 35th IMO exam in 1994)



Let MC be the real axis, MA be the imaginary axis, $b = -c$, $o = \frac{c^2}{a}$,

Solving the equation $\frac{a-f}{a-b} = \overline{\left(\frac{a-f}{a-b}\right)}$, we get $\bar{f} = -\frac{2ac+af-cf}{a+c}$;

Solve the $\frac{a-g}{a-c} = \overline{\left(\frac{a-g}{a-c}\right)}$ equations and $\frac{e-g}{e-f} = \overline{\left(\frac{e-g}{e-f}\right)}$ get

$$g = \frac{2ace - acf - c^2f + aef - cef}{ac - c^2 + ae + ce - 2cf}; \quad \bar{g} = \frac{2ac^2 + acf - c^2f - aef - cef}{ac - c^2 + ae + ce - 2cf}$$

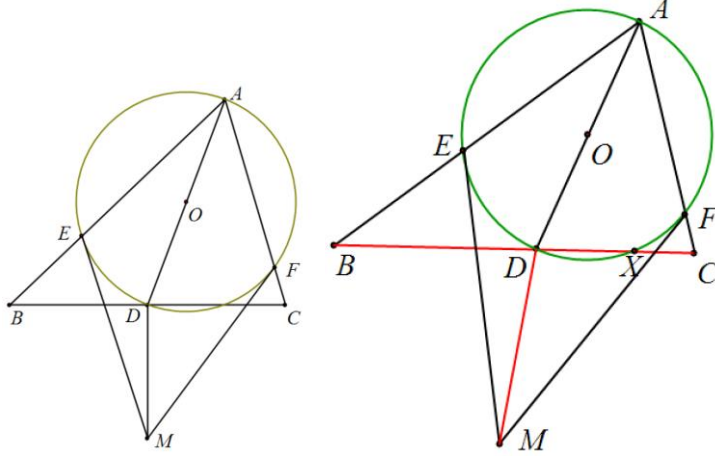
By calculating, $\frac{\frac{e-f}{e-o} + \overline{\left(\frac{e-f}{e-o}\right)}}{(e-f)\overline{(e-f)} - (e-g)\overline{(e-g)}}$ we can get the identity

$$2 \frac{a-f}{a-b} \frac{c-o}{e-b} (c-g)\overline{(c-g)}(e-f)\overline{(e-f)}(e-o)\overline{(e-o)} \left(\frac{e-f}{e-o} + \overline{\left(\frac{e-f}{e-o}\right)} \right) \\ = (c-e)^2 (b-f)\overline{(b-f)} \left((e-f)\overline{(e-f)} - (e-g)\overline{(e-g)} \right), \quad \text{so we can get the}$$

proposition:

triangle ABC , $AB = AC$. Draw $BO \perp AB$ through B , intersecting the extension of the midline AM at point O . Draw a straight line FG through point E on BC , intersecting straight lines AB and AC at points F and G respectively. Prove:

$$4 \frac{AF}{AB} \cdot \frac{BM}{EB} \cdot CG^2 \cdot EF^3 \cdot EO \cos \angle OEF = CE^2 \cdot BF^2 \cdot (EF^2 - EG^2).$$



Example 3 1 : In $\triangle ABC$, circle O passes through three points D , E , and F , $EO \perp EM$, $FO \perp FM$, prove that the necessary and sufficient condition for $BD = DC$ is $DM \perp BC$.

Proof: Let circle O be the unit circle, which intersects BC at points D and X . Let

$$a = -1, \quad d = 1, \quad b = \frac{ae(d+x) - dx(a+e)}{ae - dx}, \quad c = \frac{af(d+x) - dx(a+f)}{af - dx}, \quad m = \frac{2ef}{e+f},$$

$$\frac{m-1}{b-1} + \overline{\left(\frac{m-1}{b-1}\right)} = \frac{(e+x)(-e-f+2ef-2x+ex+fx)}{2(-1+e)(e+f)x},$$

$$b+c-2 = \frac{2x(-e-f+2ef-2x+ex+fx)}{(e+x)(f+x)}.$$

$$\frac{x - (-e)}{x - (-x)} \left(\frac{x - (-e)}{f - (-e)} \left((b-d)\overline{(b-d)} - (c-d)\overline{(c-d)} \right) = (b-c)\overline{(b-c)} \left(\frac{m-d}{b-c} + \overline{\left(\frac{m-d}{b-c} \right)} \right) \right),$$

$\triangle ABC$, circle O passes through three points D , X , E , and F , $EO \perp EM$, FO

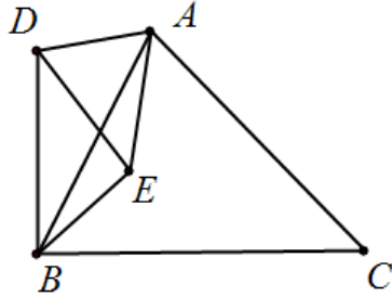
$\perp FM$, prove that: $\sin \angle OEX \frac{\sin \angle OXF}{\sin \angle OEF} (BD^2 - CD^2) = 2BC \cdot MD \cos \angle MDB$.

Note: $\frac{x - (-e)}{x - (-x)} \frac{x - (-e)}{f - (-e)} - X$ and $-E$ were introduced in the code. In order to reduce the

number of new points, they were changed to trigonometric functions.

Example 3 2 : Given $\triangle ABC \sim \triangle ADE$ (similar), $\angle DBC = 90^\circ$. Proof:

$$\frac{BE^2 - DE^2}{AC^2 - BC^2} = \frac{BD^2}{AB^2}.$$



Proof: Assume $a = 0$, from $\frac{a-c}{a-b} = \frac{a-e}{a-d}$, we get $d = be/c$, $\bar{d} = \bar{b}\bar{e}/\bar{c}$,

$$\begin{aligned} & (a-b)(\overline{a-b})((b-e)(\overline{b-e}) - (d-e)(\overline{d-e})) - (d-b)(\overline{d-b})((a-c)(\overline{a-c}) - (b-c)(\overline{b-c})) \\ &= b\bar{b}(b-c)(\overline{b-c})\left(\frac{b-d}{b-c} + \overline{\left(\frac{b-d}{b-c}\right)}\right). \end{aligned}$$

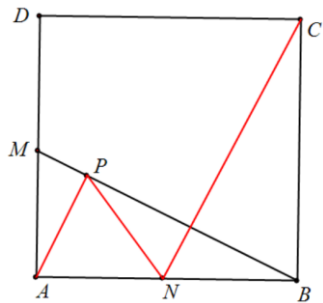
Extended proposition: Given $\triangle ABC \sim \triangle ADE$ (similar in sequence), prove:

$$AB^2(BE^2 - DE^2) - BD^2(AC^2 - BC^2) = 2AB^2 \cdot BC \cdot BD \cos \angle DBC.$$

Example 3 3 : There is a point P in the convex quadrilateral $ABCD$, which satisfies $S_{\triangle ADP} \cdot S_{\triangle BPC} = S_{\triangle ABP} \cdot S_{\triangle DPC}$. Prove that: point P is on the diagonal of the quadrilateral.

Analysis: Its essence is to explore the relationship $[APD][BCP] - [ABP][CDP]$ between $[APC]$ and $[BPD]$. After calculation, we can get $[APD][BCP] - [ABP][CDP] + [APC][BPD] = 0$, so we get an identity that all five points on the plane must satisfy.

Example 34 : Given that M and N are on the sides DA and AB of the square $ABCD$, and $AM = AN$, draw a perpendicular line through A to BM , with P as the foot of the perpendicular. Prove that $\angle APN = \angle BNC$. (2000 I Love Math Junior High School Summer Camp Mathematics Competition)



Proof: Assume $A = (0,0)$, $B = (1,1)$, $C = (1+i,1-i)$, $D = (i,-i)$, $M = (im, -im)$,

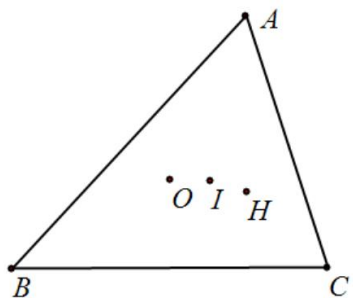
$$N = (n, n) \quad , \quad P = \left(\frac{m}{-i+m}, \frac{m}{i+m} \right) \quad , \quad \text{identity:}$$

$$(p-a)(n-c)\overline{(p-a)(n-c)} \left(\frac{\frac{p-n}{n-c}}{\frac{p-a}{n-b}} - \overline{\left(\frac{\frac{p-n}{n-c}}{\frac{p-a}{n-b}} \right)} \right) (b-mi)(b+mi)$$

$$= -2im(m-n)(b-n)^2.$$

Extended proposition: Given that M and N are on the sides DA and AB of the square $ABCD$ respectively, draw a perpendicular line to BM through A , with the foot of the perpendicular being P . Prove:

$$BM \cdot PN \cdot NC \sin(\angle APN - \angle BNC) = AB \cdot BN \cdot (AN - AM).$$



Example 35 : The circumcenter, incenter, and orthocenter of triangle ABC are points O , I , and H respectively. Prove that the necessary and sufficient condition for $OI \parallel BC$ is $AI \perp IH$.

prove: $O = (0, 0)$, $A = \left(a^2, \frac{1}{a^2}\right)$, $B = \left(b^2, \frac{1}{b^2}\right)$, $C = \left(c^2, \frac{1}{c^2}\right)$,

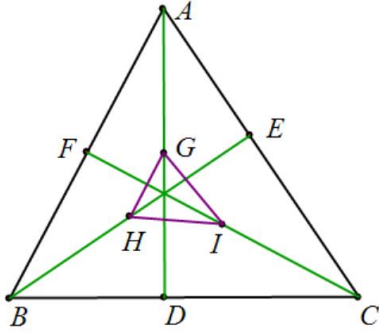
$$H = \left(a^2 + b^2 + c^2, \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right), \quad I = \left(-ab - bc - ca, -\frac{1}{ab} - \frac{1}{bc} - \frac{1}{ca}\right),$$

$$\frac{[OBC] - [IBC]}{(i-h)\overline{(i-h)} \left(\frac{i-a^2}{i-h} + \overline{\left(\frac{i-a^2}{i-h} \right)} \right)} \frac{[BCI]}{[OBC]} = -\frac{b^2 - c^2}{b^2 + c^2}.$$

Extended proposition: The circumcenter, incenter, and orthocenter of acute angle $\triangle ABC$ are points O , I , and H respectively . Prove:

$$2(S_{\triangle IBC} - S_{\triangle OBC})S_{\triangle IBC} = S_{\triangle OBC} IA \cdot IH \cdot \tan A \cos \angle AIH .$$

Example 36 : Proof: The necessary and sufficient condition for the midpoints of the three altitudes of a triangle to be collinear is that the triangle is a right triangle. (2003 54th Romanian Mathematical Olympiad Final)



Let the circumcircle of $\triangle ABC$ be the unit circle, $d = \frac{b+c+a-\frac{bc}{a}}{2}$,

$$e = \frac{c+a+b-\frac{ca}{b}}{2}, \quad f = \frac{a+b+c-\frac{ab}{c}}{2}, \quad g = \frac{a+d}{2}, \quad h = \frac{b+e}{2}, \quad i = \frac{c+f}{2},$$

$$[GHI] = -\frac{(a+b)(b+c)(c+a)(a-b)(b-c)(c-a)}{16a^2b^2c^2},$$

Generating identities $16[GHI][ABC]$

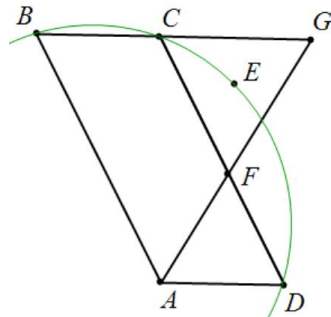
$$= |a-b|^2 \left(\frac{a-c}{a-b} + \overline{\left(\frac{a-c}{a-b} \right)} \right) |b-c|^2 \left(\frac{b-a}{b-c} + \overline{\left(\frac{b-a}{b-c} \right)} \right) |c-a|^2 \left(\frac{c-b}{c-a} + \overline{\left(\frac{b-a}{b-c} \right)} \right),$$

Right now $16[GHI][ABC] = 2AC \cdot AB \cos A \cdot 2BC \cdot BA \cos B \cdot 2CA \cdot CB \cos C$,

Right now $2[GHI] = [ABC] \cos A \cos B \cos C$.

Extended proposition: In $\triangle ABC$, the midpoints of triangles AD , BE and CF are G , H and I respectively. Prove: $2S_{\triangle GHI} = S_{\triangle ABC} \cos A \cos B \cos C$.

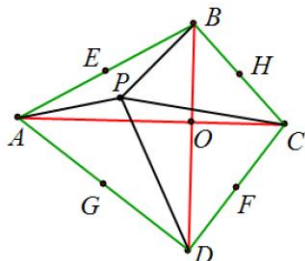
Example 37 : Let $ABCD$ be a parallelogram and $BCED$ be a quadrilateral inscribed in a circle l among the five points A, B, C, D and E . Let l be a straight line passing through point A , intersecting line segment DC at point F (F is an interior point of line segment DC), and intersecting line BC at point G . If $EF = EG = EC$, prove that l is the angle bisector of $\angle DAB$. (2007 IMO test question)



Extended proposition: Let $ABCD$ be a parallelogram l among the five points A, B, C, D , and E . Let l be a straight line passing through point A , intersecting line segment DC at point F (F is an interior point of line segment DC), and intersecting line BC at point G . If $EF = EG = EC$, prove that:

$$8S_{\triangle BFG} \cdot CE \cdot ED \cdot DB \sin(\angle CED + \angle CBD) = CF^2 \cdot (BC + BG)(BG^2 - BA^2).$$

Example 38 : In a convex quadrilateral $ABCD$, the two diagonals AC and BD are perpendicular to each other, the two sides AB and DC are not parallel, point P is the intersection of the perpendicular bisectors of line segments AB and CD , and P is inside the quadrilateral $ABCD$. Proof: The necessary and sufficient condition for $ABCD$ to be an inscribed quadrilateral is that the areas of $\triangle ABP$ and $\triangle CDP$ are equal . (1998 IMO test question)



Proof: Let AC be the real axis and BD be the imaginary axis to construct a coordinate system.

Solving the equation $(p-c)\overline{(p-c)} = (p-d)\overline{(p-d)}$,

$$(p-a)\overline{(p-a)} = \overline{(p-b)(p-b)} \text{ we } p = \frac{a^2c + b^2c - ac^2 + bc^2 - a^2d - b^2d - ad^2 + bd^2}{2(bc - ad)}$$

$$\text{get } \overline{p} = -\frac{a^2c + b^2c - ac^2 - bc^2 + a^2d + b^2d - ad^2 - bd^2}{2(bc - ad)}$$

A, B, C , and D form a circle, that is $OA \cdot OD = OB \cdot OC$, $ac + bd = 0$.

The identity

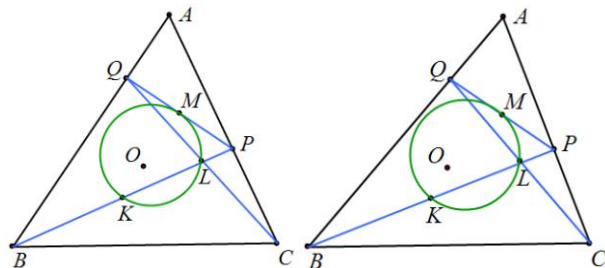
$$\frac{ad - bc}{4} \left(\begin{vmatrix} a & \bar{a} & 1 \\ p & \bar{p} & 1 \\ b & b & 1 \end{vmatrix} - \begin{vmatrix} p & \bar{p} & 1 \\ d & \bar{d} & 1 \\ c & c & 1 \end{vmatrix} \right) = \left(\frac{a+b}{2} - \frac{c+d}{2} \right) \left(\frac{a+d}{2} - \frac{b+c}{2} \right) ac + bd ,$$

Note: Through the identity equation, we can clearly see why it is agreed that " the two pairs of sides AB and DC are not parallel ".

Extended proposition: In a convex quadrilateral $ABCD$, the two diagonals AC and BD are perpendicular to each other, point P is the intersection of the perpendicular bisectors of line segments AB and CD , and P is inside the quadrilateral $ABCD$. Proof:

$$(OB \cdot OC - OA \cdot OD) \cdot (S_{\triangle APB} - S_{\triangle PDC}) = EF \cdot GH \cdot (OA \cdot OC - OB \cdot OD).$$

Example 39 : Let O be the circumcenter of triangle ABC , and let points P and Q be points on sides CA and AB respectively. Let K , L , and M be the midpoints of line segments BP , CQ , and PQ respectively, and let Γ be a circle passing through points K , L , and M . If line PQ is tangent to circle Γ , prove that: $OP = OQ$.
(2009 IMO test question)



Proof: Let the circumcircle of $\triangle ABC$ be the unit circle, $o=0$, $\bar{a}=\frac{1}{a}$, $\bar{b}=\frac{1}{b}$,

$$\bar{c}=\frac{1}{c}, \quad k=\frac{p+b}{2}, \quad m=\frac{p+q}{2}, \quad l=\frac{c+q}{2},$$

$$\text{Solve the equation } \begin{vmatrix} a & \bar{a} & 1 \\ p & \bar{p} & 1 \\ c & \bar{c} & 1 \end{vmatrix} = 0 \text{ and get } \bar{p} = \frac{a+c-p}{ac}; \text{ Solve the equation}$$

$$\begin{vmatrix} a & \bar{a} & 1 \\ q & \bar{q} & 1 \\ b & \bar{b} & 1 \end{vmatrix} = 0 \text{ and get } \bar{q} = \frac{a+b-q}{ab};$$

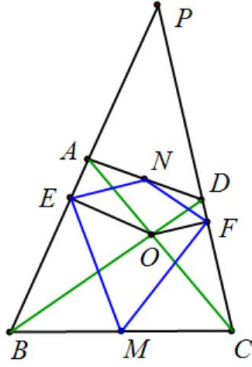
$$8abc(m-l)\overline{(m-l)}(k-m)\overline{(k-m)}\left(\frac{\frac{q-p}{k-m}}{\frac{k-l}{k-m}} - \overline{\left(\frac{\frac{q-p}{k-m}}{\frac{k-l}{k-m}}\right)}\right) = (b-c)(c-p)(b-q)(p\bar{p}-q\bar{q})$$

.

Extended proposition: Let O be the circumcenter of $\triangle ABC$, and let points P and Q be points on sides CA and AB respectively. Let K , L , and M be the midpoints of line segments

$$16OA \cdot MK \cdot KL \cdot LM \cdot PQ \sin(\angle PML - \angle MKL) = BC \cdot BQ \cdot CP \cdot (OP^2 - OQ^2)_{BP, CQ,}$$

and PQ respectively. Prove: .



Example 40 : Given an acute angle $\triangle PBC$, $PB \neq PC$. Let A and D be points on the sides PB and PC respectively, connect AC and BD , and intersect at O . Through point O , make $OE \perp AB$ and $OF \perp CD$, with the feet of perpendiculars being E and F respectively, and the midpoints of line segments BC and AD being M and N respectively.

- (1) If points A , B , C , and D are concentric circles, prove that: $EM \cdot FN = EN \cdot FM$.
- (2) If so $EM \cdot FN = EN \cdot FM$, are there four points A , B , C , and D that are in the same circle? Prove your conclusion. (Questions from the 24th China Mathematical Olympiad in 2009)

Proof: Let O be the origin, OA be the real axis, $m = \frac{b+c}{2}$, $n = \frac{a+d}{2}$, $\bar{d} = \frac{d\bar{b}}{b}$,

and we get

$$e = \frac{a(b-\bar{b})}{2(a-\bar{b})}, \quad \bar{e} = -\frac{a(b-\bar{b})}{2(a-b)}, \quad f = \frac{cd(b-\bar{b})}{2(bc-\bar{b}d)}, \quad \bar{f} = -\frac{cd(b-\bar{b})}{2b(c-d)},$$

$$16(a-b)(\overline{a-b})(c-d)(\overline{c-d})((e-m)(\overline{e-m})(f-n)(\overline{f-n})-(e-n)(\overline{e-n})(f-m)(\overline{f-m}))$$

$$+ \left(\begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \right) \left(\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} - \begin{vmatrix} d & \bar{d} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} \right) (ac - \bar{b}d)^2 = 0.$$

Extended proposition: $AB^2 \cdot CD^2 \cdot (EM^2 \cdot FN^2 - EN^2 \cdot FM^2)$

$$= (S_{\triangle BCD} - S_{\triangle ACD}) \cdot (S_{\triangle ABC} - S_{\triangle DBC}) \cdot (OA \cdot OC - OB \cdot OD)^2.$$

Note: According to the identity equation, when A , B , C , and D are cocircular, it can be deduced that $EM \cdot FN = EN \cdot FM$. Otherwise, it is not necessarily the

case. Since $S_{\triangle BCD} - S_{\triangle ACD} \neq 0$, and $S_{\triangle ABC} - S_{\triangle DBC}$ may be 0, then AD is parallel to

BC .

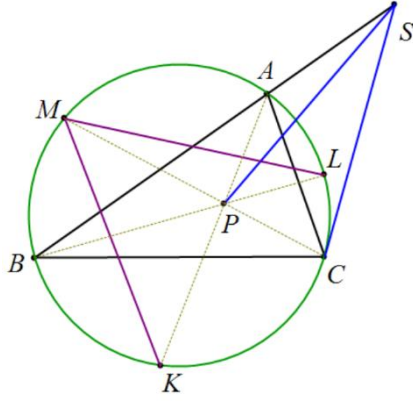


Figure 1

Example 4 1 : As shown in Figure 1, P is a point inside triangle ABC ($AC \neq BC$), AP , BP , CP intersect the circumscribed circle O of triangle ABC at K , L , M respectively, and the tangent line of circle O at point C intersects the straight line AB at S . Given that $SC = SP$, prove that: $MK = ML$. (2010 IMO test question)

Proof: Let O be the unit circle, P be on the real axis, $k = \frac{a-p}{ap-1}$, $l = \frac{b-p}{bp-1}$,

$$m = \frac{c-p}{cp-1}, \quad s = \frac{ab(c+c) - cc(a+b)}{ab-cc},$$

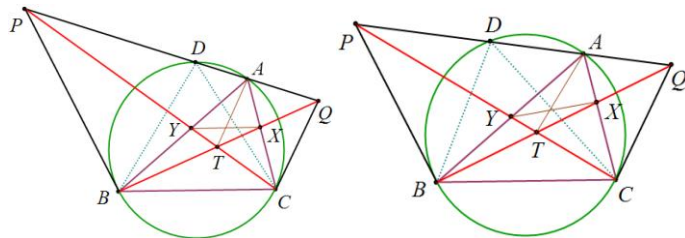
$$\begin{aligned} & [A, B, C](c-p)(\overline{p-m})((m-k)(\overline{m-k}) - (m-l)(\overline{m-l})) \\ &= [MKL]((c-a)(\overline{c-a}) - (c-b)(\overline{c-b}))((s-p)(\overline{s-p}) - (s-c)(\overline{s-c})), \end{aligned}$$

Extended proposition: As shown in Figure 1, P is a point inside $\triangle ABC$ ($AC \neq BC$), AP , BP , CP intersect the circumscribed circle O of $\triangle ABC$ at K , L , M respectively, and the tangent line of circle O at point C intersects the straight line AB at S . Proof:

$$S_{\triangle ABC} \cdot CP \cdot PM \cdot (MK^2 - ML^2) = S_{\triangle MKL} \cdot (CA^2 - CB^2)(SP^2 - SC^2).$$

Note: The identity reminds us to pay attention $CA = CB$ to the situation where S is at infinity.

Example 42 : As shown in the figure, point D is the midpoint of arc BC on the circumscribed circle O of acute angle $\triangle ABC$. Line DA and the tangents of circle O through points B and C intersect at points P and Q respectively. The intersection of BQ and AC is X , the intersection of CP and AB is Y , and the intersection of BQ and CP is T . Prove that AT bisects line segment XY . (2017 National High School Mathematics League Paper B)



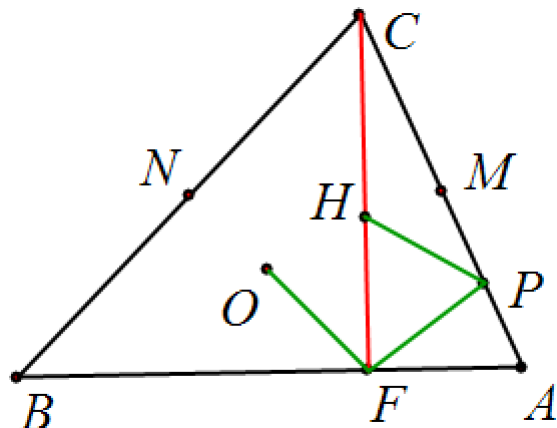
Point D is the midpoint of arc BC on the circumscribed circle O of acute angle $\triangle ABC$, which is equivalent to $DC = DB$; AT bisects segment XY , which is equivalent to $S_{\triangle AXT} = S_{\triangle AYT}$. Below, we discard the condition $DC = DB$, set circle O as the unit circle, and find the complex coordinates of P , Q , X , Y , and T according to the constraints to generate the identity

$$\begin{vmatrix} t & \bar{t} & 1 \\ x & \bar{x} & 1 \\ a & \bar{a} & 1 \end{vmatrix} - \begin{vmatrix} a & \bar{a} & 1 \\ y & \bar{y} & 1 \\ t & \bar{t} & 1 \end{vmatrix} = \begin{vmatrix} x & \bar{x} & 1 \\ y & \bar{y} & 1 \\ t & \bar{t} & 1 \end{vmatrix} \begin{vmatrix} b-0 & \bar{b}-\bar{0} & 1 \\ \frac{b-c}{c-b} & \frac{\bar{b}-\bar{c}}{\bar{c}-\bar{b}} & 1 \\ c-0 & \bar{c}-\bar{0} & 1 \end{vmatrix} \left((d-b)(\overline{d-b}) - (d-c)(\overline{d-c}) \right),$$

Extended proposition: As shown in the figure, A , B , C , and D are four points in a circle. The straight line DA and the tangent line of circle O through points B and C intersect at points P and Q respectively. The intersection point of BQ and AC is X , the intersection point of CP and AB is Y , and the intersection point of BQ and CP is T . Prove:

$$S_{\triangle AXT} - S_{\triangle AYT} = \frac{S_{\triangle XYT}}{BC^2} \frac{S_{\triangle ABC}}{S_{\triangle DBC}} (DC^2 - DB^2).$$

Example 43 : Let $\triangle ABC$ be an acute triangle, and $BC > CA$, O is its circumcenter, H is the orthocenter, and F is the foot of the height CH . Draw a perpendicular line OF through F , intersecting the side CA at P . Prove that: $\angle FHP = \angle BAC$.
(IMO 37 Pre)



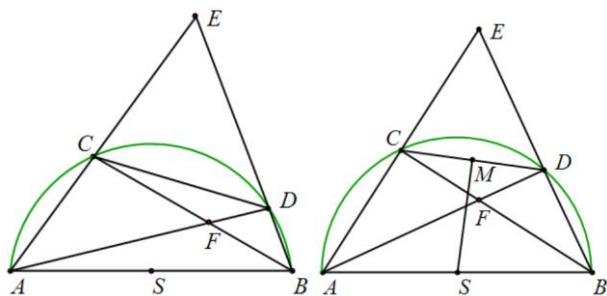
Assume the circumscribed circle of $\triangle ABC$ is the unit circle, $h = a + b + c$,
 $f = \frac{a+b+c-ab/c}{2}$, $\bar{p} = \frac{a+c-p}{ac}$, the identity equation is:

$$4\left(\frac{a+c}{2}\right)\left(\frac{b+c}{2}\right)(f-p)\overline{(f-p)}\left(\frac{f-o}{f-p} + \overline{\left(\frac{f-o}{f-p}\right)}\right)$$

$$= c(a-b)(h-f)\overline{(h-f)}\left(\frac{\frac{h-p}{h-f}}{\frac{a-b}{a-c}} - \overline{\left(\frac{\frac{h-p}{h-f}}{\frac{a-b}{a-c}}\right)}\right),$$

Extended proposition: Suppose $\triangle ABC$ is an acute triangle, and $BC > CA$, O is its circumcenter, H is the orthocenter, and F is the foot of the height CH . F is a point on $4OM \cdot ON \cdot FP \cdot FO \cos \angle PFO = OC \cdot AC \cdot HP \cdot HF \sin(\angle PHF - \angle BAC)$
 AC . Prove: .

Example 44 : There are two points C and D on a semicircle with AB as diameter and S as center, satisfying that point C is on arc AD and $\angle CSD = 90^\circ$, and E and F are the intersection points of lines AC and BD , AD and BC respectively. Prove: $EF = AB$. (2007 Croatian Mathematical Competition)



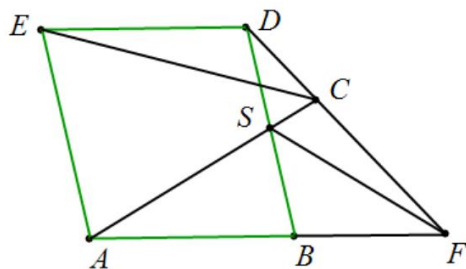
Proof: Assume $S = (0, 0)$, $A = (-1, -1)$, $B = (1, 1)$, $C = \left(c, \frac{1}{c}\right)$, $D = \left(d, \frac{1}{d}\right)$,

$$(e, \bar{e}) = \left(\frac{c-d+2cd}{c+d}, \frac{2-c+d}{c+d} \right), \quad (f, \bar{f}) = \left(\frac{-c+d+2cd}{c+d}, \frac{2+c-d}{c+d} \right),$$

$$(s-d)(\overline{s-d}) \left(\frac{s-c}{s-d} + \overline{\left(\frac{s-c}{s-d} \right)} \right) = \frac{(c+d)^2}{8cd} (4 - (e-f)(\overline{e-f})),$$

Extended proposition: As shown in the figure, there are two points C and D on the semicircle with AB as diameter and S as center, which satisfies point C on arc AD , E and F are the intersection points of lines AC and BD , AD and BC respectively, and M is the midpoint of CD . Proof: $4SB^4 |\cos \angle CSD| = SM^2 \cdot |AB^2 - EF^2|$.

Example 45 : Quadrilateral $ABDE$ is a parallelogram, AC and BD intersect at point S , and rays AB and DC intersect at point F . If points A , B , C , and D are concyclic, prove that $\angle AFS = \angle ECD$. (2010 China-Europe Mathematical Olympiad)



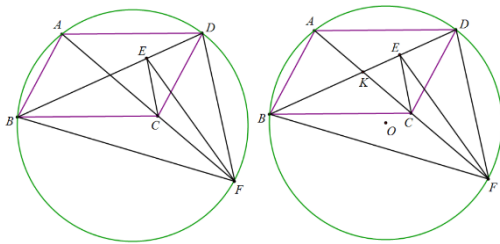
Proof: Assume that $E = A + D - B$, according to the intersection relationship S ,

$$F, (f-s)(c-e) \over (f-s)(c-e) \left(\frac{f-a}{c-e} - \frac{\overline{f-a}}{\overline{c-e}} \right) \\ = - \frac{[ACD]}{[ABC] + [ACD]} (a-b)(c-d) \over (a-b)(c-d) \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \frac{\overline{a-d}}{\overline{a-b}} \frac{\overline{c-b}}{\overline{c-d}} \right),$$

Extended proposition: Convex quadrilateral $ABCD$, quadrilateral $ABDE$ is a parallelogram, AC and BD intersect at point S , and rays AB and DC intersect at point F . Proof: $(S_{\triangle ABC} + S_{\triangle ACD}) \cdot FA \cdot FS \cdot CE \cdot CD |\sin(\angle AFS - \angle ECD)|$

$$= S_{\triangle ACD} \cdot AB \cdot BC \cdot CD \cdot DA |\sin(\angle DAB + \angle DCB)|.$$

Example 46 : In parallelogram $ABCD$, it is known that E is a point on the diagonal BD , and $\angle ECD = \angle ACB$, and the extension of AC intersects the circumscribed circle of ABD at point F . Prove: $\angle DFE = \angle AFB$. (2014 National Junior High School League Question)



$$\frac{e}{f} = \frac{b+d-e}{bd}, \quad f = \frac{bd(2a-b-d)}{ab+ad-2bd},$$

$$(f-d)(f-b) \overline{\left(\frac{f-e}{f-d} - \frac{f-d}{f-b} \right)} \left(\frac{f-e}{f-b} - \frac{f-d}{f-a} \right),$$

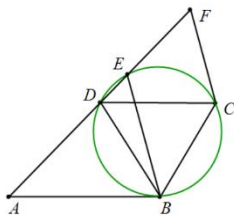
$$= \frac{a^3 b^2 d^2}{(a-b)^3 (a-d)^3 (b+d)} [ABF][BCF] (c-b)(c-d) \overline{(c-b)(c-d)} \left(\frac{c-e}{c-b} - \frac{c-d}{c-a} \right) \left(\frac{c-e}{c-b} - \frac{c-d}{c-a} \right).$$

Extended proposition: In parallelogram $ABCD$, it is known that E is a point on the diagonal BD , the extension of AC intersects the circumscribed circle of ABD at point F . R is the radius of the circumscribed circle, and K is the intersection point of the parallelogram diagonals. Prove:

$$AB^2 \cdot AD^2 \cdot OK \cdot FD \cdot FB \cdot FE \cdot FA \sin(\angle EFD - \angle BFA)$$

$$= 8R^3 \cdot AB \cdot FB \cdot CF \cdot CE \cdot CA \sin(\angle ECD - \angle BCA).$$

Example 47 : Let quadrilateral $ABCD$ be a trapezoid, $AB \parallel CD$, the circumscribed circle of triangle BCD intersects the straight line AD at points D and E (D is between points A and E), and a line parallel to BE through point C intersects AD at point F . Proof: $BC^2 = AD \cdot EF$. (2009–2010 Hungarian Mathematical Olympiad)



Proof: Let the circumscribed circle of $\triangle BCD$ be the unit circle,

$$A = \left(a, \bar{a}\right), \quad B = \left(b, \frac{1}{b}\right), \quad C = \left(c, \frac{1}{c}\right), \quad D = \left(d, \frac{1}{d}\right), \quad E = \left(e, \frac{1}{e}\right), \quad F = \left(f, \bar{f}\right),$$

$$-bcde \left((b-c) \overline{(b-c)} - (a-d) \overline{(e-f)} \right) = -abce + b^2de - bcde + c^2de + abcf - bcdf, \quad ,$$

$$\frac{bcde}{b-e} ([F, E, B] - [C, E, B]) = bcd - c^2d + bce - bde - bcf + cdf,$$

$$\frac{bcde}{c-d} ([D, A, B] - [C, A, B]) = abc - bcd - abe + b^2e - bce + cde,$$

$$A, D, \text{ and } E \text{ are collinear, } \bar{a} = \frac{-a+d+e}{de}; \quad F, D, \text{ and } E \text{ are collinear,}$$

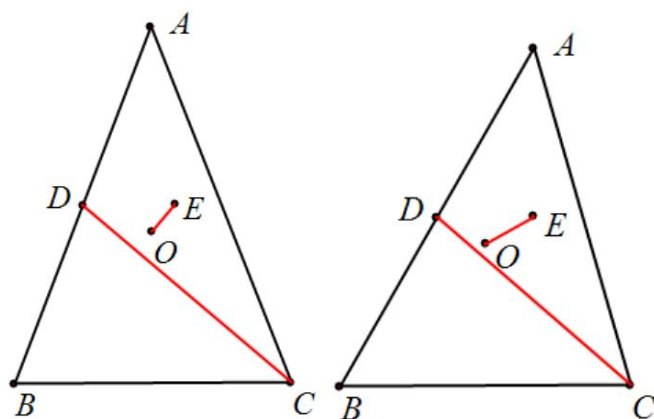
$$\bar{f} = \frac{d+e-f}{de}.$$

$$(b-d) \left((b-c) \overline{(b-c)} - (a-d) \overline{(e-f)} \right) - b \frac{a-d}{b-e} ([F, E, B] - [C, E, B])$$

$$+ d \frac{b-c}{c-d} ([D, A, B] - [C, A, B]) = 0.$$

Extended proposition: In the inscribed quadrilateral $DBCE$, point A is on the extended line of line segment ED , and point F is on the extended line of line segment DE . Proof:

$$\frac{\sin \angle BCD}{2} (BC^2 - AD \cdot EF) = \frac{AD}{BD} (S_{\triangle CEB} - S_{\triangle FEB}) + \frac{BC}{CD} (S_{\triangle CAB} - S_{\triangle DAB}).$$



Example 48 : As shown in Figure 6 , let O be the circumcenter of triangle ABC , D be the midpoint of AB , E be the centroid of triangle ACD , and $AB = AC$. Prove that: $OE \perp CD$. (1983 British Olympic Test Question)

Proof: Let the circumcircle of $\triangle ABC$ be the unit circle, $o = 0$, $d = \frac{a+b}{2}$,

$$e = \frac{a+d+c}{3}, \quad 3ee^{-\left(\frac{c-d}{o-e} + \overline{\left(\frac{c-d}{o-e}\right)}\right)} = (a-c)\overline{(a-c)} - (a-b)\overline{(a-b)},$$

Extended proposition: $6OE \cdot CD \cos \alpha = AC^2 - AB^2$, is α the angle formed by the straight line OE turning to CD .

Example 1 : As shown in Figure 6 , let O be the circumcenter of triangle ABC , D be the midpoint of AB , and E be the centroid of triangle ACD . Prove that: $6OE \cdot CD |\cos \alpha| = |AC^2 - AB^2|$, α is the angle between lines OE and CD .

Example 1 : As shown in Figure 6, let O be the circumcenter of triangle ABC , D be the midpoint of AB , E be the centroid of triangle ACD , and $AB = AC$. Prove that: $OE \perp CD$. (1983 British Olympic Test Question)

prove

:

$$6\left(O - \frac{A+C + \frac{A+B}{2}}{3}\right)\left(C - \frac{A+B}{2}\right) + 3\left(O - \frac{B+C}{2}\right)(B-C) + 3\left(O - \frac{C+A}{2}\right)(A-C) + [(A-B)^2 - (A-C)^2] = 0.$$

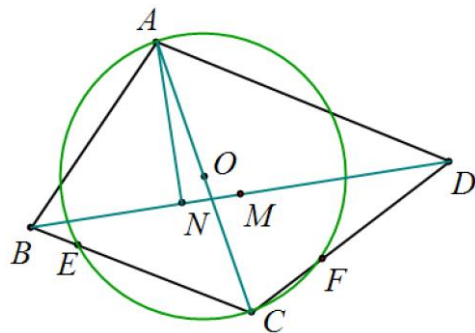
When $\left(O - \frac{B+C}{2}\right)(B-C) = \left(O - \frac{C+A}{2}\right)(A-C) = 0$, that is

$$6OE \cdot CD |\cos \alpha| = |AC^2 - AB^2|, \quad \alpha \text{ is the angle between straight lines } OE \text{ and } CD.$$

From the identity we can get a new proposition: Let O be the circumcenter of $\triangle ABC$, D be the midpoint of AB , E be the centroid of $\triangle ACD$, and $OE \perp CD$. Prove: $AB = AC$.

From the perspective of drawing, it $AB = AC \Rightarrow OE \perp CD$ is natural to draw a triangle ABC without knowing $AB = AC$. From the perspective of problem solving, we are more accustomed to using $AB = AC$, and $OE \perp CD$ isosceles triangles have many well-known properties. However $OE \perp CD$, it is difficult to use it. This reminds people of the "Steiner–Lemmers Theorem". "The two interior bisectors

of an isosceles triangle are equal " is easy to prove, but on the contrary , "a triangle with two equal interior bisectors is an isosceles triangle" is more difficult. Therefore , in traditional geometric research, similar $AB=AC \Leftarrow OE \perp CD$ properties are rare. Using the identity method, it is easy to explore such properties.



Example 49 : As shown in the figure, in quadrilateral $ABCD$, $\angle ABC = \angle ADC < 90^\circ$, the other intersection points of circle O with AC as diameter and sides BC and CD are E and F respectively. M is the midpoint of BD , and $AN \perp BD$ at point N . Proof: M, N, E , and F are in the same circle. (2020 China Southeast Mathematical Olympiad)

Proof: Let the circle be the unit circle, $c = -a$, $m = \frac{b+d}{2}$, $\bar{b} = \frac{-b+c+e}{ce}$,

$$\bar{d} = \frac{c-d+f}{cf},$$

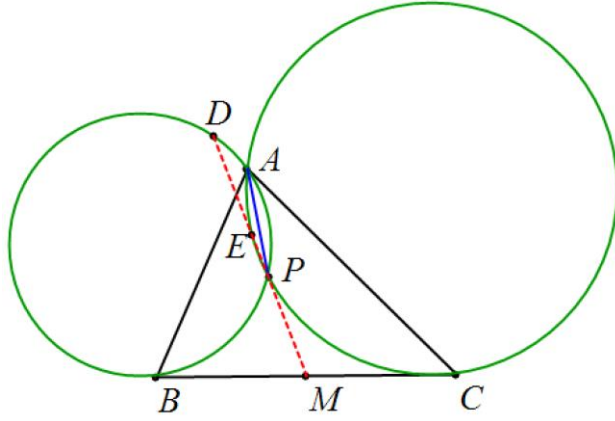
$$8(c-b)(c-d)(m-e)\overline{(m-e)}(n-f)\overline{(n-f)} \left(\frac{\frac{m-f}{n-f}}{n-e} - \overline{\left(\frac{\frac{m-f}{n-f}}{n-e} \right)} \right)$$

$$= (0-a)(e-f) \begin{vmatrix} m & \bar{m} & 1 \\ n & \bar{n} & 1 \\ c & \bar{c} & 1 \end{vmatrix} (b-c)\overline{(b-c)}(d-c)\overline{(d-c)} \left(\frac{\frac{b-a}{d-c}}{d-a} - \overline{\left(\frac{\frac{b-a}{d-c}}{d-a} \right)} \right),$$

Extended proposition: As shown in the figure, in quadrilateral $ABCD$, the other intersection points of circle O with AC as diameter and sides BC and CD are E and F respectively. M is the midpoint of BD , and $AN \perp BD$ at point N . Proof:

$$2OA \cdot MF \cdot ME \cdot NF \cdot NE \sin(\angle FME - \angle FNE)$$

$$= S_{\triangle MNC} \cdot EF \cdot AB \cdot DA \sin(\angle ABC - \angle ADC).$$



Example 50 : As shown in the figure, in the acute angle $\triangle ABC$, M is the midpoint of the side BC . Point P is inside $\triangle ABC$, so that AP bisects $\angle BAC$, and the straight line MP intersects the circumscribed circles of ABP and ACP at two points D and E respectively that are different from point P . Proof: If $DE = MP$, then $BC = 2 BP$. (2019 National High School Mathematics Joint Competition Paper A)

Proof: Take A as the origin and AP as the real axis, $m = \frac{b+c}{2}$, $\bar{c} = \frac{bc}{b}$, solve

$$\text{the equation } \frac{m-e}{m-p} = \overline{\left(\frac{m-e}{m-p}\right)}, \quad \frac{\frac{a-c}{a-p}}{\frac{e-c}{e-p}} = \overline{\left(\frac{\frac{a-c}{a-p}}{\frac{e-c}{e-p}}\right)}, \quad \text{and get } e = \frac{(\bar{b}-c)(\bar{b}p-bc)}{\bar{b}^2+bc-2\bar{b}p},$$

$$\bar{e} = -\frac{b(\bar{b}-c)(c-p)}{\bar{b}(b+c-2p)}; \quad \text{similarly } d = \frac{b(\bar{b}-c)(\bar{b}-p)}{\bar{b}^2+bc-2\bar{b}p}, \quad \bar{d} = \frac{(\bar{b}-c)(b-p)}{b+c-2p};$$

$$\text{set} \quad \quad \quad \text{up} \quad \quad \quad p=1 \quad \quad \quad ,$$

$$(b-c)\overline{(b-c)} - 4(b-p)\overline{(b-p)} = -\frac{4\bar{b} - 4b\bar{b} - 4\bar{b}^2 + 3b\bar{b}^2 + b^2c + \bar{b}^2c - bc^2}{\bar{b}} \quad ,$$

$$(e-d) - (m-p) = -\frac{4\bar{b} - 4b\bar{b} - 4\bar{b}^2 + 3b\bar{b}^2 + b^2c + \bar{b}^2c - bc^2}{2(\bar{b}^2 + bc - 2\bar{b})} \quad ,$$

If it were the traditional complex number solution, it would end here. Not only has the original proposition been proved, but also the inverse proposition.

But from the perspective of identity, we can go one step further:

$$(b-c)\overline{(b-c)} - 4(b-p)\overline{(b-p)} = \frac{2(\bar{b}^2 + bc - 2\bar{b})}{\bar{b}}((e-d) - (m-p)), \quad \text{which can be}$$

transformed into

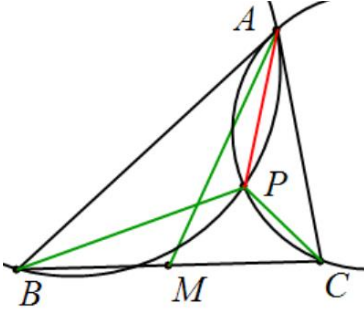
$$(b-c)\overline{(b-c)}-4(b-p)\overline{(b-p)}=4\left(\frac{b+c}{2}-p\right)\overline{\left(\frac{b+c}{2}-p\right)}((e-d)-(m-p)) , \quad \text{its geometric}$$

meaning is:

Extended proposition: As shown in the figure, in the acute angle $\triangle ABC$, M is the midpoint of the BC side. Point P is inside $\triangle ABC$, so that AP bisects $\angle BAC$, and the straight line MP intersects the circumscribed circles of ABP and ACP at two points D and E different from point P respectively . Proof:

$$BC^2 - 4BP^2 = 4MP \cdot (ED - MP) .$$

Example 5 1 : In triangle ABC , AM is the median. A circle is drawn through two points A and B and is tangent to AC . A circle is drawn through two points A and C and is tangent to AB . The two circles intersect at P . Proof: $\angle CAP = \angle BAM$.

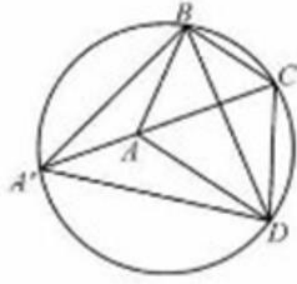


$$\begin{aligned} \text{Identity: } & 2(a-p)\overline{(a-p)}\left(a-\frac{b+c}{2}\right)\overline{\left(a-\frac{b+c}{2}\right)}\left(\frac{\frac{a-c}{a-p}}{a-\frac{b+c}{2}}-\frac{\overline{\left(\frac{a-c}{a-p}\right)}}{\overline{a-\frac{b+c}{2}}}\right) \\ & = (a-p)\overline{(a-p)}(c-a)\overline{(c-a)}\left(\frac{\frac{c-p}{a-p}}{a-b}-\frac{\overline{\left(\frac{c-p}{a-p}\right)}}{\overline{a-b}}\right) \\ & \quad + (a-p)\overline{(a-p)}(b-a)\overline{(b-a)}\left(\frac{\frac{a-c}{b-a}}{b-p}-\frac{\overline{\left(\frac{a-c}{b-a}\right)}}{\overline{b-p}}\right) . \end{aligned}$$

Extended

proposition:

$$2AM \sin(\angle PAC - \angle BAM) = CP \sin(\angle ACP - \angle BAP) + BP \sin(\angle PAC - \angle PBA) .$$



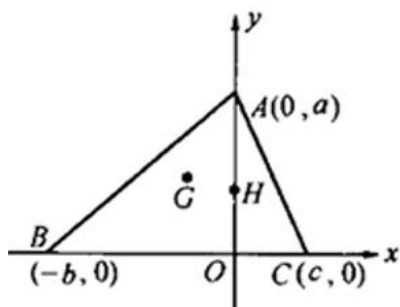
Example 52 : In convex quadrilateral $ABCD$, we have $AB \cdot CD = AD \cdot BC$. Proof:
 $\angle BAC + \angle CBD + \angle DCA + \angle ADB = 180^\circ$. (38th Russian Mathematical Olympiad,
 2012)

Identity

$$\begin{aligned} & \left((a-c)\overline{(a-c)}(b-d)\overline{(b-d)} \right)^2 \left(\frac{a-b}{a-c} \frac{b-c}{b-d} \frac{c-d}{c-a} \frac{d-a}{d-b} - \overline{\left(\frac{a-b}{a-c} \frac{b-c}{b-d} \frac{c-d}{c-a} \frac{d-a}{d-b} \right)} \right) \\ &= \left(|a-b|^2 |c-d|^2 - |a-d|^2 |b-c|^2 \right) |a-b|^2 |c-d|^2 \left(\frac{a-d}{a-b} \frac{c-b}{c-d} - \overline{\left(\frac{a-d}{a-b} \frac{c-b}{c-d} \right)} \right). \end{aligned}$$

Extended proposition: $AC^2 \cdot BD^2 \sin(\angle CAB + \angle DBC + \angle ACD + \angle BDA)$

$$= (DA^2 \cdot BC^2 - AB^2 \cdot CD^2) \sin(\angle BAD + \angle DCB)$$



Example 53 : As shown in $AB \neq AC, H, G$ Figure 54, the acute angles $\triangle ABC$ are

the orthocenter and centroid of the triangle. Given $\frac{1}{S_{\triangle HAB}} + \frac{1}{S_{\triangle HAC}} = \frac{2}{S_{\triangle HBC}}$.

Prove: $\angle AGH = 90^\circ$. (Training question for the Chinese National Training

Team in 2003)

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, then

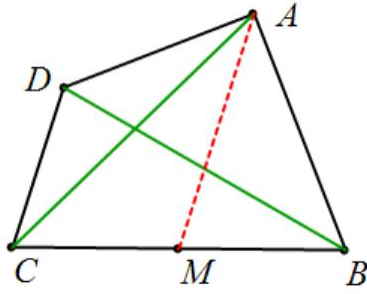
$$h = a + b + c, \quad g = \frac{a + b + c}{3}, \quad g = \frac{a + b + c}{3}$$

$$\frac{1}{\begin{vmatrix} h & \bar{h} & 1 \\ a & \bar{a} & 1 \\ b & \bar{b} & 1 \end{vmatrix}} + \frac{1}{\begin{vmatrix} h & \bar{h} & 1 \\ c & \bar{c} & 1 \\ a & \bar{a} & 1 \end{vmatrix}} - \frac{2}{\begin{vmatrix} h & \bar{h} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}}$$

$$= \frac{9a^2b^2c^2}{(a-b)(a+b)(a-c)(a+c)(b-c)(b+c)} (g-h)(\overline{g-h}) \left(\frac{g-a}{g-h} + \overline{\left(\frac{g-a}{g-h} \right)} \right),$$

Extended proposition:

$$72AB \cdot BC \cdot CA \cdot HA \cdot HB \cdot HC \left(\frac{1}{S_{\triangle HAB}} + \frac{1}{S_{\triangle HCA}} - \frac{2}{S_{\triangle HBC}} \right) = R^2 \cdot GA \cdot GH \cos \angle HGA.$$

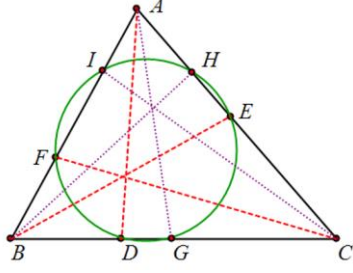


Example 54 : In convex quadrilateral $ABCD$, $\angle DAB = 90^\circ$, is M the midpoint of BC , $\angle ADC = \angle BAM$ side BC ,, prove: $\angle ADB = \angle CAM$. (42nd Russian Mathematical Olympiad 2016)

Proof: Let $a = 0$, $\bar{b} = -b$, $\bar{d} = d$,

$$\begin{aligned} & (d-b)(\overline{d-b}) \left(a - \frac{b+c}{2} \right) \overline{\left(a - \frac{b+c}{2} \right)} \left(\frac{\frac{d-a}{d-b}}{a - \frac{b+c}{2}} - \frac{\overline{\frac{d-a}{d-b}}}{\overline{a - \frac{b+c}{2}}} \right) \\ &= (d-c)(\overline{d-c})(a-b)\overline{(a-b)} \left(\frac{\frac{d-a}{d-c}}{a-b} - \frac{\overline{\frac{d-a}{d-c}}}{\overline{a-b}} \right), \end{aligned}$$

Extended proposition: In convex quadrilateral $ABCD$, $\angle DAB = 90^\circ$, M is the midpoint of side BC , , proof:
 $DB \cdot AC \sin(\angle ADB - \angle MAC) = DC \cdot AB \sin(\angle ADC - \angle BAM)$.



Example 55 : As shown in the figure, the three sides of $\triangle ABC$ intersect the circle at six points: $D, E, F, G, H,$ and I . It is known that $AD, BE,$ and CF intersect at one point. Prove that $AG, BH,$ and CI intersect at one point.

Proof: Let the circle be the unit circle, then $a = \frac{fi(h+e)-he(f+i)}{fi-he},$

$\bar{a} = \frac{e+h-f-i}{eh-fi},$ and similarly, , we can find that, $\{b, \bar{b}\}$ and $\{c, \bar{c}\}$ we have the

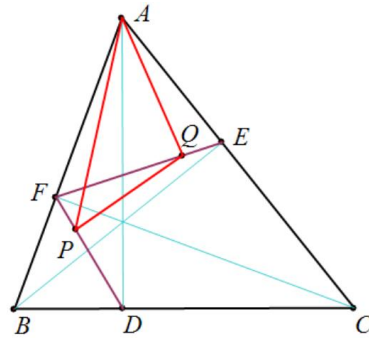
identity

$$[GHI]([AFC][DBE]-[DFC][ABE])=[DEF]([GIC][ABH]-[AIC][GBH]),$$

Extended proposition: As shown in the figure, the three sides of $\triangle ABC$ intersect the circle at six points: $D, E, F, G, H,$ and I . Prove:

$$S_{\triangle GHI} (S_{\triangle AFC} S_{\triangle DBE} - S_{\triangle DFC} S_{\triangle ABE}) = S_{\triangle DEF} (S_{\triangle GIC} S_{\triangle ABH} - S_{\triangle AIC} S_{\triangle GBH}).$$

Example 56 : Let AD , BE , and CF be the three altitudes of acute angle $\triangle ABC$, and P and Q be on line segments DE and EF respectively. Prove that: The necessary and sufficient condition for $\angle PAQ = \angle DAC$ is that AP bisects $\angle QPF$. (Extension



of the 2006 German National Team Selection Test)

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, $d = \frac{a+b+c-bc/a}{2}$, $e = \frac{a+b+c-ac/b}{2}$, $f = \frac{a+b+c-ba/c}{2}$, and solve the equation $\frac{d-p}{d-f} = \overline{\left(\frac{d-p}{d-f}\right)}$ to get $\bar{p}$; Solve the equation $\frac{f-q}{e-f} = \overline{\left(\frac{f-q}{e-f}\right)}$ to get $\bar{q}$.

$$\begin{aligned} & (a-b)^4 (a-c)^2 (p-a)^2 \overline{(p-a)}^2 \left(\frac{\frac{p-f}{p-a}}{p-q} - \overline{\left(\frac{\frac{p-f}{p-a}}{p-q} \right)} \right) \\ & = 2a^3 b^2 c [PAB]^2 (a-d)(a-q) \overline{(a-d)(a-q)} \left(\frac{\frac{a-c}{a-q}}{a-p} - \overline{\left(\frac{\frac{a-c}{a-q}}{a-p} \right)} \right), \end{aligned}$$

Extended proposition: Let AD , BE , and CF be the three altitudes of acute angle $\triangle ABC$, and let P and Q be on line segments DE and EF respectively. Prove:

$$AB^4 \cdot AC \cdot PF \cdot PA \cdot PQ \sin(\angle FPA - \angle APQ) = 32 S_{\triangle PAB}^2 \cdot R^2 \cdot AD \cdot AQ \sin(\angle CAD - \angle QAP)$$

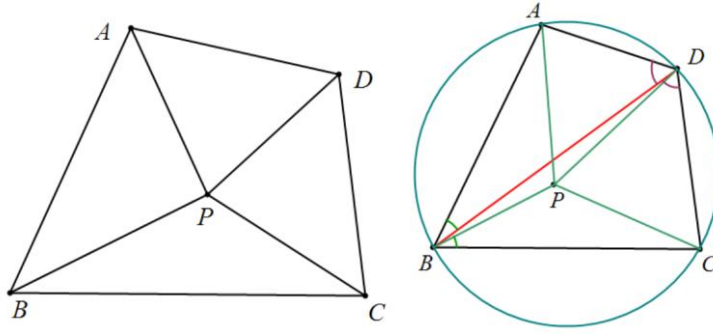


Fig. 30

Example 57 : In convex quadrilateral $ABCD$, diagonal BD is neither the bisector of $\angle ABC$ nor the bisector of $\angle CDA$. Point P in quadrilateral $ABCD$ satisfies $\angle PBC = \angle DBA$, $\angle PDC = \angle BDA$. Prove that the necessary and sufficient condition for points A, B, C , and D to be concentric is $PA = PC$. (Question from the 45th IMO exam in 2004)

Proof: Because $(p-a)(\overline{p-a}) - (p-c)(\overline{p-c}) = \frac{(a-c)(p-ac\overline{p})}{ac} = 0$, so

$$PA = PC \Leftrightarrow p - ac\overline{p} = 0.$$

First prove that points A, B, C , and D are concentric circles $\Rightarrow PA = PC$.

$$\text{From } \frac{b-p}{b-a} = \frac{1/b-1/c}{1/b-1/a}, \quad \frac{d-p}{d-b} = \frac{1/d-1/c}{1/d-1/b}, \quad \text{we can get } p = \frac{ac+bd}{b+d},$$

$$\overline{p} = \frac{ac+bd}{ac(b+d)},$$

Obviously satisfied $p - ac\overline{p} = 0$.

Prove that points A, B, C , and D are concentric circles $\Leftarrow PA = PC$. Assume that the circumscribed circle of $\triangle ABC$ is the unit circle. We only need to prove $\overline{d} = \frac{1}{d}$.

$$\text{Solve the system of equations } p - ac\overline{p} = 0, \quad \frac{b-p}{b-a} = \frac{1/b-1/c}{1/b-1/a}, \quad \frac{d-p}{d-b} = \frac{\overline{d}-1/c}{\overline{d}-1/b},$$

and we get $p = \frac{ac+bd}{b+d}$, $\overline{p} = \frac{ac+bd}{ac(b+d)}$, $\overline{d} = \frac{1}{d}$ (the other two solutions are

$p = a$ or $p = c$, which do not match the question, so they are discarded), so

$PA=PC \Rightarrow \bar{d} = \frac{1}{d}$, that is, the four points A , B , C , and D are concentric circles.

Proof: $A=(1,1)$, $B=(0,0)$, $C=(c,\bar{c})$, $D=(d,\bar{d})$, Solve the equation

$$\frac{\frac{b-p}{b-a}}{\frac{b-d}{b-a}} = \frac{\overline{\left(\frac{b-p}{b-a}\right)}}{\overline{\left(\frac{b-d}{b-a}\right)}}, \quad \frac{\frac{d-c}{d-b}}{\frac{d-a}{d-a}} = \frac{\overline{\left(\frac{d-c}{d-b}\right)}}{\overline{\left(\frac{d-p}{d-b}\right)}}, \text{ and get}$$

$$P = \left(\frac{c \left(-\bar{c}d^2 + d^2\bar{d} + \bar{c}d^2\bar{d} + c\bar{d}^2 - d\bar{d}^2 - cd\bar{d}^2 \right)}{d \left(-\bar{c}d - c\bar{c}d + \bar{c}d^2 + c\bar{d} + c\bar{c}\bar{d} - c\bar{d}^2 \right)}, \frac{\bar{c} \left(\bar{c}d^2 - d^2\bar{d} - \bar{c}d^2\bar{d} - c\bar{d}^2 + d\bar{d}^2 + cd\bar{d}^2 \right)}{\bar{d} \left(\bar{c}d + c\bar{c}d - \bar{c}d^2 - c\bar{d} - c\bar{c}\bar{d} + c\bar{d}^2 \right)} \right),$$

$$\begin{aligned} \text{Identity:} \quad & [ABD] \left(|p-a|^2 - |p-c|^2 \right) |c-d|^2 |b-d|^4 \left(\frac{\frac{b-a}{b-d}}{\frac{b-c}{b-d}} - \frac{\overline{\left(\frac{b-a}{b-d}\right)}}{\overline{\left(\frac{b-d}{b-c}\right)}} \right) \\ & = [CDP][ABCD] \left| (b-c)\overline{(b-d)} - (b-d)\overline{(b-a)} \right|^2, \end{aligned}$$

Extended proposition: There is a point P in the convex quadrilateral $ABCD$, satisfying $\angle PBC = \angle DBA$, $\angle PDC = \angle BDA$. Proof :

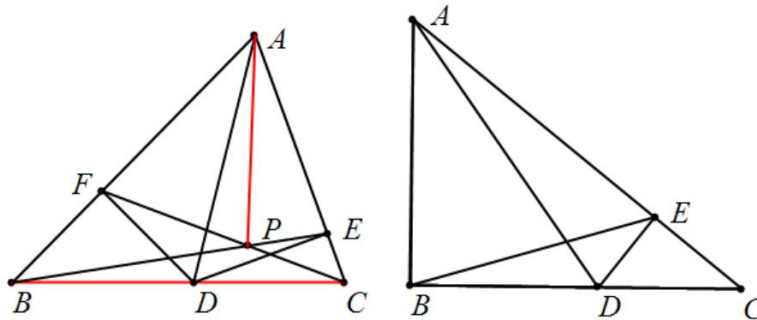
$$\begin{aligned} & S_{ABD} \cdot (PA^2 - PC^2) \cdot CD \cdot \sin(\angle ABD - \angle DBC) \\ & = S_{CDP} \cdot (BA^2 + BC^2 - 2BA \cdot BC \cos \angle DBP) \cdot AD \cdot \sin(\angle BAD + \angle BCD). \end{aligned}$$

Through the identity, we can clearly understand the intention of the question conditions.

$$\angle ABD - \angle DBC = BA^2 + BC^2 - 2BA \cdot BC \cos \angle DBP = 0, \quad \text{can it be realized}$$

$$PA^2 - PC^2 = 0 \Leftrightarrow \sin(\angle BAD + \angle BCD) = 0.$$

Example 58 : Let D be a point on the side BC of a non-right triangle ABC . Draw perpendicular lines to AC and AB from point D , with their feet at E and F respectively. Lines BF and CE intersect at point P . Prove that $AP \perp BC$ if and only if AD bisects $\angle BAC$. (2006 Romanian national team selection test questions)



Proof: Let $a = 0$, $d = 1$, $\bar{c} = \frac{b - \bar{b} - c + \bar{b}c}{b - 1}$, find the coordinates of E , F , P ,

and the identity $[BDP] |a - d|^4 \left(\frac{a - c}{a - d} - \left(\frac{a - c}{a - d} \right) \right)$

$$= \frac{(c - 1)}{2(b - c)} [BDA]^2 |b - c|^2 \left(\frac{p - a}{b - c} + \left(\frac{p - a}{b - c} \right) \right),$$

Extended proposition: In $\triangle ABC$, D is a point on side BC . Draw perpendicular lines to AC and AB from point D , with their feet at E and F respectively. Straight lines BF and CE intersect at point P . Prove

$$BP \sin \angle PBC \sin (\angle CAD - \angle DAB) = PA \sin \angle CAD \sin \angle BAD \cos \alpha.$$

Note: If $\angle B = 90^\circ$ (Figure 2), B , F , and P are combined into one point $[BDP] = 0$. Therefore, the original question emphasizes that $\triangle ABC$ is not a right triangle.

Example 59 : As shown in Figure 1 , draw squares $ABDE$ and $CAFG$ on the two right-angled sides AB and AC of right-angle triangle ABC . BG intersects AC at H , and CD intersects AB at I . Prove that $AH = AI$.

Proof: From the similarity of triangles, we can get $\frac{AH}{FG} = \frac{BA}{BF}$, $\frac{AI}{ED} = \frac{CA}{CE}$ so $AH = AI$.

Another proof: To prove $AH = AI$, we only need to prove $S_{\triangle HBF} = S_{\triangle IEC}$ (because the two triangles have the same base); and $S_{\triangle HBF} = S_{\triangle GBA} = S_{\triangle CBA}$, $S_{\triangle IEC} = S_{\triangle DAC} = S_{\triangle BAC}$, so $S_{\triangle HBF} = S_{\triangle IEC}$, $AH = AI$.

This proof may seem more complicated, but it requires less knowledge and is acceptable to elementary school students.

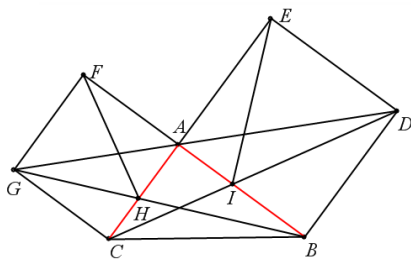


Figure 1

Proof: Let $a=0$, $b=1$, $e=i$, $d=b+e$, $f=-ci$, $g=c+f$, solution is

$$h = \frac{c(1+i)(c-i\bar{c})}{ic-i\bar{c}+2c\bar{c}}, \quad i = \frac{(1-i)(c-i\bar{c})}{-2i+c-\bar{c}}, \quad (ii-h\bar{h})[BFG]^2[CED]^2$$

$$= \left((a-c)(\overline{a-c}) \left(\frac{a-b}{a-c} + \overline{\left(\frac{a-b}{a-c} \right)} \right) \right)^2 (b^2 - c\bar{c})[ABG]^2,$$

Extended proposition:

Figure 1 , draw squares $ABDE$ and $CAFG$ outward from the two sides AB and AC of triangle ABC . BG intersects AC at H , and CD intersects AB at I . Prove that:

$$4(AH^2 - AI^2)S_{\triangle BFG}^2 S_{\triangle CDE}^2 = (AB \cdot AC \cos A)^2 (AB^2 - AC^2) S_{\triangle ABG}^2.$$

Another proof: $S_{\triangle ADC} = S_{\triangle ADI} + S_{\triangle AIC}$, yes

$$\frac{1}{2} AD \cdot AC \sin(45^\circ + A) = \frac{1}{2} AD \cdot AI \sin 45^\circ + \frac{1}{2} AI \cdot AC \sin A,$$

$$AI = \frac{\sqrt{2} AB \cdot AC \cdot \sin(45^\circ + A)}{AB + AC \cdot \sin A}, \quad AH = \frac{\sqrt{2} AB \cdot AC \sin(45^\circ + A)}{AC + AB \sin A}$$

$AP = AQ$, which is equivalent to $AB + AC \cdot \sin \alpha = AC + AB \cdot \sin \alpha$, that is

$$(AB - AC)(1 - \sin A) = 0, \text{ so } AB = AC \text{ or } \angle BAC = 90^\circ.$$

Multiple evidence and re-exploration of two vertical issues

Example 60 : In $\angle ACB = 90^\circ \triangle ABC$, its incircle O is tangent to the three sides at D , E , and F . Prove that: $S_{\triangle ABC} = AF \cdot BF$.

Proof 1: Let $AD = AF = m$, $BE = BF = n$, $BC = a$, $AC = b$, then $a^2 + b^2 = (m+n)^2$, $m - n = b - a$, the solution is $mn = \frac{1}{2}ab$.

$$S_{\triangle ABC} = \frac{1}{2}ab = AF \cdot BF \text{ So.}$$

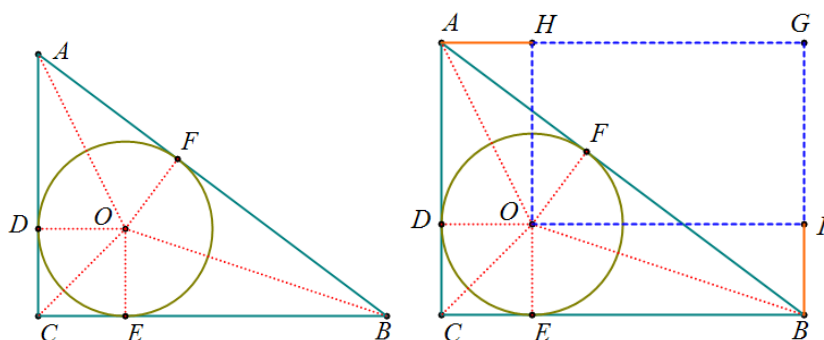
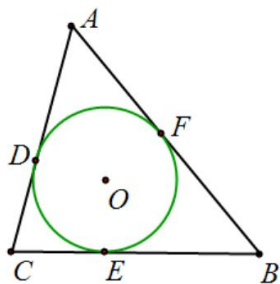


Figure 7 Figure 8

Proof 2: $S_{\triangle AOF} = S_{\triangle AOD} = S_{\triangle AOH}$, $S_{\triangle BOF} = S_{\triangle BOE} = S_{\triangle BOI}$, therefore

$$S_{\triangle ABC} + S_{OIGH} = S_{CBGA} = 2S_{\triangle ABC}, \text{ therefore } S_{\triangle ABC} = S_{OIGH} = AD \cdot BE = AF \cdot BF.$$

the two proofs, Proof 2 requires additional drawing, but Proof 2 does have its significance. Because of this intuitive understanding, the conclusion can be seen at a glance.



Proof: Let the inscribed circle be the unit circle, then $\bar{d} = \frac{1}{d}$, $\bar{e} = \frac{1}{e}$, $\bar{f} = \frac{1}{f}$,

$$a = \frac{2df}{d+f}, \bar{a} = \frac{2}{d+f}, b = \frac{2ef}{e+f}, \bar{b} = \frac{2}{e+f}, c = \frac{2ed}{e+d}, \bar{c} = \frac{2}{e+d}, \text{ identities:}$$

$$\frac{\frac{1}{4} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}}{(a-f)(\overline{f-b})} = \frac{d-e}{d+e},$$

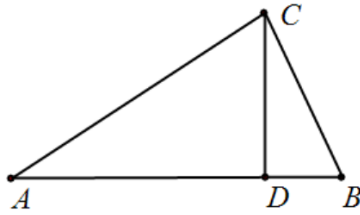
$\triangle ABC$, its incircle O is tangent to the three sides at D , E , and F . Prove that: $S_{\triangle ABC} = AF \cdot FB \cot \frac{C}{2}$.

What if $\angle C$ is not 90° ? Will there be a conclusion $S_{\triangle ABC} = AF \cdot FB \sin C$? It turns out that the guess is wrong. Among the common special trigonometric function values, in addition to $\sin 90^\circ = 1$, there are also $\tan 45^\circ = \cot 45^\circ = 1$.

Controversial Issue

Example 1 : As shown in Figure 1, in $\triangle ABC$, D is a point on line segment AB . Prove that:

- (1) If $CA \perp CB$, $AD \cdot AB = AC^2$, then $DB \cdot DA = DC^2$.
- (2) like $CA \perp CB$, $DB \cdot DA = DC^2$, but $AD \cdot AB = AC^2$.
- (3) If $DB \cdot DA = DC^2$, $AD \cdot AB = AC^2$, then $CA \perp CB$. (Complex number identity extension projection theorem)



Proof: Let $A = (a, a)$, $B = (b, b)$, $C = (c, \bar{c})$, $D = (d, d)$,

$$(c-a)\overline{(c-a)}\left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a}\right)}\right) = 2ab - ac - bc - a\bar{c} - b\bar{c} + 2c\bar{c},$$

$$(a-c)\overline{(a-c)} - (a-d)(a-b) = ab - ac - a\bar{c} + c\bar{c} + ad - bd,$$

$$(d-c)\overline{(d-c)} - (a-d)(d-b) = ab + c\bar{c} - ad - bd - cd - \bar{c}d + 2d^2,$$

First solve $2ab - ac - bc - a\bar{c} - b\bar{c} + 2c\bar{c} = 0$, we get $\bar{c} = \frac{2ab - ac - bc}{a + b - 2c}$, so

$$\frac{(a-c)\overline{(a-c)} - (a-d)(a-b)}{(d-c)\overline{(d-c)} - (a-d)(d-b)} = -\frac{a-b}{a+b-2d}.$$

$$\frac{\frac{a+b-2d}{a-b} \left((a-c)\overline{(a-c)} - (a-d)(a-b) \right) + (d-c)\overline{(d-c)} - (a-d)(d-b)}{(c-a)\overline{(c-a)}\left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a}\right)}\right)} = \frac{a-d}{a-b},$$

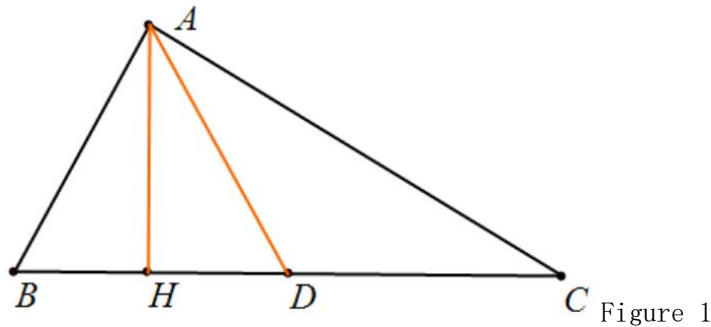
Extended proposition: As shown in Figure 1, in $\triangle ABC$, D is a point on line segment AB . Prove:

$$\frac{DA - DB}{AB} (AC^2 - AD \cdot AB) + (DC^2 - AD \cdot DB) = 2 \frac{AD}{AB} (CA \cdot CB \cos C).$$

Explanation: Through the identity equation, we find that if $CA \perp CB$

$DB \cdot DA = DC^2$, then $AD \cdot AB = AC^2$ D is the midpoint of AB .

Example 2 : In $\triangle ABC$, AD and AH are the median and height respectively. Prove that: $\angle CAD = \angle HAB \Leftrightarrow AB \perp AC$. (2004 Turkish Mathematics Competition)



Proof: Let BC be the real axis and HA be the imaginary axis.

$$\begin{aligned} & (a-d)(-a-d)(a-h)(-a-h) \begin{pmatrix} \frac{a-c}{a-b} & \frac{-a-c}{-a-b} \\ \frac{a-d}{a-h} & \frac{-a-d}{-a-h} \end{pmatrix} \\ &= \frac{a}{2(b-c)}(a-b)(-a-b) \left(\frac{a-c}{a-b} + \frac{-a-c}{-a-b} \right) ((a-b)(-a-b) - (a-c)(-a-c)), \end{aligned}$$

Note: $\angle CAD = \angle HAB \Leftrightarrow AB \perp AC$ Or $A = 90^\circ$.

the acute angle ABC , AD and AH are the median and height respectively. Prove that $2BC \cdot AD \sin(\angle CAD - \angle HAB) = (AC^2 - AB^2) \cos A$:

Why do we need to add the limit of acute angles? Because the conclusion is also true in Figure 1 and Figure 2, but not in Figure 3. Originally, H was on the right side of line AB , but it moved to the left side in Figure 3. The order of the lines has changed, so the conclusion needs to be changed

$$2BC \cdot AD \sin(\angle CAD + \angle HAB) = (AC^2 - AB^2) \cos A.$$

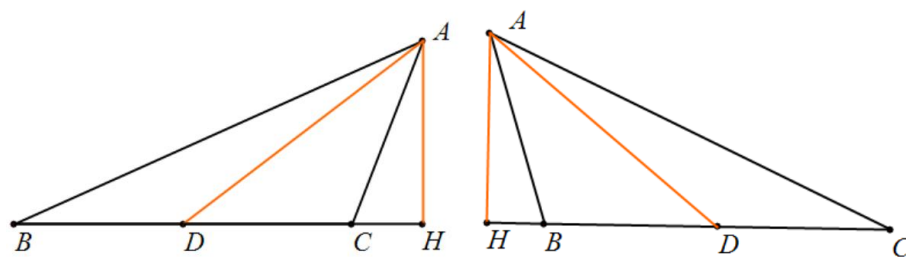


Figure 2 Figure 3

Example 3 : In triangle ABC , CH is the highest. Given $\frac{BC^2}{AC^2} = \frac{BH}{AH}$, prove: $\angle$

$C = 90^\circ$ or $\angle A = \angle B$.

Proof: Let the circumcircle of $\triangle ABC$ be the unit circle, $h = \frac{a+b+c-ab/c}{2}$,

$$4\left((a-c)^2 \overline{(a-c)}^2 (b-h) \overline{(b-h)} - (b-c)^2 \overline{(b-c)}^2 (a-h) \overline{(a-h)}\right)$$

$$= \frac{(a-b)^5 (a-c)}{a^3 b^2 (b-c)} \left(\frac{\frac{a-c}{a-b}}{\frac{b-a}{b-c}} - \frac{\overline{\left(\frac{a-c}{a-b}\right)}}{\overline{\left(\frac{b-a}{b-c}\right)}} \right) \left(\frac{\frac{a-c}{a-b}}{\frac{b-a}{b-c}} + \frac{\overline{\left(\frac{a-c}{a-b}\right)}}{\overline{\left(\frac{b-a}{b-c}\right)}} \right) \left(\frac{c-b}{c-a} + \overline{\left(\frac{c-b}{c-a}\right)} \right),$$

$\triangle ABC$, O is the circumcenter, CH is the height, prove:

$$(CA^2 \cdot BH)^2 - (CB^2 \cdot AH)^2 = 2AB \cdot OC \cdot CA^2 \cdot BC^2 \sin(\angle A - \angle B) \cos(\angle A - \angle B) \cos C.$$

Note: After establishing the identity, it is easy to find that the original question has loopholes, and it is corrected to: In $\triangle ABC$, CH is the height.

Given $\frac{BC^2}{AC^2} = \frac{BH}{AH}$, prove: $\angle C = 90^\circ$ or $\angle A = \angle B$ or $\angle A$ and $\angle B$ differ by

90° .

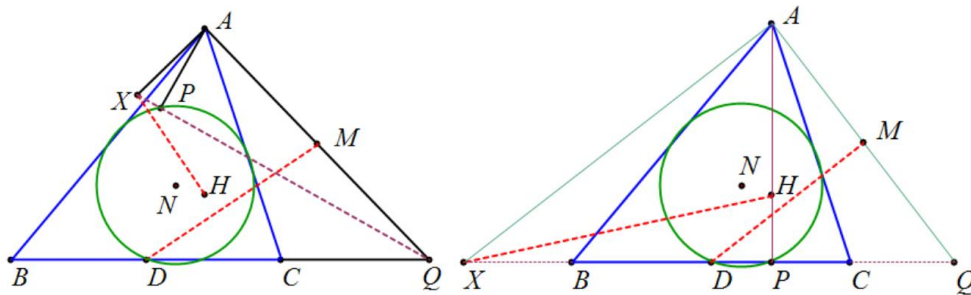


Figure 1 Figure 2

Example 4 : As shown in Figure 1 , P is a point on the nine-point circle of triangle ABC . Draw a perpendicular line through P to AP , intersecting the straight line BC at point Q , and draw a perpendicular line through A to AQ , intersecting the straight line PQ at point X . Let H be the orthocenter of triangle ABC , D and M be the midpoints of BC and AQ respectively : Prove that: $HX \perp DM$. (2025 China Training Team Selection Test Questions)

Suppose the circumscribed circle of $\triangle ABC$ is the unit circle, $O=0$, $H=A+B+C$, $N=\frac{H}{2}$, and according to the problem, find the complex coordinates of the point and generate the identity equation

$$[PBC]|d-m|^2\left(\frac{h-x}{d-m}+\overline{\left(\frac{h-x}{d-m}\right)}\right)=2[ABC]\left(|p-n|^2-|d-n|^2\right).$$

Extended proposition:

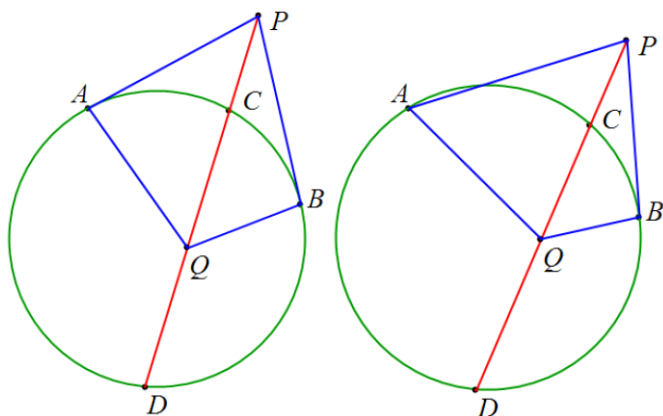
Figure 2 , in triangle ABC , a perpendicular line through P to AP intersects the straight line BC at point Q , and a perpendicular line through A to AQ intersects the straight line PQ at point X . Let H be the orthocenter of triangle ABC , N be the center of the nine-point circle of triangle ABC , D and M be the midpoints of BC and AQ respectively : Prove $S_{\triangle PBC} \cdot DM \cdot HX |\cos \alpha| = S_{\triangle ABC} \cdot |NP^2 - ND^2| :.$

Generally speaking, candidates will not use this counterexample to overturn the question.

But what if there is a candidate who gives a counterexample and proves that there is a problem with the question, will you give him points or not?

According to the correction conclusion given by the computer, there may be problems when P falls on BC . This also includes the overlap of P and D . Because at this time the right side of the equation is 0, but there are two items on the left side that may be 0, which need to be eliminated.

Example 5 : As shown in the figure, two tangent lines and one secant line are drawn through the outer point P of the circle, and the tangent points are A and B . The secant lines intersect the circle at points C and D , and C is between P and D . Take a point Q on the chord CD , so that $\angle DAQ = \angle PBC$. Prove that: $\angle DBQ = \angle PAC$. (Question from the 2003 National High School Mathematics League)



Proof: Let the circle be the unit circle, solve the equation $\frac{c-p}{c-d} = \overline{\left(\frac{c-p}{c-d}\right)}$,

and get $\overline{p} = \frac{c+d-p}{cd}$. Similarly, we get $\overline{q} = \frac{c+d-q}{cd}$

Generates the identity: $\frac{\frac{b-c}{a-c}}{a-d} |b-q|^2 |a-p|^2 \left(\frac{\frac{b-d}{a-p}}{a-c} - \overline{\left(\frac{\frac{b-d}{a-p}}{a-c} \right)} \right)$

$$= |a-d|^2 |b-c|^2 \left(\frac{\frac{a-q}{b-c}}{b-p} - \overline{\left(\frac{\frac{a-q}{b-c}}{b-p} \right)} \right).$$

Extended proposition: It is known that four points A , B , C , and D are in a circle, Q is on the line segment CD , and P is on the extended line of $AQ \cdot BP \sin(\angle DAQ - \angle PBC) = AP \cdot BQ \sin(\angle DBQ - \angle PAC)$ the line segment CD .

Prove: .

The study found that PA and PB are tangents, and this condition is redundant. This question is from the 2003 National High School Mathematics League. As a professional competition, there are very high-level question setters, and after setting the questions, there will be special testers to test the scientificity and difficulty of the questions. Even after going through layers of checks, there are still many cases of redundant test conditions. This is due to the lack of research on whether the test conditions are redundant in the past. In the

derivation process, using a certain condition to prove the conclusion does not mean that the condition is necessary. It may be possible to change the problem-solving method and deduce the conclusion without using this condition. Using the identity algorithm in this article, all conditions are separated one by one, and then reassembled, which makes it easy to identify which conditions are redundant.

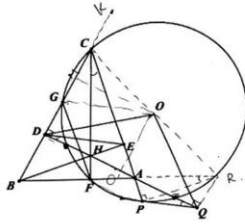
2024 年全国高中联赛北京赛区预赛二试试题

考试时间：9:40-12:30

一（40 分）设 a, b, c 是三个正数，求证：

$$\frac{2a}{\sqrt{2a^2+b^2+c^2}} + \frac{2b}{\sqrt{a^2+2b^2+c^2}} + \frac{2c}{\sqrt{a^2+b^2+2c^2}} \leq \frac{3\sqrt{2}(a+b+c)}{\sqrt{5a^2+5b^2+5c^2+ab+bc+ca}}.$$

二（40 分）如图所示，锐角 $\triangle ABC$ 的三条高线 AD, BE, CF 交于点 H ，过点 F 作 $FG \parallel AC$ 交直线 BC 于点 G ，设 $\triangle CFG$ 的外接圆为 $\odot O$ ， $\odot O$ 与直线 AC 的另一个交点为 P ，过 P 作 $PQ \parallel DE$ 交直线 AD 于点 Q ，连接 OD, OQ ，求证： $OD = OQ$ 。

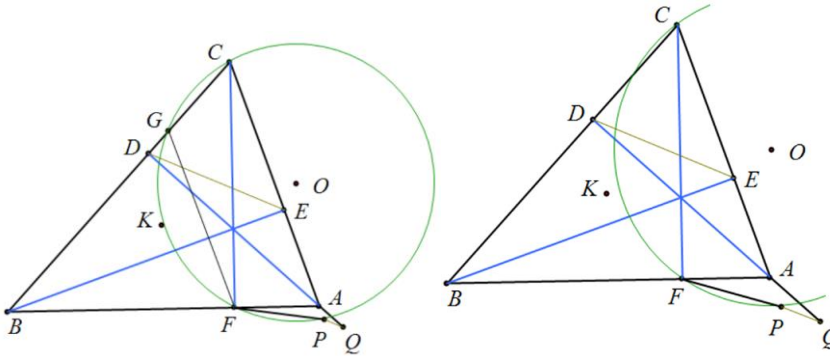


AI provided by Professor Zhu Songchun's team

from Peking University

arXiv preprint arXiv:2412.10673

Example 6 : As shown in the figure, the three altitudes of the acute angle $\triangle ABC$ are AD , BE , and CF . Through point F , draw $FG \parallel AC$ intersecting the straight line BC at point G . Let the circumscribed circle O of $\triangle CFG$ intersect the straight line AC at another point P . Through P , draw $PQ \parallel DE$ intersecting the straight line AD at point Q . Connect OD and OQ . Prove that: $OD = OQ$. (Questions from the Beijing Division of the 2024 National High School League)



Proof: Let the circumcircle of $\triangle ABC$ be the unit circle,

$$D = \left(\frac{a^2 + ab + ac - bc}{2a}, -\frac{a^2 - ab - ac - bc}{2abc} \right),$$

$$E = \left(\frac{ab + b^2 - ac + bc}{2b}, \frac{ab - b^2 + ac + bc}{2abc} \right),$$

$$F = \left(\frac{-ab + ac + bc + c^2}{2c}, \frac{ab + ac + bc - c^2}{2abc} \right) \quad , \quad P = \left(p, \frac{a + c - p}{ac} \right) \quad ,$$

$$Q = \left(q, -\frac{a^2 - bc - aq}{abc} \right) \quad , \quad O = \left(\frac{-ab + ac + bc + c^2 + 2bp}{2(b+c)}, \frac{ab + 3ac + bc + c^2 - 2cp}{2ac(b+c)} \right) \quad ,$$

$$[ABK][BCK](|o-d|^2 - |o-q|^2) = [BCQ]([QDE] - [PDE]) \quad ,$$

Extended proposition: As shown in the figure, the three altitudes of the acute angle $\triangle ABC$ are AD , BE , and CF , and point P is on the extended line of line segment CA . Let the circumscribed circle of $\triangle CFP$ be O , and take point Q on the straight line AD , prove: $S_{\triangle ABK} \cdot S_{\triangle BCK} (OD^2 - OQ^2) = S_{\triangle BCQ} \cdot KA^2 (S_{\triangle PDE} - S_{\triangle QDE})$.

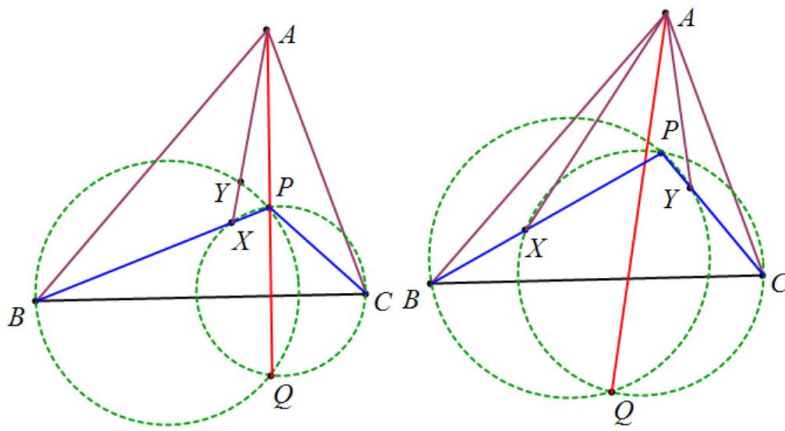
Note: If we remove $FG \parallel AC$, we can further remove G without affecting the conclusion.

Example 7 : In triangle ABC , $AB > AC$, two points X and Y in triangle ABC are both on the bisector of $\angle BAC$, and $\angle ABX = \angle ACY$. Let the extension of BX intersect the line segment CY at point P , and the circumscribed circle ω_1 of triangle BPY intersects the circumscribed circle ω_2 of triangle CPX at another point Q . Prove that points A , P , and Q are collinear. (2021 High School Mathematics League)

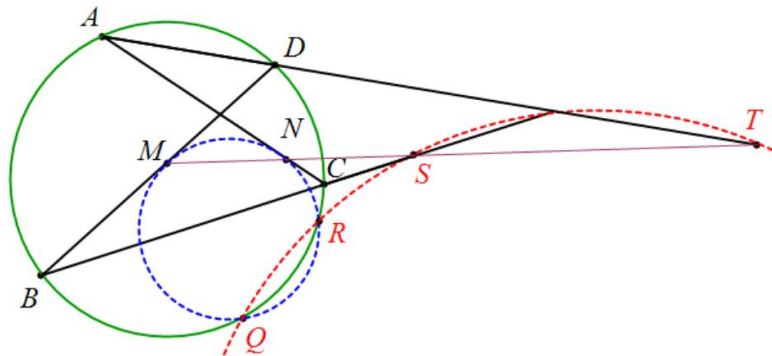
Extended proposition: In triangle ABC , there are two points X and Y , and $\angle ABX = \angle ACY$. Let the extension of BX intersect the line segment CY at point P , and the circumscribed circle ω_1 of triangle BPY intersects the circumscribed circle ω_2 of triangle CPX at another point Q . Prove:

$$S_{\triangle APQ} \cdot S_{\triangle ABY} \cdot BC \sin(\angle CAY) = S_{\triangle CYQ} \cdot S_{\triangle ABP} \cdot AX \sin(\angle CAY - \angle XAB).$$

Note: A , X , and Y do not have to be collinear.



Example 8 : As shown in the figure, circle ω_1 is the circumscribed circle of quadrilateral $ABCD$, straight lines AC and BD intersect at point E , and straight lines AD and BC intersect at point F . Circle ω_2 is tangent to line segments EB and EC at points M and N , respectively, and intersects circle ω_1 at points Q and R . Straight lines BC and AD intersect with line MN at points S and T , respectively. Prove that points Q , R , S , and T are in the same circle. (The 16th Women's Mathematical Olympiad in 2017)



Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle, let

$$M = \left(m, \frac{b+d-m}{bd}\right), \quad N = \left(n, \frac{a+c-n}{ac}\right), \quad \text{we can find } S = (s, \bar{s}), \quad T = (t, \bar{t}),$$

$$Q = \left(q, \frac{1}{q}\right), \quad R = \left(r, \frac{1}{r}\right), \quad \text{the identity}$$

$$\begin{aligned} & ([CMN] - [BMN])([AMN] - [DMN])^2 (b-m)(a-n)|s-r|^2 |q-t|^2 \left(\frac{s-t}{s-r} \frac{q-r}{q-t} - \overline{\left(\frac{s-t}{s-r} \frac{q-r}{q-t} \right)} \right) \\ &= ([NMQ] + [NRQ])[AMN][BMN](b-c)(d-a)|m-q|^2 |r-n|^2 \left(\frac{m-n}{m-q} \frac{r-q}{r-n} - \overline{\left(\frac{m-n}{m-q} \frac{r-q}{r-n} \right)} \right) \end{aligned}$$

, Extended proposition: As shown in the figure, circle ω_1 is the circumscribed circle of quadrilateral $ABCD$, straight lines AC and BD intersect at point E , straight lines AD and BC intersect at point F . Circle ω_2 intersects line segments EB and EC at points M and N respectively, and intersects circle ω_1 at points Q and R . Straight lines BC and AD intersect line MN at points S and T respectively. Prove:

$$(S_{\triangle AMN} - S_{\triangle DMN})^2 (S_{\triangle CMN} - S_{\triangle BMN}) \cdot BM \cdot AN \cdot SR \cdot QT \cdot ST \sin(\angle TSR + \angle TQR)$$

$$= S_{\triangle AMN} S_{\triangle BMN} S_{\triangle ADS} BC \cdot AD \cdot MQ \cdot MN \cdot RN \sin(\angle NMQ + \angle NRQ).$$

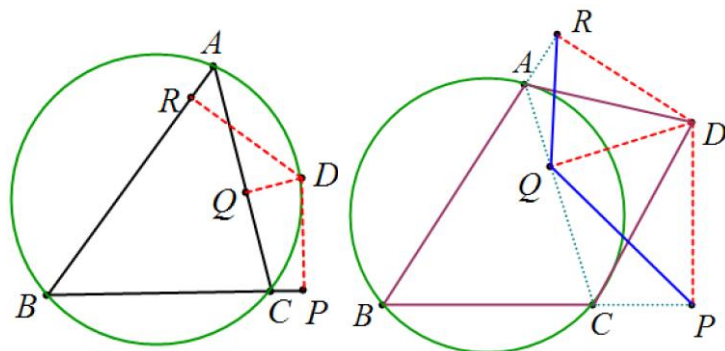
Note: Tangency in the original question is a redundant condition.

Example 9 : In a circle inscribed quadrilateral $ABCD$, points P , Q and R are projections of D to lines BC , CA and AB respectively. Proof: $PQ = QR$ if and only if the intersection of the angle bisectors of $\angle ABC$ and $\angle ADC$ is on AC . (44th IMO test question)

Proof: Let the circumscribed circle of $\triangle ABC$ be the unit circle. According to the properties of the angle bisector, the intersection of the angle bisectors of $\angle ABC$ and $\angle ADC$ on AC is equivalent to $AB \cdot CD = AC \cdot BD$.

$$r = \frac{a+b+d-ab\bar{d}}{2}, \quad p = \frac{c+b+d-cb\bar{d}}{2}, \quad q = \frac{a+c+d-ac\bar{d}}{2},$$

$$4((q-p)(\overline{q-p}) - (q-r)(\overline{q-r})) \\ = (a-b)(\overline{a-b})(c-d)(\overline{c-d}) - (c-b)(\overline{c-b})(a-d)(\overline{a-d}).$$



The study found that the four points A , B , C and D are in a circle and the conditions are redundant.

Quadrilateral $ABCD$, points P , Q and R are projections of D to lines BC , CA and AB respectively. Proof: $4R^2(QP^2 - QR^2) = (AB^2 \cdot CD^2 - BC^2 \cdot AD^2)$, where R is the radius of the circumcircle of $\triangle ABC$.

12. 如图, 在凸四边形 $ABCD$ 中, M 为边 AB 的中点, 且 $MC=MD$. 分别过点 C 、 D 作边 BC 、 AD 的垂线, 设两条垂线的交点为 P . 过点 P 作 $PQ \perp AB$ 于 Q . 求证: $\angle PQC = \angle PQD$.

证明: 如图, 联结 PA 、 PB , 分别取 PA 、 PB 的中点 E 、 F ,

联结 EM 、 ED 、 FM 、 FC .

则四边形 $PEMF$ 为平行四边形.....5 分

从而, $\angle PEM = \angle PFM$.

由 $ME = \frac{1}{2}BP = CF$, $MF = \frac{1}{2}AP = DE$, $MD = MC$

所以, $\triangle DEM \cong \triangle MFC$10 分

即, $\angle DEM = \angle MFC$.

所以, $\angle PED = \angle DEM - \angle PEM = \angle MFC - \angle PFM = \angle PFC$.

又, $\angle PED = 2\angle PAD$, $\angle PFC = 2\angle PBC$.

得 $\angle PAD = \angle PBC$.

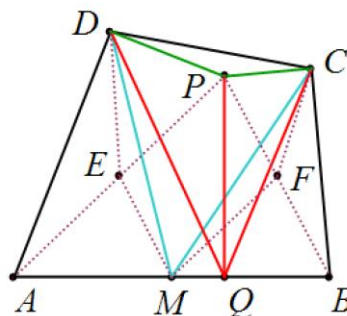
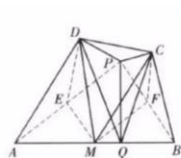
由于 $\angle PQA = \angle PDA = 90^\circ$,

$\angle PQB = \angle PCB = 90^\circ$,

则 P 、 Q 、 A 、 D 和 P 、 Q 、 B 、 C 分别四点共圆.....15 分

故 $\angle PQD = \angle PAD$, $\angle PQC = \angle PBC$.

所以, $\angle PQC = \angle PQD$20 分



Example 10 : As shown in the figure, in the convex quadrilateral $ABCD$, M is the midpoint of side AB , and $MC = MD$. Draw perpendicular lines to sides BC and AD through points C and D respectively, let the intersection of the two perpendicular lines be P , and draw $PQ \perp AB$ at Q through point P . Prove: $\angle PQC = \angle PQD$. (2018 Hunan High School Mathematics Competition)

Proof: Let AB be the real axis, QP be the imaginary axis, $q = 0$, $m = \frac{a+b}{2}$,

solve the equation $\frac{p-d}{a-d} = \overline{\left(\frac{p-d}{a-d}\right)}$, and we get $\bar{p} = \frac{ad + a\bar{d} - 2d\bar{d}}{d - \bar{d}}$. Solve the

equation $\frac{p-c}{b-c} = -\overline{\left(\frac{p-c}{b-c}\right)}$, and we get $\bar{c} = -\frac{c(ad - bd + a\bar{d} + b\bar{d} - 2d\bar{d})}{-ad - bd + 2cd - a\bar{d} + b\bar{d} - 2c\bar{d} + 2d\bar{d}}$.

$$\begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix} \left((m-c)\overline{(m-c)} - (m-d)\overline{(m-d)} \right)$$

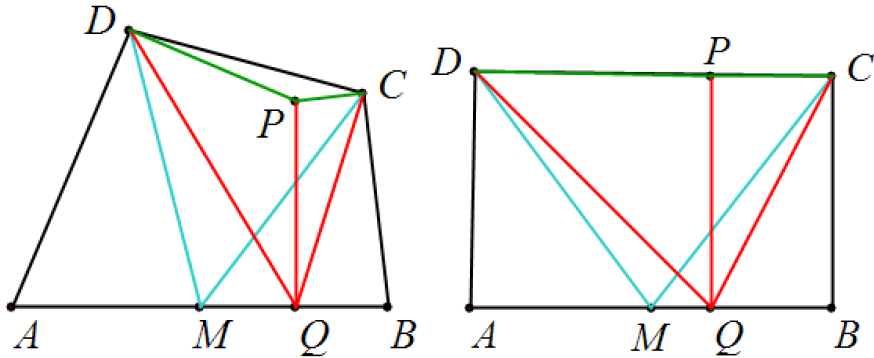
$$= \frac{a-b}{d-\bar{d}} \left(\begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ a & \bar{a} & 1 \end{vmatrix} - \begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} \right) (q-p)\overline{(q-p)} \left(\frac{q-d}{q-p} - \overline{\left(\frac{q-d}{q-p}\right)} \right).$$

As shown in the figure, in the convex quadrilateral $ABCD$, M is the midpoint of side AB . Draw perpendicular lines to sides BC and AD through points C and D respectively, let the intersection of the two perpendicular lines be P , and draw $PQ \perp AB$ at Q through point P . Prove:

$$S_{\triangle ABC} S_{\triangle ABD} (MC^2 - MD^2) = (S_{\triangle BCA} - S_{\triangle BCD}) \cdot AB^2 \cdot QC \cdot QD \sin(\angle PQC - \angle PQD).$$

Note: The identity reminds you of $S_{\triangle BCA} = S_{\triangle BCD}$ the situation you should pay attention to. As shown in Figure 2, it fully meets all the conditions of the

question, but at this time $S_{\triangle BCA} = S_{\triangle BCD}$, and $\angle PQC \neq \angle PQD$.



$$\begin{aligned} & (m-c)(\overline{m-c}) - (m-d)(\overline{m-d}) \\ &= \frac{(ad+bd-2cd+a\bar{d}+b\bar{d}-2d\bar{d})(-2ac+ad+bd+a\bar{d}-b\bar{d}+2c\bar{d}-2d\bar{d})}{2(ad+bd-2cd+a\bar{d}-b\bar{d}+2c\bar{d}-2d\bar{d})}, \end{aligned}$$

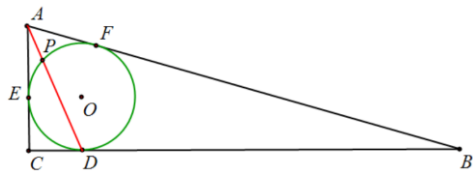
$$\frac{\frac{q-d}{q-p} - \left(\frac{\overline{q-d}}{\overline{q-p}} \right)}{\frac{q-p}{q-c} - \left(\frac{\overline{q-p}}{\overline{q-c}} \right)} = \frac{c(d-\bar{d})^3(ad+bd-2cd+a\bar{d}+b\bar{d}-2d\bar{d})}{(ad+a\bar{d}-2d\bar{d})^2(ad+bd-2cd+a\bar{d}-b\bar{d}+2c\bar{d}-2d\bar{d})},$$

$$\text{Need to be excluded } S_{\triangle BCA} - S_{\triangle BCD} = \begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ a & \bar{a} & 1 \end{vmatrix} - \begin{vmatrix} b & \bar{b} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix}$$

$$= \frac{(c-b)(d-\bar{d})(-2ac+ad+bd+a\bar{d}-b\bar{d}+2c\bar{d}-2d\bar{d})}{ad+bd-2cd+a\bar{d}-b\bar{d}+2c\bar{d}-2d\bar{d}}, \text{ that is to exclude } AD //$$

BC .

$x^5 x^{10} + (-x^6 + x^4) x^9 = 0$
 $4 x^{6^2-4} x^4 x^{6+4} x^{5^2+2} x^{4^2} = 0$
 $x^{4^2} = 0$
GB method deduce $x^5 x^{10} + (-x^6 + x^4) x^9 = 0$
Wu method did not find

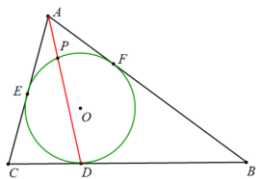


Example 1 1 : In right triangle ABC , $\angle ACB = 90^\circ$, the inscribed circle O of triangle ABC is tangent to the sides BC , CA , and AB at points D , E , and F respectively. Connect AD and intersect the inscribed circle O at point P . Connect BP and CP . If $\angle BPC = 90^\circ$, prove that: $AE + AP = PD$. (2006 China Mathematical Olympiad)

Proof: Let circle O be the unit circle, $a = \frac{2ef}{e+f}$, $b = \frac{2df}{d+f}$, $c = \frac{2ed}{e+d}$, solve

the equation $\frac{a-p}{a-d} = \overline{\left(\frac{a-p}{a-d}\right)}$ to get $p = \frac{de+df-2ef}{2d-e-f}$,

$$\begin{aligned} & [F, E, O][D, E, F] \left((a-e)\overline{(a-e)} - ((p-d)-(a-p))\overline{((p-d)-(a-p))} \right) \\ &= [D, F, O][D, E, O] (e-f)\overline{(e-f)} (p-c)\overline{(p-c)} \left(\frac{p-b}{p-c} + \overline{\left(\frac{p-b}{p-c}\right)} \right). \end{aligned}$$



$\triangle ABC$, the inscribed circle O of $\triangle ABC$ is tangent to the sides BC , CA , and AB at points D , E , and F respectively. Connect AD and it intersects with the inscribed circle O at point P . Connect BP and CP . R is the radius of the inscribed circle. Prove that:

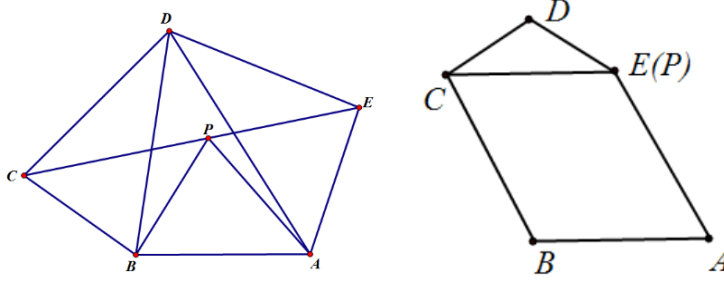
$$\begin{aligned} & R^2 \cdot OE \cdot FD \cdot EF \cdot (AE^2 - (PD - AP)^2) \\ &= 2EF^2 \cdot OD \cdot EO \cdot DF \cdot PC \cdot PB \cos \angle BPC. \end{aligned}$$

Note: According to the identity equation, the original condition $\angle C = 90^\circ$ is redundant.

Failure Question

Example 1 : As shown in the figure, in *the* convex pentagon $ABCDE$, $CD = DE$, $\angle EDC \neq 2\angle ADB$. Point P is inside the pentagon, satisfying $AP = AE$, $BP = BC$.

Prove: P is on the diagonal CE if and only if $S_{\triangle CBD} + S_{\triangle AED} = S_{\triangle BAD} + S_{\triangle BAP}$. (2019 IMO pre -selected question)



$$D = (0, 0), K = (1, 1), P = (p, \bar{p}), C = (c, 2 - c), E = (2 - c, c),$$

$$A = \left(-\frac{2\bar{a} - 2c - \bar{a}c + c^2 - \bar{a}p + p\bar{p}}{c - \bar{p}}, \bar{a} \right), B = \left(\frac{-2c + \bar{b}c + c^2 - \bar{b}p + p\bar{p}}{-2 + c + \bar{p}}, \bar{b} \right),$$

$$\text{Identity}([CBD] + [AED] - [BAD] - [BAP])([BDC] - [BPD])(c - p)(\overline{c - p})(e - c)$$

$$= 2[CPE][CPB](d - c)(\overline{d - c})(d - a)(\overline{d - a}) \left(\frac{\frac{d - k}{d - a} - \frac{\overline{\frac{d - k}{d - a}}}{d - b} \right),$$

Extended proposition: As shown in the figure, in the convex pentagon $ABCDE$, $CD = DE$, point P is inside the pentagon, satisfying $AP = AE$, $BP = BC$. Prove:

$$2S_{\triangle CPB} \cdot DC \cdot DA \cdot DB \sin \left(\frac{\angle EDC}{2} - \angle ADB \right) \sin \angle PCE$$

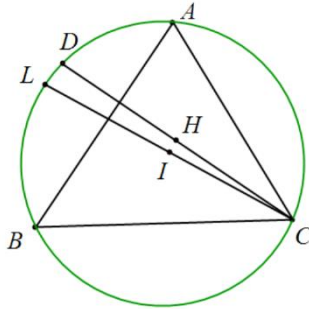
$$= CP \cdot (S_{\triangle BPD} - S_{\triangle BDC}) (S_{\triangle CBD} + S_{\triangle AED} - S_{\triangle BAD} - S_{\triangle BAP}). \quad (\text{Note that it is a directed area and a directed angle})$$

This problem cannot search for non-degenerate conditions $\angle EDC = 2\angle ADB$. This special situation is rarely encountered.

According to the question, after manual addition, a complex number identity is generated.

According to the identity, we construct a counterexample, that is, quadrilateral $ABCD$ is a rhombus, $\angle A = 60^\circ$, $DC = DE$, P and E coincide, and P is on the diagonal CE , which does not hold. However, this counterexample is refuted by

the author's argument that " $S_{\triangle CBD} + S_{\triangle AED} = S_{\triangle BAD} + S_{\triangle BAP}$ point P is inside the pentagon ". So, this author is really watertight!



Example 2 : Given that the orthocenter of acute angle $\triangle ABC$ is H , the incenter is I , and $AC \neq BC$, extend CH and CI to intersect the circumcircle of $\triangle ABC$ at D and L respectively . Prove: $\angle CIH = 90^\circ \Leftrightarrow \angle IDL = 90^\circ$.

the circumscribed circle of $A = a^2 \triangle ABC$ be the unit circle, , $B = b^2$, $C = c^2$,

$$h = a^2 + b^2 + c^2, \quad i = -ab - bc - ca, \quad L = -ab, \quad d = -\frac{a^2 b^2}{c^2}, \quad \text{the inner circle fails}$$

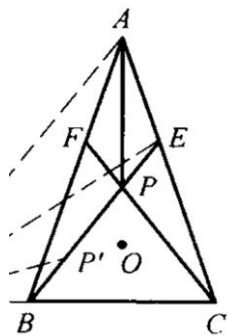
From $IC \perp IH$, we can get :

$$\frac{i-h}{i-c^2} + \overline{\left(\frac{i-h}{i-c^2}\right)} = \frac{(a+b)(a^2 b + ab^2 + 2abc + ac^2 + bc^2)}{ab(a+c)(b+c)} = 0$$

$$\text{It is obtained } \frac{d-i}{d+ab} + \overline{\left(\frac{d-i}{d+ab}\right)} = \frac{a^2 b + ab^2 + 2abc + ac^2 + bc^2}{abc} = 0 \text{ from } DI \perp DL.$$

Failed to generate complex identities and could not find the geometric meaning of the non-degenerate condition.

Example 3 : Let P be a point on the straight line where the bisector of $\angle A$ in $\triangle ABC$ lies, and straight lines BP and CP intersect AC and AB at E and F respectively .
 Proof : If $BE = CF$, then $\triangle ABC$ is an *isosceles* triangle .



Failed to generate complex identities and could not find the geometric meaning of the non-degenerate condition.